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Master's Degree in Industrial Engineering

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Master's Thesis

"Design and modeling of blade geometry in Kaplan and Deriaz turbines: software development and a retrofit proposal."

Juan Fernández Lozano

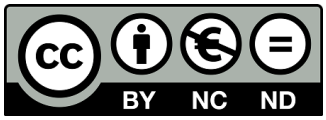
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ABSTRACT

At present, no open-source software exists that can automatically generate the preliminary three-dimensional blade geometry of a Deriaz turbine from global hydraulic parameters alone. This gap is especially relevant given the role that this diagonal-flow turbine type can play in renewable energy integration challenges, both in Spain and worldwide: thanks to its double-regulation mechanism, the Deriaz turbine maintains high efficiency over a much wider load range than the conventional Francis turbine, making it a particularly suitable solution for reversible hydroelectric power plants tasked with offsetting the intermittency of wind and photovoltaic generation.

Starting from this gap, this Master's Thesis lays the computational groundwork for a future open-source software tool capable of designing the optimal blade geometry of Deriaz turbines from their design hydraulic parameters. To this end, the complete analytical derivation of the Euler equation for turbomachines is first developed from the cylindrical coordinate framework and the concept of angular momentum, without assuming any starting kinematic relations. Since no specialized geometric design software exists for the purely axial-flow case either, the methodology is first applied to the Kaplan turbine, which, in addition to covering this second gap, constitutes a simpler and more pedagogical intermediate step before tackling the added complexity of the diagonal case. The formulation is then extended to the Deriaz turbine, incorporating the parametrization through the conical angle γ and the derivation of the meridional velocity in conical-spherical channels, making it possible to generate the complete three-dimensional geometry of the runner without requiring a starting geometry or CFD-based optimization loops.

The methodology is applied to a real case study, the Saucelle II Hydroelectric Power Plant (Duero River, Salamanca), yielding a dimensionally consistent design (150 rpm synchronous speed, spherical radii of 2.0 and 2.6 m, 9 blades according to the solidity criterion, $\sigma = 1.40$). The resulting geometry is validated through STL export to a professional CAD environment and through a physical prototype produced by 3D printing, confirming its geometric consistency and kinematic feasibility, without validating its hydraulic or structural behavior under real load, aspects that remain as future work.

Finally, all of the code developed is released as free and open-source software under the GNU GPLv3 license in the KaplanDeriaz3D open-source project, available on GitHub and linked to MATLAB File Exchange, with the aim of allowing the tool to evolve through community contributions beyond the scope of this work.

Keywords: Deriaz turbine; Kaplan turbine; preliminary blade design; hydraulic turbomachinery; Euler equation; reversible hydroelectric power plant; free open-source software.

DEDICATION

To those who understand knowledge as a common good rather than a commodity. To all those who, altruistically and quietly, give their ideas, discoveries, and tools to the world, making it possible for society's progress to truly belong to everyone.

CONTENTS

1. INTRODUCTION 1
1.1. Motivation: The Storage Challenge in Renewable Integration 1
1.2. Types of hydraulic turbines and their suitability for pumping 3
1.3. The Deriaz turbine: fundamentals and hydrodynamic design philosophy 5
1.3.1. Reversibility, operational efficiency, and selection criteria 7
1.3.2. Relevant Deriaz turbine installations worldwide 9
1.4. State of the art in Deriaz turbine design 10
1.4.1. Turbomachinery design software and its limitations for Deriaz turbine design 12
2. THEORETICAL FOUNDATIONS OF HYDRAULIC TURBOMACHINERY . . 16
2.1. Analysis of the Geometric Framework 16
2.1.1. Kinematic Reference Frame: Cylindrical Coordinates 16
2.1.2. Fundamentals of Angular Momentum and Local-Basis Algebra 18
2.1.3. Analytical Derivation of the Angular Momentum Vector 19
2.1.4. Definition of Mechanical Torque and Derivation of the Euler Equation . . . 20
2.2. Considerations on free vortex and forced vortex flow 23
2.3. Degree of Reaction (R) 24
2.4. Specific Speed (n_q) 25
2.5. Flow Kinematics 25
2.5.1. Velocity Triangles in Radial and Axial Turbomachines 25
3. BLADE DESIGN IN AN AXIAL TURBOMACHINE (KAPLAN TURBINE) . 29
3.1. Relative velocity angle: distribution $\beta(r, z)$ along the axial blade 29
3.2. Evolution of the azimuthal position angle θ along the blade 33
3.3. Construction of the blade's three-dimensional geometry in an axial turbomachine 38
3.4. Determination of blade count and sizing of the inter-blade spacing 40
3.4.1. Justification of the Mean Radius Choice for Chord Integration (L_{chord}) and Determination of Blade Count (Z) 41

3.5. Head and Flow Rate Considerations for the Velocity Triangle 42
3.5.1. Head Losses and Energy Exchange (H_∞) 42
3.5.2. Conservation of Useful Flow Rate ($Q_{\text{runner,actual}}$) 42
3.5.3. Associated Power Values 42
3.5.4. Derivation of the velocity $V_{\theta 1}(r)$ through the Euler Equation 43
3.5.5. Justification of the Calculation Procedure: Euler's Hypothesis 43
3.6. Code developed for calculating and generating the three-dimensional geometry of the blade's mean surface through MATLAB programming 44
3.6.1. Block 1: Energy, geometric, and efficiency parameters 45
3.6.2. Block 2: Fluid kinematics and effective velocities 45
3.6.3. Block 3: Spatial discretization and data matrices 46
3.6.4. Block 4: Velocity triangles and 3D integration (Euler's Hypothesis) 47
3.6.5. Block 5: Calculation of the Blade Count (Z) 48
3.6.6. Blocks 6 and 7: 3D Visualization and Geometric Reconstruction 50
3.7. Geometric optimization of the blade and considerations on the double-regulation mechanism 50
4. NUMERICAL DESIGN OF THE DERIAZ TURBINE 55
4.1. Transition Considerations toward the Deriaz Diagonal Model 55
4.1.1. Transition in the construction of the velocity triangle's components 55
4.2. Distribution of the Meridional Velocity in Curved Channels and Diagonal Turbines. 57
4.3. Analytical calculation of the mean meridional velocity in the Deriaz runner . . 57
4.3.1. Justification of the variation of velocity V_m with the spherical radius (R_e) . 60
4.3.2. Expression of the meridional velocity vector in cylindrical components . . 67
4.3.3. Justification of the kinematic decoupling in the transition toward the draft tube 69
4.4. Geometric Transformation of the Velocity Triangles in the Deriaz Diagonal Turbine 70
4.5. Three-Dimensional Geometric Construction of the Blade 71
4.5.1. Fundamental Differential Equation of the Diagonal Blade 71
4.5.2. Blade Angle Distribution: Cubic Hermite Interpolation (C^1) 72
4.5.3. Discretization and Step-by-Step Numerical Integration 73

4.5.4. Generation of the 3D Cartesian Mesh 74
4.6. Code developed for calculating and generating the three-dimensional geometry of the blade's mean surface through MATLAB programming 74
4.6.1. Block 1: Energy, geometric, and efficiency parameters 74
4.6.2. Block 2: Kinematics and conversions 75
4.6.3. Block 3: Spatial discretization and data matrices 76
4.6.4. Block 4: 3D numerical integration in the conical-spherical channel (Methodological core) 77
4.6.5. Block 5: Calculation of the blade count (Z) 80
4.6.6. Blocks 6 and 7: Three-dimensional rendering of the blade's mean surface and assembly of the complete runner 82
4.7. Geometric Basis and Blade Trimming Post-processing 83
4.7.1. Technical Justification of CAD/CAM Trimming 84
5. PRELIMINARY PROPOSAL TO REPLACE A FRANCIS UNIT WITH A DERIAZ RUNNER 85
5.1. Site selection 85
5.2. Current site characteristics 85
5.2.1. Justification of the Designated Parameters and Runner Design Criteria . . . 86
5.3. Preliminary nature of the design and future validation directions 88
5.4. CAD modeling of the Deriaz turbine's three-dimensional blade shape for the chosen site 88
6. CONCLUSIONS, BUDGET, AND FUTURE WORK 91
6.1. General conclusions 91
6.2. Release as free software: the KaplanDeriaz3D project 92
6.3. Budget 94
6.3.1. Personnel costs 94
6.3.2. Software license costs 94
6.3.3. Other costs 94
6.4. Future work 95
6.4.1. Numerical and experimental validation 95
6.4.2. Refinement of the hydrodynamic model 96
6.4.3. Pump-mode behavior and regulation 96

6.4.4. Structural analysis 97
6.4.5. Control and automation 97
6.4.6. Extension of the software tool 97
BIBLIOGRAPHY. 98
A. APPENDIX A: COMPLETE MATLAB CODE FOR KAPLAN BLADE GENERATION . . . . . .
B. APPENDIX B: COMPLETE MATLAB CODE FOR DERIAZ BLADE GENERATION . . . . . .
C. APPENDIX C: DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GAI) . . . . . .
C.1. Part 1: declaration on legal, ethical, and responsible behavior. . . . . .
C.2. Part 2: declaration of technical use. . . . . .
C.2.1. Documentation and writing . . . . . .
C.2.2. Developing specific content. . . . . .
C.3. Part 3: reflection on usefulness . . . . . .

LIST OF FIGURES

1.1 Cumulative generation structure of the peninsular electrical system 2

1.2 Runners of the three classic types of hydraulic turbine. 4

1.3 Representative diagrams of the flow path in Kaplan, Francis, and Pelton turbines. 4

1.4 Runner of a Deriaz turbine 6

1.5 Flow diagram in a Deriaz turbine 6

1.6 Efficiency curves versus partial load for the main hydraulic turbines . . . 8

1.7 Operating ranges of the main hydraulic turbines 9

1.8 Deriaz pump-turbine of the Takami Hydroelectric Power Plant, Hokkaido (Japan) [12]. 10

1.9 Examples of results obtained with general-purpose computational tools used in turbomachinery design and analysis. 13

1.10 Interface of the commercial software Alab Turbine Designer 14

2.1 Schematic representation of the elements of cylindrical coordinates. Own elaboration. 18

2.2 Schematic representation of the elements of angular momentum. Own elaboration. 20

2.3 Tangential velocity profiles in a free vortex and in a forced vortex.[26] . . 24

2.4 Schematic representation of the velocity triangles in a radial turbine. Own elaboration. 27

2.5 Schematic representation of the velocity triangles in an axial turbine. Own elaboration. 28

3.1 Schematic representation of the decrease in the angle β in an axial velocity triangle as the fluid's tangential component (V_θ) decreases. Own elaboration. 30

3.2 Graph of the cubic interpolation of the angle $\beta(z)$ along the axial axis. Own elaboration. 32

3.3 Schematic representation of three fluid molecules at different angular positions and heights within the runner. Own elaboration. 34

3.4 Schematic representation of the kinetic triangles of the same particle for three values of θ. Own elaboration. 34

3.5 Schematic representation of the fundamental planes that define the blade's orientation in space. Own elaboration. 36

3.6 Schematic representation of the fundamental planes of the same particle for three values of θ and z. Own elaboration. 36

3.7 Three-dimensional rendering results: (a) mean surface of the isolated blade, and (b) runner configuration with the distribution of Z_{optimal} blades in an axial turbomachine. Own elaboration. 50

3.8 Visual validation of the trimming operations on a real runner [27]. Dashed lines delimit the base theoretical volume; green traces the leading-edge translation (sweep); pink marks the flat radial trim of the trailing edge (constant θ_{rel}); and red highlights the mechanical root fillet. Own elaboration. 53

3.9 Three-dimensional visualization in the MATLAB environment of the formulated blade, and delimitation of the industrial cutting boundaries. Own elaboration. 53

4.1 Geometric evolution of the flow-through cross-section along the meridional channel in a Deriaz turbine. The flow-through surface takes the form of a truncated cone orthogonal to the streamlines, centered at C_e and bounded by the spherical radii $R_{e,\text{int}}$ and $R_{e,\text{ext}}$. As the fluid advances and the angle γ increases (from the horizontal $\gamma = 0°$ toward the vertical $\gamma = 90°$), the cylindrical radius's projection decreases, which gives the flow area an intrinsically shrinking nature. Own elaboration. 59

4.2 Balance of pressure forces on a differential fluid element in the direction normal to the path. Own elaboration. 61

4.3 Schematic representation, in the meridional plane, of the evolution of the meridional velocity profile $V_m(R_e, \gamma)$ through the Deriaz turbine's spherical flow-through channel. Own elaboration. 66

4.4 3D rendering of the blade mean-surface geometry and of the diagonal Deriaz runner generated in MATLAB. Own elaboration. 83

5.1 Solid Edge modeling of the geometry generated by the numerical algorithm. Own elaboration. 89

5.2 3D-printed physical prototype of the Deriaz runner, showing the blade articulation on bearings and the absence of interference with the spherical walls of the hub and casing during orientation rotation. Own elaboration. 90

6.1 Graphical interface of the KaplanDeriaz3D application for MATLAB, developed with MATLAB App Designer. Own elaboration. 92

6.2 Graphical interface of the KaplanDeriaz3D application for Python (PySide6 + PyVista). Own elaboration. 93

LIST OF TABLES

1.1 Comparison of design parameters by turbine type. 5

6.1 Estimated budget for the Master's Thesis. 95

1. INTRODUCTION

1.1. Motivation: The Storage Challenge in Renewable Integration

The world's electrical system is undergoing a profound transformation. The penetration of renewable energy sources continues to increase in a growing number of countries, driven both by decarbonization targets and by the progressive improvement in the economic competitiveness of these technologies compared with conventional sources. However, solar photovoltaic and wind power share a structural limitation: their generation depends directly on a variable and uncontrollable primary resource, the sun and the wind, and they lack the intrinsic capacity to store the energy they produce. When the resource is unavailable, whether at night in the case of solar power or during periods of atmospheric calm in the case of wind power, generation is interrupted regardless of the demand existing at that moment.

This intermittency requires the use of energy storage systems that can temporarily decouple generation from consumption. Among the available options are electrochemical batteries, whose deployment has grown markedly in recent years thanks to falling costs, although their large-scale storage capacity over extended periods remains limited compared with other alternatives. Faced with this limitation, pumped-hydro energy storage (PHES) remains the most mature and economically competitive large-scale energy storage technology available today, with decades of commercial operation and a storage capacity that can extend over hours or even days.

The operating principle of pumped-hydro energy storage consists of taking advantage of renewable generation surpluses, once national demand and export commitments to neighboring countries have been met, to pump water from a lower reservoir to an upper one. Subsequently, when the available renewable generation is insufficient to meet demand, or simply when it is economically advantageous, that energy stored as potential energy is released by running the stored water through the turbines, returning electricity to the grid.

This operating pattern can be observed directly in real generation data from the Spanish peninsular electrical system [1]. Figure 1.1 shows the cumulative generation structure for April 1, 2026. During the central hours of the day, coinciding with the peak of solar photovoltaic generation, the contribution of hydropower generation drops markedly, even reaching negative values: this indicates that, instead of generating electricity, reversible hydroelectric power plants are consuming energy from the grid to pump water to the upper reservoir, taking advantage of the solar surplus. Conversely, during nighttime hours, when photovoltaic generation disappears and wind power is not always sufficient to meet demand, the hydropower contribution grows markedly, running the previously stored water through the turbines.

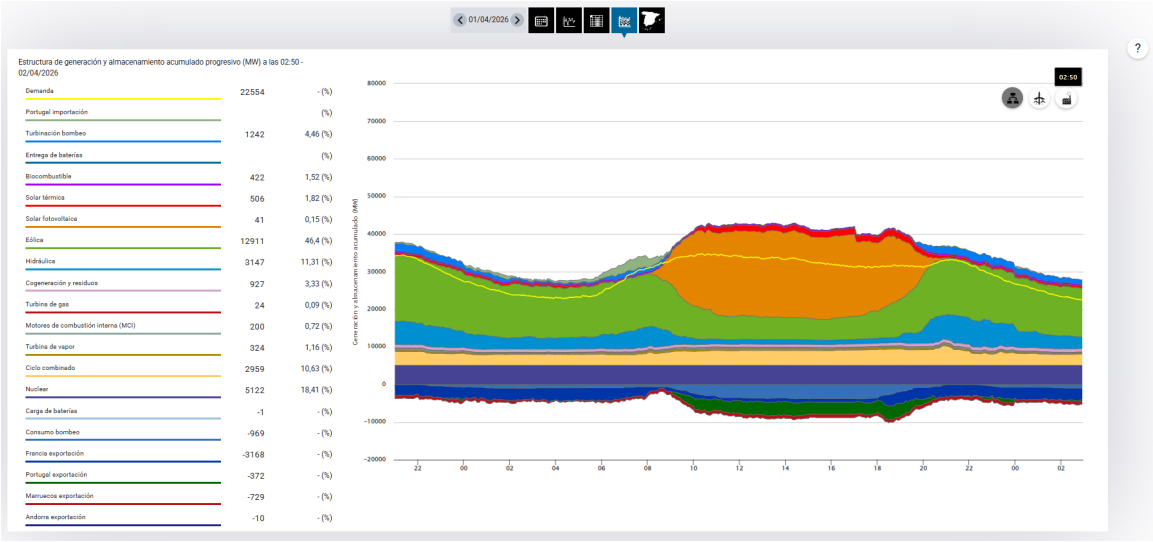


Fig. 1.1. Cumulative generation and storage structure of the Spanish peninsular electrical system, April 1, 2026. The reduction and subsequent negative value of the hydropower contribution can be seen during the hours of peak solar generation, as well as its recovery during the night.

A more detailed analysis of Red Eléctrica's own data viewer for that same day also makes it possible to precisely distinguish the behavior of the categories directly related to pumped-hydro storage, which Red Eléctrica accounts for independently of one another. *Pumping consumption* (energy consumed by reversible power plants to raise water to the upper reservoir) is shown in a muted dark blue always located below the zero line, since by definition it is a consumption rather than a generation, and it is concentrated during the central hours of the day, coinciding with the photovoltaic generation surplus. *Pumped-hydro generation* (energy generated by running previously pumped water through the turbines), by contrast, is shown in a saturated dark blue above the zero line, and its value remains practically zero during those same central hours: this is consistent with the fact that, during that period, reversible power plants are devoting their entire available capacity to pumping water, with no generation surplus available to simultaneously run through the turbines. *Conventional hydropower* (a category accounted for independently of the previous one, shown in a lighter blue at the base of the chart), which corresponds to generation obtained from natural water inflows rather than from prior pumping, also shows a profile with lower generation during the central hours of the day: since it does not depend on prior pumping, these plants do not need to "recharge" anything, but they can modulate their output by reducing the flow run through the turbines during hours of solar surplus, storing in the reservoir the river and rainfall inflow that keeps arriving naturally, so as to release a larger flow during nighttime hours of lower renewable generation. Nevertheless, this reduction in the flow run through the turbines during the day is never reduced all the way to zero, since hydropower generation, being a renewable, dispatchable, and fast-responding source, plays an essential role in the fine adjustment between generation

and demand needed to keep the system frequency within its safety margins. Likewise, by using synchronous generators mechanically coupled to the grid, this type of power plant provides real rotational inertia to the electrical system, a contribution that is increasingly necessary as the penetration of sources such as photovoltaic and, to a large extent, wind power grows, since these connect to the grid through power electronics and provide hardly any natural inertia. These three energy profiles can be verified directly by consulting the original interactive chart at the link provided in reference [1].

For a hydroelectric power plant to be able to operate both in turbine mode and in pump mode, it is necessary to reverse the direction of rotation of the turbine-generator unit, so that the machine, which in its usual operation extracts energy from the water flow to convert it into rotational mechanical energy, instead behaves as a pump that consumes electrical energy to raise water to the upper reservoir. Although pumped-hydro storage as a concept has been in use for decades, its large-scale deployment as an indispensable complement to an electrical system with high renewable penetration is a relatively recent phenomenon, driven precisely by the need to offset the intermittency described above. This circumstance has renewed technical and academic interest in the turbine types best suited to operate reversibly and efficiently in both modes of operation.

This context of operational reversibility is especially relevant in Spain, where the construction of new reservoirs today faces increasingly strict environmental, social, and administrative restrictions, which in practice severely limit the feasibility of new large-scale hydroelectric developments. Given this scenario, the most realistic way to increase the contribution of hydropower to the system is not to build new infrastructure, but to maximize the efficiency and operational flexibility of existing sites through the modernization of their generating units. In this context, replacing fixed-geometry Francis turbines with double-regulation Deriaz runners in existing power plants, as proposed in Chapter 5 for the case of Saucelle II, does not constitute merely a one-off efficiency improvement, but a strategy consistent with the structural need to extract the maximum possible value from the current Spanish hydroelectric fleet.

1.2. Types of hydraulic turbines and their suitability for pumping

Hydraulic turbines are usually classified into two major families according to the physical principle by which they extract energy from the flow. Impulse turbines, whose only relevant commercial representative is the Pelton turbine, operate at atmospheric pressure: the water is first accelerated through one or more injectors until it forms a high-velocity free jet, which strikes tangentially on the bucket-shaped blades of the runner, transferring its kinetic energy by impulse. Reaction turbines, which include the Francis, the Kaplan, and the Deriaz turbine studied in this work, instead operate fully flooded, so that both the pressure and the velocity of the flow vary continuously as it passes through the runner.

Within reaction turbines, the predominant direction of the flow as it passes through

the runner makes it possible to distinguish between radial-flow machines, in which water enters and leaves the runner in directions perpendicular to the axis of rotation, axial-flow machines, in which the flow runs parallel to the axis along its entire path, and diagonal- or mixed-flow machines, in which the flow path combines radial and axial components. The Francis turbine is the characteristic example of mixed flow, with radial inlet and progressively axialized outlet; the Kaplan turbine represents the purely axial-flow case; and the Deriaz turbine, as will be developed in the following sections, occupies an intermediate, diagonal position between the two.

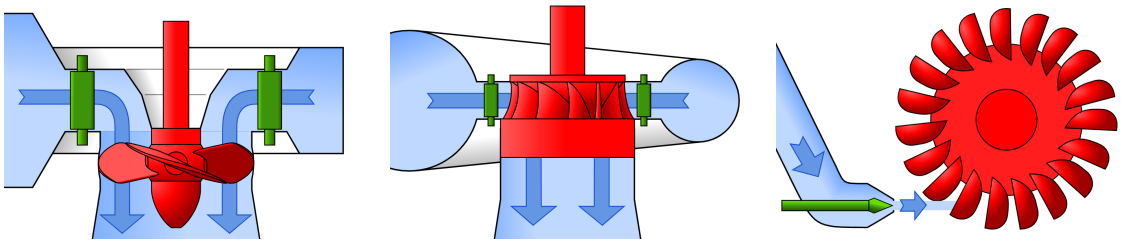
Figure 1.2 shows the runners of the three historically most widely adopted types of hydraulic turbine: Kaplan, Francis, and Pelton.



(a) Kaplan runner [2] (b) Francis runner [3] (c) Pelton runner [4]

Fig. 1.2. Runners of the three classic types of hydraulic turbine.

Figure 1.3 schematically illustrates the flow path inside Kaplan, Francis, and Pelton runners. The purely axial character of the flow in the Kaplan turbine can be seen, as can the mixed nature of the Francis turbine, with radial inlet and progressively axialized outlet, and the free-jet action striking tangentially on the buckets in the Pelton turbine.



(a) Flow diagram in a Kaplan turbine [5] (b) Flow diagram in a Francis turbine [6] (c) Flow diagram in a Pelton turbine [7]

Fig. 1.3. Representative diagrams of the flow path in Kaplan, Francis, and Pelton turbines.

The Pelton turbine, owing to its impulse nature and its operation at atmospheric pressure, is structurally unsuited to reversibility: there is no commercially viable equivalent impulse-pump configuration, so installations using Pelton turbines do not usually offer pumping capacity with the same unit, instead relying on separate pumps when that func-

tionality is required. Its range of application is also restricted to very high heads and comparatively low flow rates.

The Francis turbine is, by far, the most widely used type of reversible pump-turbine in the world today, since its reaction nature allows the direction of rotation to be reversed and the runner to be adapted, usually with a slightly larger diameter and fewer blades than in its turbine-only version, so that it can also operate as a centrifugal pump. However, being a fixed-blade machine, its efficiency degrades markedly away from its design point, especially under significant flow-rate variations, which limits its operational flexibility.

The Kaplan turbine, axial-flow with adjustable blades, is used in its reversible form mainly in low-head applications, where its high specific speed makes it especially suitable. Its ability to adjust the blade angle gives it a flatter efficiency curve than a fixed-blade Francis turbine, although its suitability is limited to low heads, with its structural feasibility penalized as the available head increases.

Table 1.1 summarizes the characteristic ranges of hydraulic head, specific speed, and flow nature of these three types, together with the Deriaz turbine presented below.

Turbine Type	Hydraulic Head (H)	Specific Speed (n_q)	Flow Nature
Pelton	Very high (100 – 1800 m)	Very low	Impulse / Free jet
Francis	Medium - High (30 – 500 m)	Low - Medium	Radial / Mixed
Deriaz	Medium (20 – 120 m)	Medium - High	Diagonal
Kaplan	Low (2 – 40 m)	High	Axial

TABLE 1.1. COMPARISON OF DESIGN PARAMETERS BY TURBINE TYPE.

As can be seen in the table, the Deriaz turbine occupies an intermediate position between the Francis and the Kaplan in both head and specific speed, which, together with its double-regulation capability, makes it a particularly attractive candidate for medium-head reversible applications, as developed in detail in the following sections.

1.3. The Deriaz turbine: fundamentals and hydrodynamic design philosophy

The Deriaz turbine, photographed in Figure 1.4 and shown schematically in Figure 1.5, is classified as a diagonal-flow hydraulic machine with adjustable blades, and represents an intermediate technological solution between the Francis and Kaplan turbines described above. Its architecture allows it to operate reversibly, acting as a turbine for electricity generation and as a pump for pumped-storage, while maintaining high efficiency in both regimes. Unlike fixed-pitch turbines, the ability to vary the angular position of the runner blades allows kinematic adaptation to variations in net head (H) and flow rate (Q), optimizing the velocity triangle at the runner inlet and outlet.

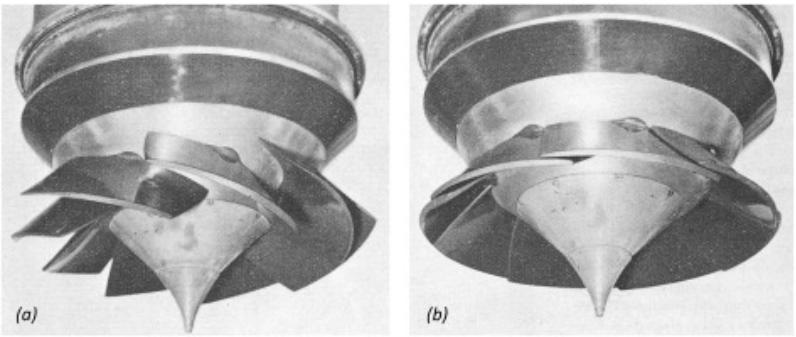


Fig. 1.4. Views of the runner of a Deriaz turbine, showing the diagonal orientation of the blades.[8]

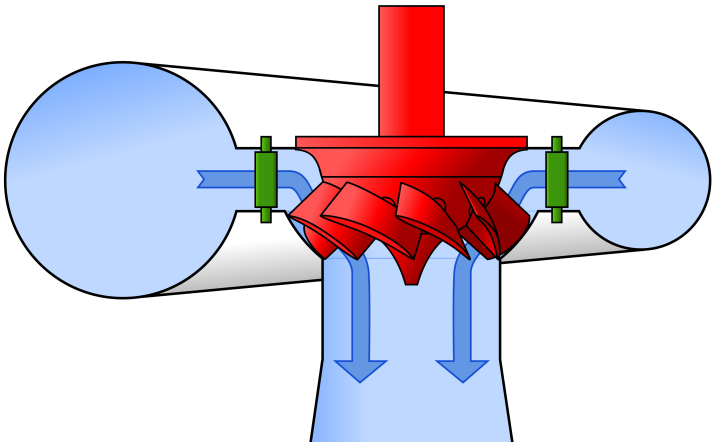


Fig. 1.5. Schematic representation of the flow path in a Deriaz turbine.[9]

Unlike wind turbomachines, whose operating principle is based on slender airfoils acting in an unconfined, effectively infinite medium through dominant lift forces ($L = \frac{1}{2}\rho V^2 C_L A$), the design of mixed- and axial-flow hydraulic turbines, such as the Deriaz and Kaplan geometries, is governed by physics dominated by significant pressure variations and the high density of the fluid ($\rho \approx 1000$ kg/m^3). In this scenario, the runner does not operate in isolation, but as a collapsed cascade of blades through which the flow is forcibly guided through inter-blade channels.

Consequently, energy transfer is not entrusted solely to optimizing the kinematic coefficients of a catalog airfoil profile, but mainly to the controlled angular deflection of the water mass flow. This deflection is strictly governed by the general Euler equation and is represented geometrically through the velocity triangles along each streamline. The distribution of blade relative-velocity angles $\beta(r, z)$ defines the camber line, or skeleton, of the blade, ensuring that the fluid enters without shock, that is, with optimal kinematic adaptation at the inlet, and discharges with minimum residual kinetic energy. The thickness of the airfoil profile thus becomes a second-order variable at this design stage, intended to

provide the blade with structural stiffness and to mitigate local pressure gradients so as to reduce the risk of cavitation.

One aspect that admits no design alternative is the spherical geometry of the inner and outer cavity of the Deriaz runner. This constraint does not stem from an aesthetic preference or a computational simplification, but from an unavoidable geometric condition imposed by the double-regulation mechanism itself: for each blade to be able to rotate freely about its trunnion axis, changing its constructive angle β_1 without colliding with the hub or casing walls, every point on its inner edge must remain at a constant distance from the center of rotation as it rotates, and the same must hold simultaneously for the outer edge with respect to the casing. The only surface of revolution that satisfies this condition for any orientation angle is the sphere, precisely because all of its points are equidistant from a single center. A conical or hyperbolic geometry, such as the one that usually defines the meridional contour of a Francis runner, does not offer this property: as the blade rotates about its trunnion, different points on its edge would trace paths at varying distances from the wall, resulting in mechanical interference with the hub or casing at certain pitch angles, or excessive clearances at others, incompatible with efficient sealing of the flow. For this reason, the spherical hub and the concentric enclosing casing are not an arbitrary feature of the Deriaz turbine, but the only geometric solution that makes the simultaneous articulation of the blades mechanically feasible without compromising the runner's sealing.

1.3.1. Reversibility, operational efficiency, and selection criteria

The superiority of the Deriaz turbine in reversible hydroelectric power plants, compared with the Francis or Kaplan types, is based on two critical mechanical principles.

First, the contribution of the centrifugal field. In purely axial-flow machines (Kaplan), the inlet radius is equivalent to the outlet radius ($r_1 \approx r_2$), which implies that the peripheral velocities are equal ($U_1 \approx U_2$) and, therefore, that the centrifugal term of the Euler equation ($U_1^2 - U_2^2)/2$ (explained in Section 2.1.4) is zero: the entire head must be obtained from the absolute and relative velocity components of the velocity triangle. In the diagonal configuration of the Deriaz, by contrast, the fluid moves toward larger radii as it passes through the runner in pump mode, so that this centrifugal term is non-zero and contributes directly to the head, at no additional cost in terms of relative velocity. This allows the Deriaz's velocity triangle, for the same target head, to be less demanding than that of an equivalent axial machine, reducing the maximum relative velocity and, with it, the cavitation risk associated with the blade.

Second, regulation through blade orientation. The main limitation of fixed-blade Francis turbines is the degradation of efficiency under variations in nominal flow rate (Figure 1.6): as the flow rate decreases, the velocity triangle becomes distorted, causing shock phenomena and boundary-layer separation. The Deriaz turbine solves this problem through double regulation, combining the adjustment of the distributor, which controls

the absolute incidence angle (α_1), with the adjustment of the runner itself, which changes the constructive blade angle (β_1) to ensure that the relative velocity ($\mathbf{w}_1$) remains tangential to the profile over a wider load range. This mechanism results in a noticeably flatter efficiency curve, minimizing energy losses under partial-load regimes, in both turbine and pump modes.

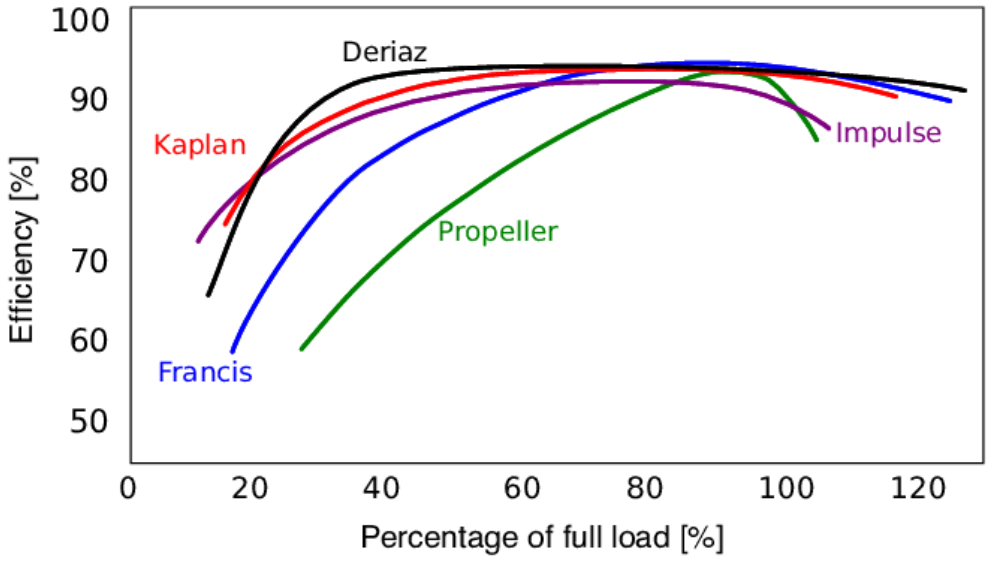


Fig. 1.6. Comparison of efficiency as a function of the percentage of full load for the main types of hydraulic turbine (Deriaz, Kaplan, Francis, Propeller, and Impulse). The Deriaz turbine can be seen to maintain high efficiency over a noticeably wider load range than the Francis, whose efficiency drops abruptly below 70% of full load. [8]

The selection of the Deriaz turbine is justified technically and economically in terms of the specific speed (n_q) and the available hydraulic head, as summarized in Table 1.1. Figure 1.7 places the Deriaz's range of application in more detail against the other types, both for large and small installations, as a function of the design head and flow rate.

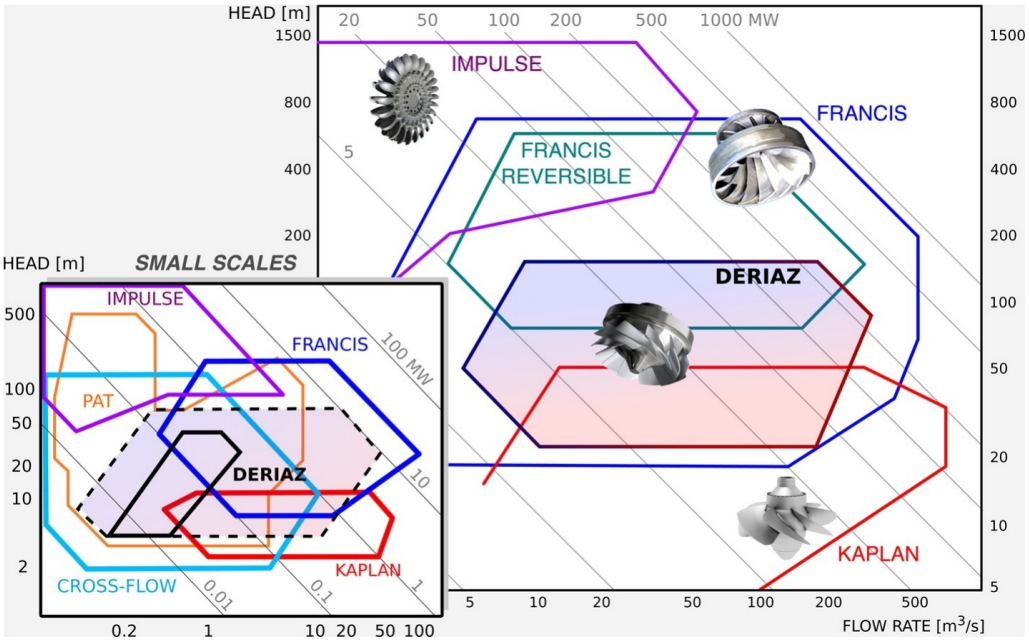


Fig. 1.7. Operating ranges of the main hydraulic turbines as a function of head and flow rate, including detail for small installations.[8]

Compared with the Francis turbine, at sites with high seasonal variability in reservoir levels, the Deriaz mitigates the cavitation risks and structural vibrations that affect fixed-blade machines away from their design point. Compared with the Kaplan turbine, at heads above 50 m axial machines present structural and rotational-speed limitations, whereas the Deriaz's diagonal design provides greater robustness and better handling of high pressures thanks to its radial-flow component.

1.3.2. Relevant Deriaz turbine installations worldwide

Despite being a minority type compared with the Francis or the Kaplan, the Deriaz turbine has real installations documented from the late 1950s to the present day, confirming both the technical maturity of the concept and its continued relevance in recent modernization projects.

One of the first applications at industrial scale was at the Sir Adam Beck pumping complex, on the Niagara River, Ontario, Canada, operated by the Hydro-Electric Power Commission of Ontario. Aeberli [10] describes the commissioning of reversible Deriaz units with adjustable blades capable of operating under unusually wide head variations, switching between pump mode and turbine mode several times a day, which constituted one of the first practical demonstrations of the fundamental advantage of double regulation described in this section.

Another relevant precedent of large-scale industrial application is found in Japan, where various manufacturers developed high-power Deriaz turbines during the 1960s and 1970s. Fuji Electric documents the commissioning of a double-speed Deriaz pump-

turbine for the Kuromatagawa No. 2 Power Plant, operated by Electric Power Development Co. [11]. More recently, Mitsubishi Heavy Industries has supplied Hokkaido Electric Power Co. with two vertical Deriaz pump-turbines of 103 MW unit output for the Takami Power Plant [12], one of the highest-output Deriaz installations in service worldwide. Figure 1.8 shows precisely the Deriaz turbine of the Takami Hydroelectric Power Plant, Hokkaido.



Fig. 1.8. Deriaz pump-turbine of the Takami Hydroelectric Power Plant, Hokkaido (Japan) [12].

Especially relevant to this work is the case of the Valdecañas Hydroelectric Power Plant, located on the Tagus River, in the province of Cáceres, Spain, and owned by Iberdrola. According to Morabito et al. [8], Valdecañas has had Deriaz pump-turbines since 1964 (three 75 MW units), making it one of the oldest and highest-output industrial precedents of this type in Europe. In 2020, the manufacturer Voith signed a contract with Iberdrola for the modernization of the pumping units at the Torrejón and Valdecañas power plants [13], under which the three existing Deriaz units at Valdecañas were replaced with three new ones of 83 MW unit output, commissioned in 2025. This project confirms that the Deriaz type has not only demonstrated reliable, continuous operation in Spain for more than six decades, but that the Spanish hydroelectric industry itself continues to actively invest in it in its current modernization processes.

1.4. State of the art in Deriaz turbine design

The Deriaz turbine, patented by Paul Deriaz in the mid-20th century and conceived using traditional graphical flow-tracing methods [14], was the first diagonal-flow hydraulic turbomachine developed, combining adjustable-pitch blades with a configuration intermediate between the axial flow of a Kaplan turbine and the radial flow of a Francis turbine. Despite its theoretical advantages over the Kaplan and Francis in the medium-head range, its mechanical complexity and the scarcity of open technical literature have historically limited its adoption and study, both from the standpoint of academic research and of the computational tools available for its design.

In recent years, academic interest in the Deriaz turbine has resurged, driven mainly by two factors: its suitability for small- and medium-scale reversible pumping installations in pumped-hydro energy storage (PHES), and its lower blade count compared with an equivalent Francis runner, which reduces the risk of harm to fish fauna passing through the turbine [8], [15].

Morabito et al. presented the design and experimental and numerical validation of a small-scale Deriaz pump-turbine prototype aimed at micro-scale PHES applications, describing in detail the theory and design procedure used so that it could be reproduced by other researchers, and validating the machine's behavior in both turbine and pump modes through CFD simulations benchmarked against experimental data [8]. This is one of the few references that openly document a design methodology for this type of machine; however, that methodology is presented as a procedure described within the article itself, without being accompanied by a computational implementation or publicly available software tool that automates the process.

Kassanos et al. developed a numerical optimization methodology for the design of a reversible Deriaz turbine, aiming to simultaneously maximize energy efficiency and minimize the impact on fish fauna in both modes of operation [15]. The procedure starts from an initial design based on the machine's reference characteristics (power, head, and rotational speed) and defines the runner geometry using Bezier curves and conformal mapping techniques mapped onto the blade surface. The blade angles at inlet and outlet, both at the hub and at the shroud, are treated as free design variables bounded within a predefined range, rather than as quantities derived analytically from the project's hydraulic parameters. A general evolutionary optimization algorithm explores this variable space, evaluating the runner's behavior at each generation using commercial CFD software. This therefore constitutes an indirect solution to the design problem, obtained by repeatedly evaluating the direct problem (computing the flow over a proposed geometry) within an optimization loop, rather than a closed-form formulation that obtains the blade angles and geometry directly from the design hydraulic parameters through the Euler equation and the velocity triangles. Furthermore, the process depends on commercial simulation and optimization software, neither open nor freely accessible.

In industry, modernization interventions on existing Deriaz runners using CFD have also been documented, such as one reported for a hydroelectric power plant in which the blade profile was corrected to improve the hydraulic load distribution and cavitation characteristics, starting from the machine's original geometry and supported by physical model tests [16]. Similarly, Schiffer et al. (Jaberg & Partner) documented the process of designing a double-regulation mixed-flow turbine from scratch for a small-hydro application, explicitly noting that the optimization process for a Deriaz turbine differs substantially from that of a Kaplan or Francis, and that the scarcity of available design recommendations makes it notably difficult to obtain a suitable starting geometry [17].

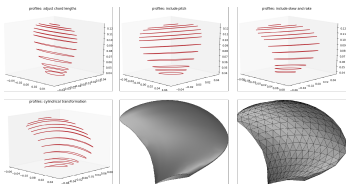
Taken together, these references confirm the resurgence of academic and industrial

interest in the Deriaz turbine in the current context of decarbonization and energy storage, but they also reveal a common gap: the procedures described are methodologies documented in scientific articles or one-off application cases, not reusable, openly accessible software implementations. This absence is not exclusive to the specific literature on Deriaz turbines, but also extends to the broader landscape of turbomachinery design software, as analyzed below.

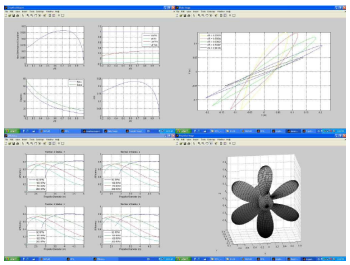
1.4.1. Turbomachinery design software and its limitations for Deriaz turbine design

Alongside the scientific literature reviewed, there are also various computational tools, both open-source and commercial, aimed at turbomachinery design and analysis. However, most of these developments are focused on gas turbines, aeronautical compressors, marine propellers, or Francis or Kaplan turbines, or else they constitute geometric or numerical analysis tools used after the hydraulic design stage, there being a notable absence of tools specifically aimed at the integral hydrodynamic design of diagonal-flow hydraulic turbines of the Deriaz type.

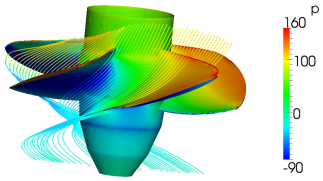
Among the general-purpose tools, *NASA Turbo-Design* stands out, a platform developed in Python for the preliminary design of turbomachinery using radial equilibrium equations, velocity distributions, and aerodynamic analysis, especially useful for axial compressors and gas turbines [18]. In a different area is *BladeX*, conceived as a geometric generator of three-dimensional blades from radial distributions of chord, twist, and airfoil profiles previously defined by the user [19]. On the other hand, *OpenProp*, developed at the Massachusetts Institute of Technology (MIT) and distributed under an open-source (GPL) license, is a well-established tool for the design and analysis of propellers and axial turbines using lifting-line theory and vortex lattices, allowing the blade geometry to be optimized from specified thrust or torque conditions, and even exporting the resulting geometry to CAD tools [20]. Finally, open-source CFD platforms such as OpenFOAM allow pressure distributions, cavitation, hydraulic losses, or energy efficiency to be evaluated through advanced numerical simulation [21]. Figure 1.9 illustrates the type of result provided by each of these three tools.



(a) Blade surface generated with BladeX from radial chord and twist distributions [19].



(b) Geometry of a propeller optimized using lifting-line theory in OpenProp [20].



(c) CFD simulation of the pressure field on the rotor of a Kaplan turbine performed with OpenFOAM [21].

Fig. 1.9. Examples of results obtained with general-purpose computational tools used in turbomachinery design and analysis.

None of these four tools, however, solves the inverse hydrodynamic problem characteristic of the design of an enclosed hydraulic turbine. NASA Turbo-Design does not incorporate specific hydraulic models for Kaplan, Francis, or Deriaz turbines; BladeX does not compute the optimal distribution of blade angles from the design energy conditions, but instead requires these to be provided beforehand; OpenProp, despite optimizing the blade geometry, is formulated for unconfined axial rotors (propellers and horizontal-axis hydrokinetic turbines), without considering the diagonal kinematics or the volute and draft-tube confinement characteristic of a power-plant hydraulic turbine; and OpenFOAM is not strictly a design tool but a subsequent validation tool, so the runner geometry must already be defined using CAD software or external algorithms.

In the specific field of hydraulic turbine design, the *Khoj* tool stands out, developed by the Norwegian University of Science and Technology (NTNU), which allows the hydraulic design of elements such as the spiral casing, guide vanes, runner, and draft tube of Francis turbines to be obtained from the power plant's operating parameters, also providing for connection to CAD and numerical simulation tools [22]. Its scope of application is restricted, however, to Francis turbines of certain specific speeds, using geometric correlations and design criteria specific to conventional radial and mixed-flow machines, without incorporating the diagonal kinematics characteristic of a Deriaz turbine.

A more advanced approach is the one developed by the École Polytechnique Fédérale de Lausanne (EPFL), where an open implementation of a three-dimensional inverse design method for Francis runners and reversible pump-turbines was published: the designer specifies the desired hydraulic load distribution or specific work, and the algorithm computes the runner surface capable of producing that fluid-dynamic behavior [23]. Although it represents one of the most advanced approaches within free software for hydraulic turbomachinery, its formulation is limited to Francis runners and pump-turbines derived from this technology, and it requires the prior definition of the meridional geometry of the hydraulic channel, making it a geometric optimization method applied to an already-defined configuration, rather than a direct design tool based on global hydraulic parameters.

In the industrial sphere, the *Alab Turbine Designer* software also stands out, a commercial platform capable of performing the hydraulic sizing of Francis, Kaplan, Bulb, Pelton turbines, and reversible pump-turbines, estimating losses, generating performance curves, and exporting information to CAD and CFD tools [24]. Unlike the previous tools, it is a proprietary platform and not open-source, and although it covers various types of hydraulic turbines, it likewise does not include specific methodologies for diagonal-flow turbines of the Deriaz type, nor does it provide public information on its internal geometry-generation algorithms.

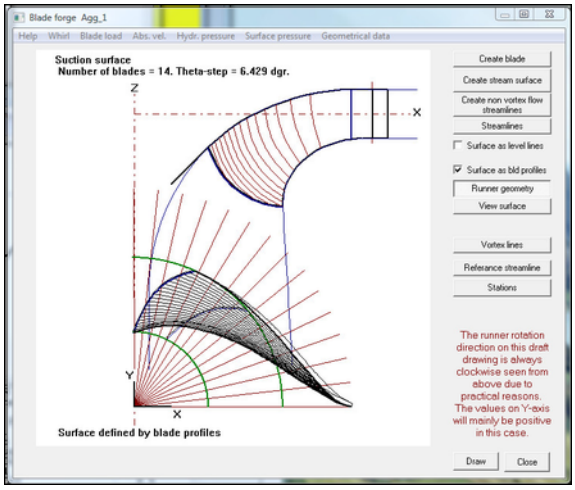


Fig. 1.10. Runner geometry generation module of the commercial software Alab Turbine Designer. [24]

From the combined review of the specific literature on Deriaz turbines and of existing turbomachinery design software, a common conclusion emerges: none of the analyzed references provides an open, publicly accessible computational tool that can obtain the complete runner geometry of a Deriaz turbine from the project's global hydraulic parameters (net head, flow rate, rotational speed, efficiencies, and main dimensions), without a prior geometry or aerodynamic distributions imposed by the user. The procedures described in the literature are methodologies documented in scientific articles, not reusable software implementations; the cases that do employ computational optimization tools depend on commercial, non-free CFD and optimization software; and the existing software tools, whether open or commercial, either do not incorporate the diagonal kinematics characteristic of the flow in a converging conical-spherical channel, or require an already-defined starting geometry. Faced with these limitations, the methodology developed in this Master's Thesis starts directly from the project's global hydraulic variables to compute the velocity distribution through the Euler equation and the velocity triangles, transform the kinematic triangles to the diagonal configuration characteristic of the Deriaz turbine, determine the spatial law of the blade geometry, and finally reconstruct the complete three-dimensional geometry of the runner, additionally implementing this procedure as an open, automated software tool freely accessible to the community, something that, to the

author's knowledge, is not available in either the literature or the equivalent open-source software reviewed.

2. THEORETICAL FOUNDATIONS OF HYDRAULIC TURBOMACHINERY

2.1. Analysis of the Geometric Framework

2.1.1. Kinematic Reference Frame: Cylindrical Coordinates

The kinematic and dynamic analysis of turbomachines is best approached from the cylindrical coordinate reference frame. Since these mechanisms base their operating principle on continuous rotational motion, this system takes advantage of the machine's axial symmetry, simplifying the description of the fluid paths through the runner.

The parametrization of three-dimensional space using cylindrical coordinates is built from an auxiliary reference Cartesian coordinate system, defined by the orthonormal basis $\{\vec{i}, \vec{j}, \vec{k}\}$ and the origin of coordinates located on the machine's axis of rotation. From this global frame, the spatial position of any fluid particle is unified through three scalar variables:

- Radius (r): Represents the minimum radial distance measurable from the fluid particle to the machine's physical axis of rotation.
- Azimuthal angle (θ): Defines the angular position of the particle measured with respect to the chosen reference Cartesian axis in the transverse plane.
- Axial coordinate (z): Determines the longitudinal position of the particle along the generating axis of the cylinder.

From these variables, a moving orthonormal local basis associated with the particle's instantaneous position is defined, denoted by the unit vectors $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_z\}$. The mathematical transformation relation between this local basis and the fixed Cartesian basis is formally established through the following trigonometric projection expressions:

$$\vec{e}_r(\theta) = \cos\theta\,\vec{i} + \sin\theta\,\vec{j} \tag{2.1}$$

$$\vec{e}_\theta(\theta) = -\sin\theta\,\vec{i} + \cos\theta\,\vec{j} \tag{2.2}$$

$$\vec{e}_z = \vec{k} \tag{2.3}$$

It is critical to highlight the local character of this basis: while the axial unit vector $\vec{e}_z$ remains invariant in space, the radial vector $\vec{e}_r(\theta)$ and the azimuthal vector $\vec{e}_\theta(\theta)$ exhibit a direct functional dependence on the angular coordinate θ. Consequently, as the particle undergoes rotation within the machine, the orientation of these vectors varies continuously.

Finally, using the local basis defined above, the generic position vector $\vec{r}$ that geometrically locates the fluid particle at any instant is formulated vectorially as:

$$\vec{r} = r\,\vec{e}_r(\theta) + z\,\vec{e}_z \tag{2.4}$$

From the position vector above, the absolute velocity vector $\vec{V}$ of the particle is derived through direct time differentiation of its position in space, applying the chain rule because of the moving basis's dependence on time:

$$\vec{V} = \frac{d\vec{r}}{dt} = \dot{r}\,\vec{e}_r(\theta) + r\dot{\theta}\,\vec{e}_\theta(\theta) + \dot{z}\,\vec{e}_z \tag{2.5}$$

In this expression, the terms denoted with an overdot represent the ordinary time-variation rates (or first-order time derivatives) of each of the kinematic variables of the fluid-dynamic particle:

- $\dot{r} = \frac{dr}{dt}$ represents the radial velocity of the particle, associated with the expansion or contraction of the streamline with respect to the axis.
- $\dot{\theta} = \frac{d\theta}{dt}$ is the instantaneous azimuthal angular velocity of the particle (directly linked to the peripheral or runner-drag velocity).
- $\dot{z} = \frac{dz}{dt}$ defines the axial or longitudinal velocity along the generating lines of the machine's boundary.

In this way, the absolute velocity vector is orthogonally decomposed into its three native cylindrical components: the radial component ($V_r = \dot{r}$), the azimuthal or tangential component ($V_\theta = r\dot{\theta}$), and the axial component ($V_z = \dot{z}$).

The geometric and physical correspondence of all the scalar variables named above (r, θ, z), as well as the three-dimensional orientation of the unit vectors of the moving local basis and the components of the absolute velocity vector ($\vec{V}$), can be observed schematically in Figure 2.1. This graphical illustration makes it possible to visualize how the orthonormal triad moves and reorients together with the fluid particle's path along the contours of the machine's reference cylinder.

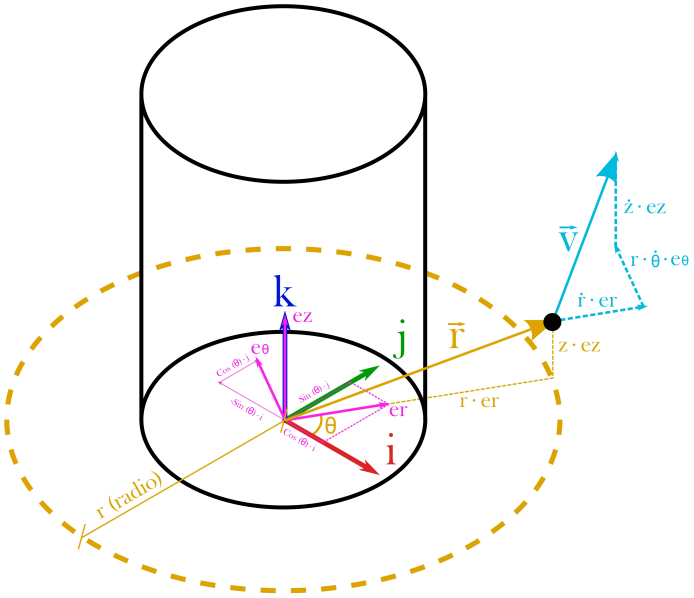


Fig. 2.1. Schematic representation of the elements of cylindrical coordinates. Own elaboration.

2.1.2. Fundamentals of Angular Momentum and Local-Basis Algebra

To analytically derive the fundamental turbomachinery equation, or Euler equation, it is imperative to formally establish the kinematic concept of angular momentum (or moment of momentum) associated with a fluid particle of mass m. The angular momentum vector, denoted $\vec{L}$, is mathematically defined as the cross product between the particle's position vector $\vec{r}$ and its linear momentum vector ($m\vec{V}$):

$$\vec{L} = \vec{r} \times (m\vec{V}) \tag{2.6}$$

From a geometric perspective, the magnitude of the angular momentum vector, $\|\vec{L}\|$, represents the area of the parallelogram bounded by the vectors $\vec{r}$ and $m\vec{V}$. Consequently, the magnitude of the angular momentum is directly proportional to the particle's distance from the origin, to its absolute velocity, and to the sine of the angle between the two vectors; that is, it is maximized when the velocity vector is strictly perpendicular to the position vector.

In the context of turbomachinery, the angular momentum quantifies the fraction of the fluid particle's momentum involved in a displacement that, instantaneously, neither brings the fluid mass closer to nor moves it farther from the machine's reference axis. It is imperative to note that the existence of a velocity perpendicular to the position vector at a specific instant does not presuppose a continuous orbital or circular trajectory around the axis. Therefore, as will be demonstrated analytically below, the velocity components associated with axial (V_z) and radial (V_r) effects, even though they mathematically make

a net contribution to the total angular momentum vector $\vec{L}$ because of this instantaneous perpendicularity, do not contribute at all to the net mechanical torque on the runner shaft. This is because, mechanically, only the components that contribute angular momentum along the axis of rotation $\vec{e}_z$ are taken into account, since all other stresses and vector tilts are compensated for and structurally absorbed by the machine's own bearings and support systems. Consequently, the effective energy transfer and useful work exchange in the runner are linked exclusively to the azimuthal, or tangential, effects of the flow.

Since the analysis is developed in cylindrical coordinates, evaluating the cross product requires defining the algebraic relations that govern the moving orthonormal local basis $\{\vec{e}_r, \vec{e}_\theta, \vec{e}_z\}$. Applying the right-hand rule for right-handed systems, the cross products of the unit vectors cyclically satisfy the following identities:

$$\vec{e}_r \times \vec{e}_\theta = \vec{e}_z, \qquad \vec{e}_\theta \times \vec{e}_z = \vec{e}_r, \qquad \vec{e}_z \times \vec{e}_r = \vec{e}_\theta \tag{2.7}$$

It is necessary to recall the anticommutative character of the cross product, whereby any reversal in the order of the factors changes the sign of the resulting orthogonal vector (for example, $\vec{e}_\theta \times \vec{e}_r = -\vec{e}_z$). Likewise, the cross product of any unit vector with itself is zero ($\vec{e}_r \times \vec{e}_r = \vec{0}$, $\vec{e}_\theta \times \vec{e}_\theta = \vec{0}$, $\vec{e}_z \times \vec{e}_z = \vec{0}$).

2.1.3. Analytical Derivation of the Angular Momentum Vector

Substituting the previously obtained expressions for the position vector $\vec{r} = r\,\vec{e}_r + z\,\vec{e}_z$ and for the absolute velocity vector $\vec{V} = \dot{r}\,\vec{e}_r + r\dot{\theta}\,\vec{e}_\theta + \dot{z}\,\vec{e}_z$, the particle's angular momentum is formulated by expanding the cross product using the distributive property:

$$\vec{L} = (r\,\vec{e}_r + z\,\vec{e}_z) \times m\left(\dot{r}\,\vec{e}_r + r\dot{\theta}\,\vec{e}_\theta + \dot{z}\,\vec{e}_z\right) \tag{2.8}$$

Expanding term by term and applying the properties of the moving local basis, the following is obtained:

$$\begin{aligned}\vec{L} = m\Big[&r\dot{r}\,(\vec{e}_r \times \vec{e}_r) + r(r\dot{\theta})\,(\vec{e}_r \times \vec{e}_\theta) + r\dot{z}\,(\vec{e}_r \times \vec{e}_z) \\ &+ z\dot{r}\,(\vec{e}_z \times \vec{e}_r) + z(r\dot{\theta})\,(\vec{e}_z \times \vec{e}_\theta) + z\dot{z}\,(\vec{e}_z \times \vec{e}_z)\Big]\end{aligned} \tag{2.9}$$

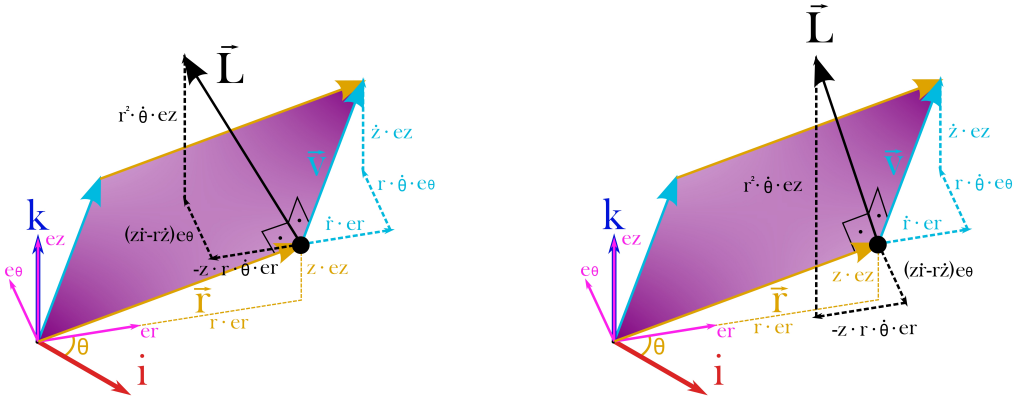
Evaluating the unit cross products according to the right-hand relations, the expression reduces to:

$$\vec{L} = m\Big[r^2\dot{\theta}\,\vec{e}_z - r\dot{z}\,\vec{e}_\theta + z\dot{r}\,\vec{e}_\theta - zr\dot{\theta}\,\vec{e}_r\Big] \tag{2.10}$$

Grouping the vector terms according to the axes of the moving local basis, the fully expanded equation of the angular momentum vector for the fluid particle is obtained:

$$\vec{L} = m\left[-zr\dot{\theta}\,\vec{e}_r + (z\dot{r} - r\dot{z})\,\vec{e}_\theta + r^2\dot{\theta}\,\vec{e}_z\right] \tag{2.11}$$

A schematic representation of the elements of angular momentum according to equation 2.11, including the radial, azimuthal, and axial components, is illustrated in Figure 2.2.



(a) Case where $z\dot{r} > r\dot{z}$. The effect of radial velocity predominates over the azimuthal component of the mechanical angular momentum.

(b) Case where $z\dot{r} < r\dot{z}$. The effect of axial velocity predominates over the azimuthal component of the mechanical angular momentum.

Fig. 2.2. Schematic representation of the elements of angular momentum. Own elaboration.

Using the definitions of the absolute velocity components in cylindrical coordinates ($V_r = \dot{r}$, $V_\theta = r\dot{\theta}$, $V_z = \dot{z}$), the previous equation can alternatively be expressed as:

$$\vec{L} = m\left[-zV_\theta\,\vec{e}_r + (zV_r - rV_z)\,\vec{e}_\theta + rV_\theta\,\vec{e}_z\right] \tag{2.12}$$

2.1.4. Definition of Mechanical Torque and Derivation of the Euler Equation

To quantitatively evaluate the energy transferred or absorbed along the axis of rotation of a turbomachine, it is necessary to introduce the dynamic variable of torque, or moment of a force ($\vec{\tau}$). According to the moment-of-momentum theorem for a particle, torque is formally defined as the time variation of the angular momentum vector:

$$\vec{\tau} = \frac{d\vec{L}}{dt} \tag{2.13}$$

As justified in the preceding sections, the turbomachine's runner is subject to severe structural constraints imposed by its bearings and support systems, which fully absorb any stress or bending in the radial and azimuthal planes. Since only one axis is free for rotation (the machine's physical shaft), the useful mechanical torque, denoted τ_z, is

obtained strictly by projecting the total torque vector onto the axial unit vector $\vec{e}_z$ through a dot product:

$$\tau_z = \vec{\tau} \cdot \vec{e}_z = \left(\frac{d\vec{L}}{dt} \right) \cdot \vec{e}_z = \frac{d}{dt} \left(\vec{L} \cdot \vec{e}_z \right) = \frac{dL_z}{dt} \tag{2.14}$$

Because the dot product filters out and cancels the orthogonal components of the local basis ($\vec{e}_r \cdot \vec{e}_z = 0$ and $\vec{e}_\theta \cdot \vec{e}_z = 0$), the only component that survives the operator is the projection of the angular momentum onto the axial axis (L_z). Substituting the value obtained in the preceding analytical derivation isolates the component:

$$L_z = m\, r^2 \dot{\theta} \tag{2.15}$$

In this expression, it is highly important to establish a rigorous kinematic distinction between two angular variables that are often confused in flow analysis:

- $\dot{\theta}$ represents the instantaneous angular velocity of the *fluid molecule* in its three-dimensional motion.
- ω defines the constant angular velocity of rotation of the *turbomachine's mechanical runner*.

From the kinematics of the system, as indicated in 2.1.3, the absolute tangential linear velocity of the fluid particle is defined as $V_\theta = r\dot{\theta}$. This component shares its vector direction with the machine blade's drag velocity, traditionally denoted U, whose magnitude is $U = r\omega$. Introducing the variable V_θ, the useful component of the angular momentum simplifies to:

$$L_z = m\, r\, V_\theta \tag{2.16}$$

To evaluate the time derivative $\frac{dL_z}{dt}$ in the context of a continuous flow passing through the runner, the Reynolds Transport Theorem is applied over the machine's control volume [25, Cap. 2]. Under the steady-state assumption, the local variation of properties within the runner itself vanishes over time. Consequently, the time variation of the system's angular momentum reduces strictly to the net macroscopic balance between the flow of that property leaving through the outlet section and that entering through the inlet section:

$$\tau_z = \frac{dL_z}{dt} = \dot{m}_{\text{out}}(r_2 V_{\theta 2}) - \dot{m}_{\text{in}}(r_1 V_{\theta 1}) \tag{2.17}$$

Owing to the principle of mass conservation under steady-state conditions, the mass flow rate passing through the turbine remains constant throughout its path, so the inflowing mass is identical to the discharged mass ($\dot{m}_{\text{in}} = \dot{m}_{\text{out}} = \dot{m}$). Extracting that term as a

common factor and identifying the inlet with subscript 1 and the outlet with subscript 2, the definitive expression for the ideal shaft torque is obtained:

$$\tau_z = \dot{m}\,(r_2 V_{\theta 2} - r_1 V_{\theta 1}) \tag{2.18}$$

It is critical to note that in diagonal-flow turbomachines, such as the Deriaz-type turbine, the channel's geometric boundary produces a continuous change in diameter along the machine's longitudinal axis. Consequently, the inlet radius (r_1) differs noticeably from the outlet radius (r_2), preventing that variable from being extracted as a common factor outside the momentum balance. This change in nominal radius is, fundamentally, the direct cause of the energy transfer through centrifugal pressure effects that characterizes the operating behavior of this type of runner.

The definitive step in evaluating the overall energy exchange in the machine is to determine the mechanical power ($\dot{W}$) developed at the shaft. From rotational mechanics, the power transferred from the fluid to the runner (or vice versa) is defined as the product of the useful shaft torque and the constant angular velocity of the mechanical runner (ω):

$$\dot{W} = \tau_z\,\omega \tag{2.19}$$

It is important to note that, while torque was derived from the kinematic behavior of the fluid particles, the angular velocity ω belongs strictly to the solid structure of the turbomachine's runner. Substituting the expression for the ideal torque obtained through the Reynolds Transport Theorem, the internal, or Eulerian, power is rewritten as:

$$\dot{W} = \dot{m}\,(r_2 V_{\theta 2} - r_1 V_{\theta 1})\,\omega = \dot{m}\,(\omega r_2 V_{\theta 2} - \omega r_1 V_{\theta 1}) \tag{2.20}$$

Recalling the definition of the blade's drag velocity ($U = r\omega$), it can be seen that this local property can be evaluated specifically for the geometric conditions of the inlet section ($U_1 = r_1\omega$) and the outlet section ($U_2 = r_2\omega$). Introducing these macroscopic variables into the general balance yields the power equation in terms of the tangential velocities:

$$\dot{W} = \dot{m}\,(U_2 V_{\theta 2} - U_1 V_{\theta 1}) \tag{2.21}$$

Finally, dividing the total power by the system's constant mass flow rate ($\dot{m}$) isolates the specific work per unit mass ($w = \frac{\dot{W}}{\dot{m}}$) exchanged inside the runner. This derivation leads directly to the well-known Euler equation for turbomachines:

$$w = U_2 V_{\theta 2} - U_1 V_{\theta 1} \tag{2.22}$$

This fundamental expression analytically demonstrates that the work transferred in a diagonal-flow machine, such as the Deriaz turbine, is independent of the fluid's interme-

diate thermodynamic paths, being governed exclusively by the kinematic conditions and the blade radii at the runner's intake and discharge sections.

Equation (2.22), although compact, expresses the specific work solely in terms of the tangential velocity components, hiding the individual contribution of each physical mechanism involved in the energy exchange. To make it explicit, it is convenient to introduce the runner's velocity triangle, in which the relative velocity of the fluid with respect to the blade, W, is related to the absolute velocity, V, and the drag velocity, U, through $W = V - U$. Since U is purely tangential, the dot product $U \cdot V = UV_\theta$, from which the following geometric identity is obtained:

$$UV_\theta = \frac{U^2 + V^2 - W^2}{2} \tag{2.23}$$

Substituting identity (2.23), applied to sections 1 and 2, into equation (2.22) gives the so-called three-heads form of the Euler equation:

$$w = \underbrace{\frac{U_2^2 - U_1^2}{2}}_{\text{centrifugal}} + \underbrace{\frac{V_2^2 - V_1^2}{2}}_{\text{absolute}} + \underbrace{\frac{W_1^2 - W_2^2}{2}}_{\text{relative}} \tag{2.24}$$

This second formulation, mathematically equivalent to equation (2.22), is especially useful for the comparative analysis of runner types, since it decomposes the total specific work into three physically independent contributions: the variation in kinetic energy associated with the centrifugal field (dependent solely on the change in radius between inlet and outlet), the variation in the fluid's absolute kinetic energy, and the variation in the kinetic energy relative to the blade.

Note that the expression $w = U_2V_{\theta 2} - U_1V_{\theta 1}$ defines the specific work transferred to the fluid (the usual convention for pumps). For turbine mode, where the fluid gives up energy to the runner, the positive convention is adopted for the extracted work: $w_{\text{turbine}} = -w = U_1V_{\theta 1} - U_2V_{\theta 2}$.

2.2. Considerations on free vortex and forced vortex flow

In axial- and diagonal-flow turbomachinery, such as Kaplan and Deriaz turbines, the fluid's behavior is governed by the sequential transition between two fundamental kinematic regimes. In the transition space between the distributor outlet and the runner's inlet edge, the water flows following a free-vortex regime, conserving its angular momentum with no external torque acting on it. By contrast, once the fluid comes into contact with the runner and is guided by the blades, its kinematics become governed by a forced vortex imposed by the solid's rotation at constant angular velocity (ω).

Figure 2.3 shows a graphical representation of the velocity profile for both types of vortex (free and forced) using streamlines represented by concentric circles of increasing

radius. On the one hand, in a free vortex, the fluid rotates with no external torque acting on it. This implies that the liquid particle satisfies the law of conservation of angular momentum [25, Cap. 6]:

$$V_\theta \cdot r = \text{constante} \implies V_\theta \propto \frac{1}{r} \tag{2.25}$$

As can be seen on the left side of Figure 2.3, due to its zero vorticity, the fluid's tangential velocity (V_θ) is inversely proportional to the radius, reaching its maximum values near the center and decreasing toward the periphery. This behavior exactly describes the kinematics experienced by a water molecule from the moment it leaves the distributor vanes until it reaches the runner's inlet edge, moving freely through the space between the two.

On the other hand, once the fluid comes into contact with the runner, its kinematic behavior changes substantially. Since the blades are fixed to an axis of rotation, the assembly behaves as a rigid body driven at a constant angular velocity (ω). This gives rise to a forced vortex, where the blades' tangential drag velocity (U) follows the expression:

$$U = \omega \cdot r \implies U \propto r \tag{2.26}$$

As illustrated in the right-hand section of Figure 2.3, in a forced vortex the tangential velocity U increases directly proportional to the radius r. Therefore, the blade's geometric design must take this linear variation of blade velocity along its radial span into account to ensure proper kinematic matching of the flow.

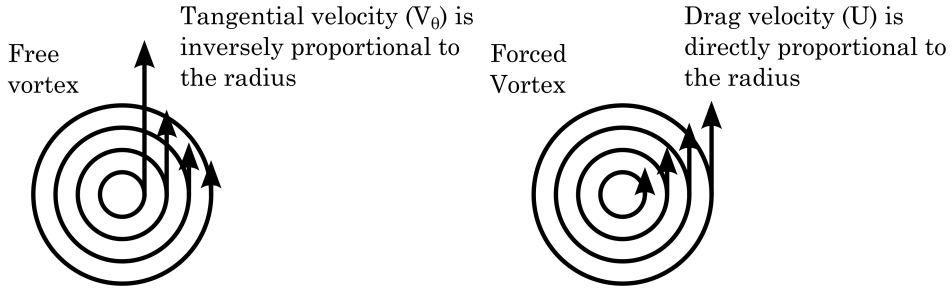


Fig. 2.3. Tangential velocity profiles in a free vortex and in a forced vortex.[26]

2.3. Degree of Reaction (R)

The Deriaz turbine is classified as a reaction machine: unlike impulse turbines (such as the Pelton), in which the entire available pressure drop occurs upstream of the runner, which then only exchanges kinetic energy, in reaction turbines the fluid reaches the runner still at

a pressure above atmospheric and continues giving up static pressure as it passes through the blades. The degree of reaction (R) quantifies this characteristic: it is defined as the fraction of the stage's total static pressure drop (distributor plus runner) that is specifically transferred in the runner,

$$R = \frac{\Delta p_{\text{runner}}}{\Delta p_{\text{distributor}} + \Delta p_{\text{runner}}}, \tag{2.27}$$

such that $R = 0$ would correspond to a purely impulse machine (the entire pressure drop occurs in the distributor) and $R = 1$ to a machine in which the entire pressure drop occurs in the runner. The degree of reaction usually falls within the range $0.5 < R < 0.8$, values similar to those of a high-specific-speed Francis turbine.

2.4. Specific Speed (n_q)

The kinematic specific speed (n_q) of a hydroelectric installation is a dimensionless variable that correlates the algorithm's input operating conditions through the following expression:

$$n_q = n\frac{\sqrt{Q}}{(gH)^{3/4}} \tag{2.28}$$

Where n represents the rotational speed in revolutions per minute, Q denotes the design volumetric flow rate in cubic meters per second, g is the acceleration of gravity, and H is the plant's net head in meters.

2.5. Flow Kinematics

The study of flow kinematics in this work will be developed from the analysis of radial and axial turbomachines, taking the Francis turbine and the Kaplan turbine as classic references, respectively. This approach is fundamental because both configurations serve as the theoretical basis and starting point for subsequently building the kinematics specific to the diagonal turbomachine (Deriaz type), which is the main subject of this project.

2.5.1. Velocity Triangles in Radial and Axial Turbomachines

In turbomachinery design, velocity triangles are the fundamental tool that allows the curvature of the blades responsible for transferring energy between the shaft and the fluid to be calculated precisely. These vector diagrams are therefore the key to optimizing the geometry and achieving the highest possible efficiency. The geometric structure of these triangles is conceptually identical whether the machine is radial or axial, and it is made up of the following main vectors:

- Drag velocity (U): Defined earlier, it is the linear velocity due to the machine's rotation, expressed as $U = r \cdot \omega$. Its direction is always tangential, that is, perpendicular to the radius at the point where the fluid particle is located.

- Absolute velocity (V): Represents the fluid's actual velocity with respect to a fixed reference frame (stationary outside the machine). This vector is decomposed into two orthogonal components:

  - *Meridional velocity (V_m):* This is the component that guides the fluid through the machine, driving it from the inlet section toward the outlet section. It is essential to note that the geometric nature of this velocity depends directly on the type of turbomachine. Thus, in an axial machine such as the Kaplan turbine, the meridional velocity is purely axial ($V_m = V_z$); in a radial machine such as the Francis turbine, it is oriented in the radial direction ($V_m = V_r$); finally, in a mixed- or diagonal-flow machine such as the Deriaz turbine, this component has a mixed character, combining both axial and radial projections according to the channel's spherical path.
  - *Azimuthal or tangential velocity (V_θ):* This is the component that quantifies the rotational, or swirl, motion that the fluid itself possesses in the direction of the runner's rotation.

- Relative velocity (W): This is the fluid's velocity as measured by an observer rotating together with the blade. To ensure that the water slides smoothly over the metal profile without causing destructive hydraulic shocks, the vector W must be strictly tangent to the blade along its entire path. This velocity is what defines the physical angle that the metal blade will form with respect to the drag velocity (angle β).

The fundamental relationship governing the behavior of the velocity triangle is defined by the following vector sum:

$$\vec{U} + \vec{W} = \vec{V} \tag{2.29}$$

This kinematic equation mathematically establishes that the fluid's absolute velocity ($\vec{V}$) is the direct result of combining the drag velocity due to the runner's own rotation ($\vec{U}$) with the relative velocity of the flow as it slides over the blade surface ($\vec{W}$).

The angles that make up the velocity triangle are defined as follows:

- Angle β, the angle between the relative velocity vector (W) and the drag velocity vector (U).

- Angle α, the angle formed between the absolute velocity vector (V) and the drag velocity vector (U).

Figure 2.4 shows the velocity triangles corresponding to a radial turbomachine. The left side represents operation in pump mode, while the right side shows turbine mode. As can be seen, a clockwise direction of rotation has been assumed for pump mode and counterclockwise for turbine mode, this being the arbitrary convention that will be maintained throughout this work. Likewise, points 1 and 2 indicate, respectively, the fluid's inlet and outlet sections in the machine. A similar approach is shown in Figure 2.5, which presents the equivalent diagram for an axial turbomachine. From both representations, it is possible to confirm the fundamental geometric difference between the two designs: whereas in the radial turbomachine (Figure 2.4) the meridional velocity is purely radial, in the axial configuration (Figure 2.5) that component is strictly parallel to the machine's axis of rotation.

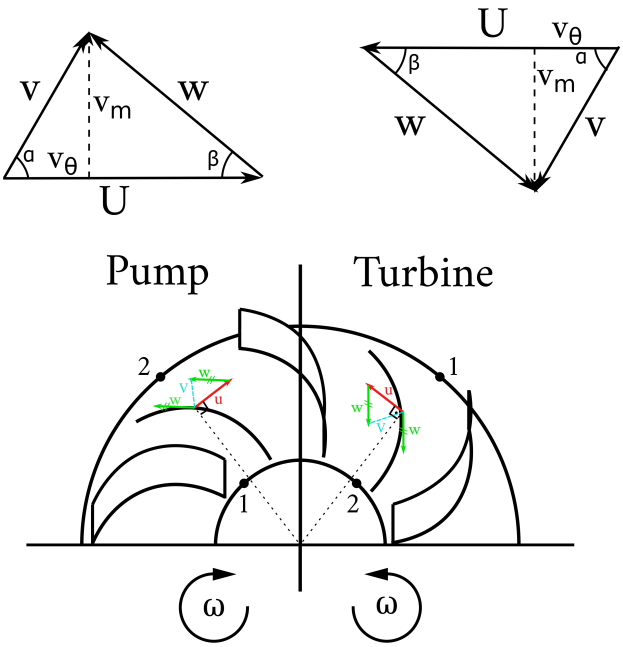


Fig. 2.4. Schematic representation of the velocity triangles in a radial turbine. Own elaboration.

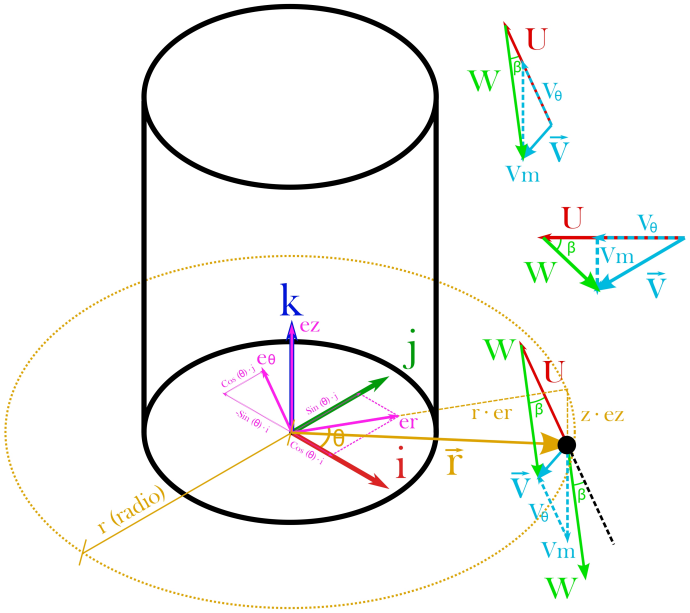


Fig. 2.5. Schematic representation of the velocity triangles in an axial turbine. Own elaboration.

On the other hand, the meridional velocity (V_m) is calculated differently depending on the turbine's geometry:

- In radial configuration: The value of V_m depends directly on the radius at which the particle is located, calculated as:

$$V_{m,\text{radial}} = \frac{Q}{A_{\text{radial section}}} = \frac{Q}{2\pi \cdot r \cdot a_t} \tag{2.30}$$

  Where Q is the flow rate and a_t represents the height of the distributor or of the turbine at that section.

- In axial configuration: The velocity is calculated through the area of the circular flow-through ring using the relation:

$$V_{m,\text{axial}} = \frac{Q}{A_{\text{axial section}}} = \frac{Q}{\pi\left(r_{\text{max}}^2 - r_{\text{min}}^2\right)} \tag{2.31}$$

  Where r_{max} and r_{min} represent the maximum and minimum radii of the annular channel, respectively. In this case, since the flow-through section is constant along the entire runner, it is crucial to understand that the meridional velocity remains unchanged at any point along the axial path.

3. BLADE DESIGN IN AN AXIAL TURBOMACHINE (KAPLAN TURBINE)

3.1. Relative velocity angle: distribution $\beta(r, z)$ along the axial blade

In order to subsequently address the design of the Deriaz turbine's blades, it is useful to first understand how this process is carried out in a purely axial turbomachine. The constant direction of the meridional velocity in this type of machine considerably simplifies the analytical calculation and makes it possible to establish the theoretical foundations that will later need to be modified for the case of a diagonal-flow turbomachine. It is of the utmost importance to highlight the spatial reference frame in which the angle β is defined. This angle is always located on the two-dimensional plane formed by the meridional velocity vector ($\vec{V}_m$) and the drag velocity vector ($\vec{U}$). The orientation of this plane changes completely depending on the turbomachine's architecture:

- In radial turbomachines: The plane containing β is strictly perpendicular to the axis of rotation (a horizontal plane), since both the drag velocity and the meridional velocity (which is radial, $V_m = V_r$) act in that direction.
- In axial turbomachines: The angle lies in a vertical plane perpendicular to the radius at the particle's position. In this case, the drag velocity remains tangential, but the meridional velocity is purely vertical ($V_m = V_z$).
- In the diagonal turbine (Deriaz type): Since this is a mixed geometry, the plane containing the angle β is tilted with respect to the machine's axis. This implies that the meridional velocity is no longer purely horizontal or vertical, but simultaneously has radial and axial components (V_r and V_z) as the fluid slides over the runner's spherical surface.

As a consequence of this diagonal nature, the blade geometry of a Deriaz turbine cannot be described by a simple planar projection. The designer is forced to handle a three-dimensional geometry in which the blade curves in space to adapt both to the runner's rotation and to the channel's changing slope, requiring a more complex mathematical treatment than that of traditional configurations.

However, before proceeding with the detailed analysis of the angle β in a diagonal turbomachine, it is useful to study its behavior in a purely axial configuration. This methodological approach makes it possible to start from a solid, simplified theoretical basis, facilitating the subsequent understanding of the three-dimensional geometric effects that occur in the diagonal runner.

In traditional configurations, the angle β of the velocity triangle, shown previously in Figures 2.4 and 2.5, is obtained analytically through the following trigonometric expressions for the flow inlet and outlet:

$$\tan(\beta_1) = \frac{V_{m1}}{U_1 - V_{\theta 1}} \tag{3.1}$$

$$\tan(\beta_2) = \frac{V_{m2}}{U_2 - V_{\theta 2}} \tag{3.2}$$

From the trigonometric relations of the velocity triangle, it is possible to analyze the direct impact that variation of the kinematic components has on the blade's geometry. Specifically, if a constant radius is considered, which implies that the drag velocity (U) remains unchanged, any decrease in the fluid's tangential velocity (V_θ) will cause a gradual reduction in the angle β. This effect is illustrated schematically in Figure 3.1, which shows how the relative velocity vector (W) tilts progressively to the right as the fluid's tangential component decreases, closing the kinematic triangle and reducing the angle β.

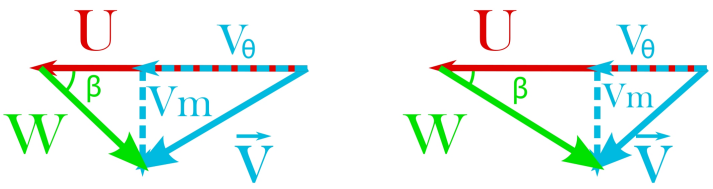


Fig. 3.1. Schematic representation of the decrease in the angle β in an axial velocity triangle as the fluid's tangential component (V_θ) decreases. Own elaboration.

Physically, this phenomenon is explained by the fact that as the fluid's rotational component (V_θ) decreases, the relative velocity vector (W) is forced to tilt further in order to close the kinematic triangle against the constant drag velocity. As a result, the angle β becomes more acute, dictating that the blade must adopt a more closed design in those sections where the fluid has a lower azimuthal velocity.

Since the main purpose of the Deriaz turbine to be designed in this project is electricity generation, with its secondary use being as a pump for pumped-hydro storage, priority will be given in the blade design to its behavior in turbine mode. Nevertheless, the possibility of adjusting the design in those positions where a large benefit is obtained in pump mode will be considered, even at the cost of a slight decrease in turbine efficiency.

According to the turbomachinery Euler equation, the optimal design for a turbine (which likewise benefits pump mode) consists of making the tangential velocity at the outlet in turbine mode, which corresponds to the inlet in pump mode, exactly equal to zero ($V_{\theta 2} = 0$). This maximizes the specific power transfer between the fluid and the

runner. Applying this design condition and solving for the angle β, the equations at the machine's boundaries are defined as a function of the radius (r) as follows:

$$\beta_1(r) = \arctan\left(\frac{V_{m1}}{U_1(r) - V_{\theta 1}(r)}\right) \tag{3.3}$$

$$\beta_2(r) = \arctan\left(\frac{V_{m2}}{U_2(r)}\right) \tag{3.4}$$

Knowing that the meridional velocity (V_m) remains constant along the entire channel in the axial case, and that the drag velocity varies linearly with the radius ($U = r \cdot \omega$), the values of the inlet angle β_1 (at the axial position $z = 0$) and the outlet angle β_2 (at the position $z = -L_z$) are fully determined for any runner radius.

However, the runner has a physical extent that goes beyond its inlet and outlet sections. Therefore, once the boundary angles $\beta_1(r)$ and $\beta_2(r)$ are known, it is necessary to interpolate their values within the spatial interval $z \in [0, -L_z]$. A negative value of the axial length L_z is used to designate the runner outlet because, since the design is based mainly on the turbine-mode criterion, the flow moves downward and is therefore negative along the adopted coordinate axis.

To obtain the continuous distribution of the angle $\beta(r, z)$ as a function of all possible values of z, a third-degree cubic interpolation is used, equivalent to a four-control-point Bézier curve. This transition law is chosen specifically because it minimizes the risk of cavitation when the machine operates in pump mode. The physical reason is that this curve smooths the blade's geometric variation both at the fluid's inlet and at its outlet, since the polynomial equation imposes a derivative equal to zero (horizontal tangent) at both ends of the z axis. In this way, the highest rate of change of the angle is concentrated exclusively in the intermediate stages of the blade, reducing the formation of vortices and mitigating the adverse hydrodynamic effects linked to the aggressive power transfer in the runner's transition zones.

An example of the shape taken by the cubic interpolation is shown in Figure 3.2. In that graph, the vertical axis (y axis) represents the angle $\beta(z)$, with a range from 9° to 6°, while the horizontal axis (x axis) shows the axial length z, between 0 and 10 meters. The illustration clearly shows how, both at the origin (0 m) and at the final end (10 m), the function, shown in green, has a derivative equal to zero. This mathematical condition is a direct demonstration of the smoothness in the change of orientation that the fluid will experience as it enters and leaves the runner's boundaries, minimizing hydraulic disturbances.

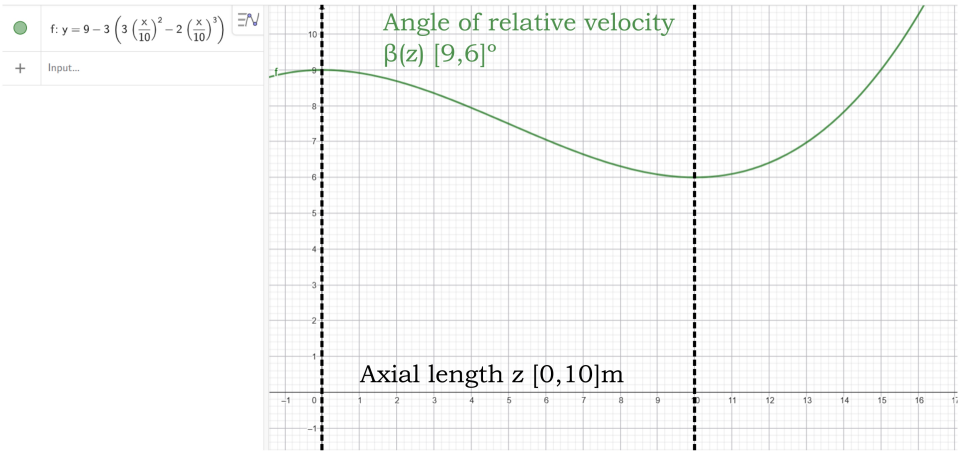


Fig. 3.2. Graph of the cubic interpolation of the angle $\beta(z)$ along the axial axis. Own elaboration.

To mathematically define the angle β along the axial axis, the following dimensionless change of variable is introduced for the height:

$$t = \frac{z}{-L_z} \tag{3.5}$$

Where the physical variable z is bounded within the interval $[0, -L_z]$, making t vary in a normalized way between $[0, 1]$. Therefore, for a constant radius r, the analytical expression for the blade angle's interpolation law is defined as:

$$\beta(t) = \beta_1 + (\beta_2 - \beta_1) \cdot \left(3t^2 - 2t^3\right) \tag{3.6}$$

Finally, taking the original variable into account, for a constant radius r, the analytical expression for the blade angle's interpolation law as a function of the height coordinate z is defined as:

$$\beta(z) = \beta_1 + (\beta_2 - \beta_1) \cdot \left[3\left(\frac{z}{-L_z}\right)^2 - 2\left(\frac{z}{-L_z}\right)^3\right] \tag{3.7}$$

In contrast to the methodology of Morabito et al. [8], the only public methodology on Deriaz turbine design found in the literature, the interpolation of β is carried out according to the expression above (3.7), rather than by manually adjusting each streamline on a CAD model until manufacturing-related geometric criteria are satisfied, as described in that work. Clearly, cubic interpolation is not the only option available, and in fact the software published on GitHub and explained in Section 6.2 allows the user to select among a small variety of β interpolations that distribute the pressure in different ways (linear, evenly distributed along the entire blade chord; with greater variation of β, that is, greater energy exchange, concentrated at the inlet or at the outlet; among others). This methodology, therefore, unlike any other found in the literature reviewed, including that

of Morabito et al. and the more recent proposal based on evolutionary algorithms coupled to CFD simulations by Kassanos et al. [15], is fully defined by a closed-form mathematical expression, resulting in a process that is entirely reproducible and automatable, without requiring manual adjustments on a CAD model or numerical optimization loops coupled to CFD simulation to obtain the final blade geometry.

Nevertheless, by applying the same interpolation law to all streamlines regardless of radius, the blade does not necessarily acquire a chord of uniform length along its span. In particular, the streamlines at larger radii start from a smaller angle β_1, a consequence of a higher drag velocity U and a smaller tangential component V_θ at those sections, and, moreover, must travel a greater azimuthal distance to complete the same relative variation of the angle, which inevitably results in a longer chord and, with it, a larger wake at the trailing edge, potentially giving rise to structural problems. This effect is the geometric price of requiring that all streamlines lose the same specific kinetic energy along the axial or conical parameter (z in the Kaplan case, γ in the Deriaz case).

Faced with this limitation, much of the literature reviewed, such as Morabito et al.'s own work [8], solves the problem by fixing only the inlet and outlet angles, and then manually adjusting, on a CAD model, both the chord and the intermediate angles of each streamline until the desired geometric criteria are satisfied. In this work, a mathematical, replicable solution has deliberately been chosen over such manual adjustment, hence the choice of cubic interpolation. It would nonetheless be of interest to explore, in future work, interpolation laws that depend explicitly on the radius, so as to equalize the chord length between streamlines regardless of their radius, even at the cost of introducing local pressure differences between radii for the same z or γ position. Determining which criterion should take priority in that case — structural, hydraulic, or otherwise — would require a specific study, beyond the scope of this work (see Section 6.4).

3.2. Evolution of the azimuthal position angle θ along the blade

For the final construction of the blade, it is not enough to know the behavior of the angle β as a function of radius and axial length. It is mathematically essential to determine how the angle θ, which represents the position around the machine's axis, will vary as the fluid molecule moves in the axial direction. In order to understand this spatial variation of θ, it is very useful to analyze the path of a single molecule through three consecutive states as it advances through the runner, characterized by the angles θ_1, θ_2, and θ_3. To do this, the velocity triangle corresponding to each of these positions is first calculated, where the second angle is defined as the first plus a differential increment of θ, and the third angle as the second plus that same differential increment. In this way, it is possible to study three states of the molecule at three different points, both in height z and in angle θ, where the first state has the smallest angular travel and the greatest height, while the last state has the greatest angular advance and the lowest height, which takes a negative value as it is located at a lower position in the channel. This arrangement of the three

states is illustrated in Figure 3.3, which shows how the fluid descends along the runner while rotating about the turbine's axis, describing a helical path that combines rotational motion with axial displacement.

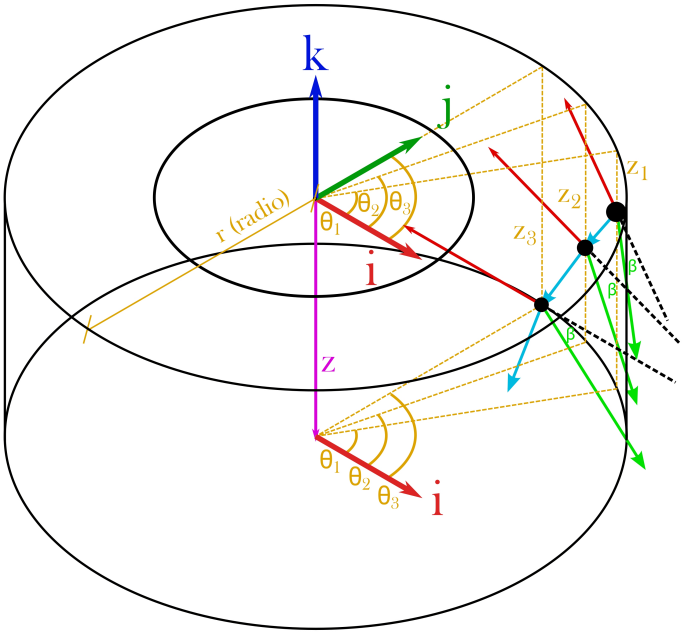
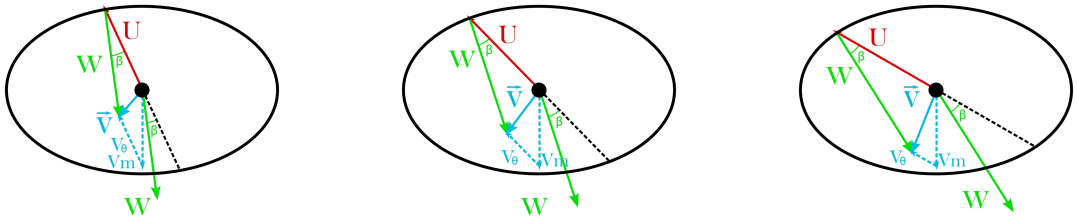


Fig. 3.3. Schematic representation of three fluid molecules at different angular positions and heights within the runner. Own elaboration.



(a) Caso θ_1 (b) Caso θ_2 (c) Caso θ_3

Fig. 3.4. Schematic representation of the kinetic triangles of the same particle for three values of θ. Own elaboration.

When evaluating the kinematics of these three states, it can be seen that the molecule's first position naturally has a larger magnitude in its azimuthal velocity vector, while the third position has a smaller magnitude. This reduction occurs because, as the fluid descends through the turbine, its tangential velocity progressively decreases as a direct consequence of the transfer of angular momentum to the machine's shaft. From this phenomenon, the three kinetic triangles are constructed, Figure 3.4, which share exactly the same magnitude for the drag velocity since they are at the same radius, but show different

vector directions. This variation in the orientation of the drag velocity occurs because this vector depends strictly on the particle's spatial position, which is governed by the cylindrical coordinate system defined by the unit vectors $\vec{e}_\theta$, $\vec{e}_r$, and $\vec{e}_z$.

Since the axial unit vector $\vec{e}_z$ undergoes no change with variations in the angle θ, and owing to the purely cylindrical nature of the Kaplan turbine, the meridional velocity remains constant in magnitude and direction for the three states evaluated, again observable in Figure 3.4. By contrast, the unit vectors $\vec{e}_\theta$ and $\vec{e}_r$ do undergo a change in orientation as a function of θ, which explains why both the drag velocity and the azimuthal velocity vary in direction along the flow's angular path.

It is worth clarifying that Figure 3.3 is a spatial representation in which the change in angle for each differential element has been visually exaggerated for illustrative purposes, since if analyzed in a strictly differential manner the graphical variation would be imperceptible; this makes it possible to model the passage of a molecule through the runner at a constant radius through three states of θ and z that maintain a proportionality defined by the distribution law of the angles β.

Finally, the three-dimensional arrangement of the profile is completed through the interaction of two fundamental planes in space. The plane that contains the angle β and rotates around the radial position vector is strictly perpendicular to the plane that describes the surface tangent to the turbomachine at that point. Geometrically, this tangent-surface plane is characterized by having the molecule's own radius as its normal vector, which is completely perpendicular to it. For its part, the plane containing the flow kinematics is defined as a plane perpendicular to the previous one, rotated by exactly the number of degrees corresponding to the profile's angle β, thereby establishing the final spatial orientation that the blade's metal surface must adopt in order to guide the flow optimally.

The orientation of these fundamental planes can be observed in Figure 3.5, which represents, in order to facilitate geometric understanding, the three particles studied previously but artificially placed at the same height z. This arrangement makes it possible to focus attention on the fact that the planes containing the relative velocity $\vec{W}$ and allowing the angle β to be visualized have a specific spatial configuration. The plane formed by the relative velocity vector $\vec{W}$ and the radial position vector, shown in green, has a tilt equivalent to a negative angle β with respect to the purple plane, which is defined by the drag velocity vector $\vec{U}$ and the meridional velocity vector $\vec{V}_m$, recalling that in the case of a purely axial turbomachine the equality $V_m = V_z$ always holds. Furthermore, Figure 3.6 independently shows the three cases corresponding to each particle for each angular position θ and its corresponding height z, making it possible to see how their respective planes intersect in space owing to the angular orientation dictated by the angle β.

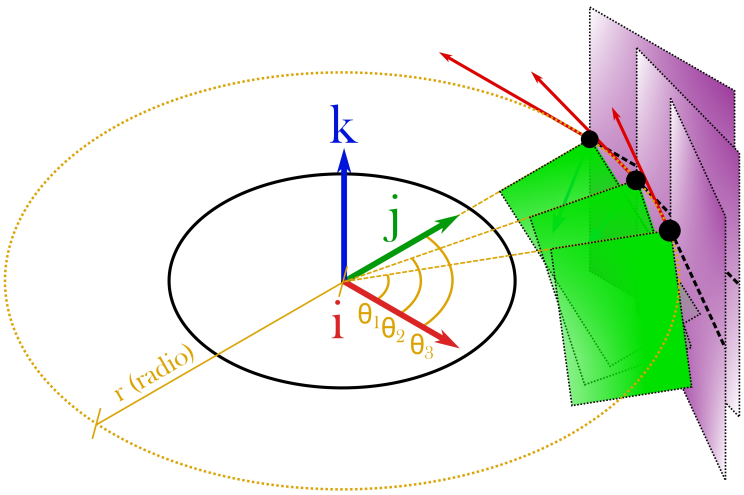
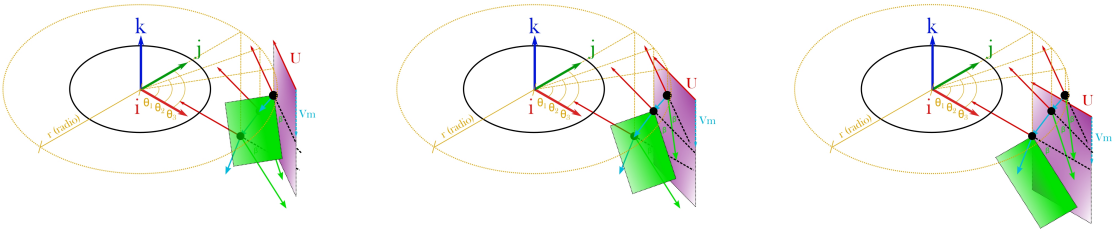


Fig. 3.5. Schematic representation of the fundamental planes that define the blade's orientation in space. Own elaboration.



(a) Caso θ_1 (b) Caso θ_2 (c) Caso θ_3

Fig. 3.6. Schematic representation of the fundamental planes of the same particle for three values of θ and z. Own elaboration.

To understand the need to evaluate the fluid's angular position, it is useful to analyze a specific scenario. If a molecule enters the runner with an azimuthal velocity exactly equal to the drag velocity, the relative tangential component vanishes, causing the relative velocity vector to equal the meridional velocity. From the perspective of the blade, which rotates together with the runner, the fluid molecule will appear to move vertically downward with zero angular velocity relative to the metal surface. However, since under real operating conditions these magnitudes differ along the path ($\vec{U} \neq \vec{V}_\theta$), the fluid will experience a continuous angular offset relative to the blade. To formally demonstrate the geometric equivalence governing the profile's design, the flow kinematics must be established under a global coordinate system in which the axial axis z points upward. In this reference frame, the fluid's downward motion implies that the height differential dz

is a net negative quantity, so that the axial velocity component acts opposite to the axis and, although represented mathematically as positive V_z, will take exclusively negative values when the turbomachine is studied as a turbine. From the velocity triangle in the tangential-axial plane, and assuming an operating condition in which the fluid's azimuthal velocity is lower than the drag velocity ($V_\theta < U$), the relative azimuthal velocity component will be negative (W_θ will point in the direction $-\vec{e}_\theta$). In this way, the tangent of the blade's angle β, in an axial turbomachine, is geometrically defined as:

$$\tan(\beta) = \frac{V_m}{W_\theta} = \frac{V_z}{W_\theta} \tag{3.8}$$

On the other hand, the infinitesimal path traced by a fluid particle on the blade surface at a constant radius r is determined by its spatial displacement within the rotating reference frame. While the molecule descends a differential distance dz, it simultaneously experiences a peripheral, or arc, displacement equal to $r \cdot d\theta_{\text{rel}}$, where θ_{rel} represents the angle, measured with respect to the axis of rotation, experienced by the particle within the blade's moving reference frame. The value of this parameter will be strictly positive if the relative displacement occurs in the direction of the unit vector $\vec{e}_\theta$, while it will take a negative sign if that advance occurs in the direction of the vector $-\vec{e}_\theta$, a physical phenomenon that will occur in those regions of the channel where the kinematic condition $\vec{V}_\theta < \vec{U}$ holds. Since the relative velocity vector $\vec{W}$ is, by definition, strictly tangent to the blade surface at every point, the relationship between its kinematic components and the differential spatial displacements must maintain an identical proportion, such that:

$$\frac{V_z}{W_\theta} = \frac{dz}{r \cdot d\theta_{\text{rel}}} \tag{3.9}$$

By equating both expressions thanks to this shared tangency condition, the kinematic relation linking the profile's slope to the blade's angular orientation in space is obtained:

$$\tan(\beta) = \frac{dz}{r \cdot d\theta_{\text{rel}}} \tag{3.10}$$

This physical constraint directly gives rise to the differential equation that governs the fluid's streamline within the turbomachine's flow channel:

$$\frac{r \cdot d\theta_{\text{rel}}}{dz} = \frac{W_\theta}{V_z} = \frac{V_\theta - U}{V_z} \tag{3.11}$$

This differential equation makes it possible, through its analytical or numerical integration, to precisely calculate the relative angular coordinate θ_{rel} needed to trace the blade's three-dimensional curvature at each height z. Evaluating this behavior shows that the sign of the relative azimuthal component has a direct physical interpretation for the trajectory. If the counterclockwise direction is taken as the machine's positive direction,

the fluid progressively decreases its absolute azimuthal velocity V_θ as it descends and transfers its angular momentum to the shaft, consistent with Euler's principle.

Once this component falls below the drag velocity ($V_\theta < U$), the relative term becomes negative. From the perspective of the moving blade itself, this mathematical reversal means that the molecule falls behind the structure's rotation and begins to move in the direction of the unit vector $-\vec{e}_\theta$, undergoing a relative clockwise setback on the metal surface as it slows down. Since dz is negative because of the fluid's descent, the differential $d\theta_{\text{rel}}$ will take the sign consistent with this lag. Consequently, the relative angle is not an arbitrary parameter and cannot be omitted from the design process, since it defines the exact blade geometry necessary to avoid destructive hydraulic shocks and to ensure that the flow slides smoothly and efficiently.

3.3. Construction of the blade's three-dimensional geometry in an axial turbomachine

Continuing with the analysis carried out in the previous section, and generalizing the procedure for any radius r, determining the blade's three-dimensional geometry starts from the flow's tangency condition on the blade surface in a rotating reference frame. The profile's local slope at any cylindrical section at constant radius r is defined through the relationship between the differential spatial displacements as:

$$\tan[\beta(r,z)] = \frac{dz}{r \cdot d\theta_{\text{rel}}} \tag{3.12}$$

Isolating the angular differential term gives the differential equation governing the fluid particle's relative streamline within the flow channel:

$$d\theta_{\text{rel}} = \frac{1}{r \cdot \tan[\beta(r,z)]} dz = \frac{\cot[\beta(r,z)]}{r} dz \tag{3.13}$$

Instead of imposing an arbitrary distribution for the velocities, the blade's geometric profile is governed directly by a cubic interpolation law for the angle $\beta(r,z)$ along the axial axis. This law guarantees a smooth, continuous transition from the leading edge ($z = 0$) to the trailing edge ($z = -L_z$):

$$\beta(r,z) = \beta_1(r) + [\beta_2(r) - \beta_1(r)] \cdot \left[3\left(\frac{z}{-L_z}\right)^2 - 2\left(\frac{z}{-L_z}\right)^3 \right] \tag{3.14}$$

Where the boundary conditions for the blade angles at inlet and outlet are determined from the kinematic velocity triangles corresponding to each local radius r:

$$\beta_1(r) = \arctan\left(\frac{V_z}{\omega \cdot r - V_{\theta 1}(r)}\right) \tag{3.15}$$

$$\beta_2(r) = \arctan\left(\frac{V_z}{\omega \cdot r}\right) \tag{3.16}$$

It is important to note that in the equations above the meridional velocity has been replaced by the axial velocity ($V_{m1} = V_{m2} = V_z$), in accordance with the constant-axial-velocity assumption established for the channel. Likewise, it is assumed that at the runner outlet the flow has no swirl component ($V_{\theta 2} = 0$), satisfying the turbine's optimal discharge condition.

To calculate the complete relative angular coordinate $\theta_{\text{rel}}(r, z)$ at any point in space, both sides of the differential equation are integrated. The channel's physical limits extend from the inlet plane $z = 0$ (where the initial reference angle is set as $\theta_{\text{rel}} = 0$) to a generic negative height z:

$$\int_0^{\theta_{\text{rel}}} d\theta_{\text{rel}} = \int_0^z \frac{\cot[\beta(r, z)]}{r} dz \tag{3.17}$$

Since the radius r remains a constant, invariant parameter throughout the integration along a cylindrical streamline, it is pulled out of the linear operator. The final closed-form mathematical expression for obtaining the relative angular position is defined as follows:

$$\theta_{\text{rel}}(r, z) = \frac{1}{r} \int_0^z \cot\left[\beta_1(r) + [\beta_2(r) - \beta_1(r)] \cdot \left(3\left(\frac{z}{-L_z}\right)^2 - 2\left(\frac{z}{-L_z}\right)^3\right)\right] dz \tag{3.18}$$

Substituting the angular boundary conditions in terms of the requested operational and design variables, the explicit distribution law takes its final form:

$$\theta_{\text{rel}}(r, z) = \frac{1}{r} \int_0^z \cot\left[\arctan\left(\frac{V_z}{\omega \cdot r - V_{\theta 1}(r)}\right) + \left[\arctan\left(\frac{V_z}{\omega \cdot r}\right) - \arctan\left(\frac{V_z}{\omega \cdot r - V_{\theta 1}(r)}\right)\right] \cdot \left(3\left(\frac{z}{-L_z}\right)^2 - 2\left(\frac{z}{-L_z}\right)^3\right)\right] dz \tag{3.19}$$

This systematic analytical formulation fully determines the differential equation for the angular coordinate without the need for intermediate linear approximations. Owing to the strongly nonlinear nature of the cotangent function combined with the cubic distribution of its argument, directly solving this integral for each design radius lacks an exact closed-form analytical solution. Consequently, the trajectory is calculated through numerical solution of the integral in the runner's calculation code, discretizing the channel into small axial steps to guarantee absolute geometric precision in the blade's 3D modeling.

3.4. Determination of blade count and sizing of the inter-blade spacing

Sizing the inter-blade hydrodynamic channel is one of the most critical factors in the inverse numerical design, since it directly governs the correct deflection of the fluid's velocity vector. Physically, control of the free space between blades is not defined through a constant metric distance, but through the optimal selection of the integer blade count (Z). This variable establishes a hydrodynamic trade-off between guiding the flow and viscous friction losses. An excess of blades restricts the flow-through area and sharply increases surface resistance, whereas an insufficient number of profiles prevents the central streamlines from undergoing angular momentum transfer, inducing recirculation and boundary-layer separation.

To rigorously establish this geometric boundary, the algorithm adopts the runner solidity criterion (σ) proposed by Stepanoff and Pfleiderer. Solidity defines the dimensionless ratio between the profile's chord length (L_{chord}) and the tangential, or circumferential, pitch (t), which represents the circumferential distance between two consecutive blades at a given radius:

$$\sigma = \frac{L_{\text{chord}}}{t} \tag{3.20}$$

Since the tangential pitch varies linearly as a function of the domain's local radius, its geometric expression is defined as:

$$t = \frac{2\pi r}{Z} \tag{3.21}$$

Substituting the tangential pitch into the solidity equation yields the direct mathematical formulation linking the blade count to the spatial dimensions computed by the inverse numerical model:

$$Z = \frac{2\pi r \sigma}{L_{\text{chord}}} \tag{3.22}$$

The design of axial turbomachine blade cascades usually relies on analogous optimal blade-spacing criteria; as a reference, the Zweifel criterion for axial turbines sets an optimal dimensionless tangential loading value of around 0.8 [25, Cap. 4]. For reversible, mixed-flow, diagonal-geometry hydraulic turbomachines such as the Deriaz model studied here, this work adopts as its own design criterion that the optimal solidity range for ensuring efficient deflection without flow separation must be strictly confined to the interval $\sigma \in [1.2,\ 2.0]$.

3.4.1. Justification of the Mean Radius Choice for Chord Integration (L_{chord}) and Determination of Blade Count (Z)

For the analytical determination of the optimal blade count (Z) using the hydrodynamic solidity criterion ($\sigma = L_{\text{chord}}/t$), it is essential to evaluate the actual 3D chord length (L_{chord}) at a representative cross-section of the blade. The required tangential pitch value t and the subsequent discretization of the perimeter into Z blades depend directly on the characteristic radius selected.

Owing to the marked differences in flow paths and internal kinematics between purely axial meridional-flow turbines (Kaplan) and diagonal or mixed-flow turbines (Deriaz), the definition of the equivalent mean radius must be grounded in the representative physics of each type.

Criterion for Kaplan Turbines: Hydraulic Mean Radius (R_m)

In axial-flow Kaplan-type turbines, the flow-through space between the hub and the periphery is defined by a constant-section ring bounded by the hub radius (R_{hub}) and the tip radius (R_{tip}). For this flow regime, the axial velocity component is ideally assumed uniform along the radius.

Under this assumption, the hydraulic mean radius R_m is defined as the one that divides the annular flow area into two sections of identical surface area:

$$\pi\left(R_{\text{tip}}^2 - R_m^2\right) = \pi\left(R_m^2 - R_{\text{hub}}^2\right) \tag{3.23}$$

Solving for the radius R_m gives the exact quadratic formulation implemented in the code:

$$R_m = \sqrt{\frac{R_{\text{tip}}^2 + R_{\text{hub}}^2}{2}} \tag{3.24}$$

This choice guarantees the strict conservation of the annular flow-through area and of the mean kinetic energy associated with the total flow rate Q. Evaluating the integrated 3D chord ($L_{\text{chord_actual}}$) at the position R_m provides a balanced sample of the hydrodynamic loads across the entire blade span.

Criterion for Deriaz Turbines: Logarithmic Mean Spherical Radius ($R_{e,\text{mean}}$)

In diagonal-flow Deriaz-type turbines, the flow runs over conical or spherical streamlines, with a significant variation in radius observed along the blade's mean line. In this type of machine, the blade orientation mechanism requires larger hubs, increasing the ratio between the outer and inner radius.

In an axisymmetric diagonal flow, the circulation and the losses due to friction or cascade overlap behave inversely proportional to the radius ($1/r$), consistent with the free-

vortex law. Integrating a scalar property distributed as a function of $1/r$ over the interval $[R_{\text{int}}, R_{\text{ext}}]$ gives:

$$\int_{R_{\text{int}}}^{R_{\text{ext}}} \frac{1}{r}\, dr = \ln\left(\frac{R_{\text{ext}}}{R_{\text{int}}}\right) \tag{3.25}$$

From this differential relation, the equivalent mean spherical radius is derived through the exact logarithmic mean:

$$R_{e,\text{mean}} = \frac{R_{\text{ext}} - R_{\text{int}}}{\ln\left(\frac{R_{\text{ext}}}{R_{\text{int}}}\right)} \tag{3.26}$$

Using $R_{e,\text{mean}}$ in Deriaz turbines makes it possible to weight, with rigorous precision, the continuous variations in tangential pitch ($t = 2\pi r/Z$) and hydrodynamic solidity along the conical/spherical profile, avoiding the geometric overestimation that a conventional quadratic or arithmetic mean would produce in highly conical geometries.

3.5. Head and Flow Rate Considerations for the Velocity Triangle

To determine the distribution of the tangential component of the absolute velocity ($V_{\theta 1}$) along the radius (r_1) at the runner inlet from the primary head parameters, it is necessary to apply the hydraulic and volumetric efficiency corrections to the nominal variables [25, Cap. 2].

3.5.1. Head Losses and Energy Exchange (H_∞)

Given the reservoir's net head (H_n), the theoretical, or ideal Euler, head actually exploited by the runner (H_∞) is obtained by taking into account the hydraulic efficiency (η_h):

$$H_\infty = H_n \cdot \eta_h \tag{3.27}$$

3.5.2. Conservation of Useful Flow Rate ($Q_{\text{runner,actual}}$)

The gross flow rate circulating through the plant is Q_0. Owing to volumetric leakage in the clearance between the casing and the runner, the effective flow rate circulating between the blades ($Q_{\text{runner,actual}}$) is given by the volumetric efficiency (η_v):

$$Q_{\text{runner,actual}} = Q_0 \cdot \eta_v \tag{3.28}$$

3.5.3. Associated Power Values

The nominal, or gross, power available at the head ($\dot{W}_n$) and the total useful power delivered to the shaft ($\dot{W}_t$) are respectively defined as:

$$\dot{W}_n = \rho \cdot g \cdot Q_0 \cdot H_n \tag{3.29}$$

$$\dot{W}_t = \eta_t \cdot \dot{W}_n = (\eta_h \cdot \eta_v \cdot \eta_o) \cdot \rho \cdot g \cdot Q_0 \cdot H_n \tag{3.30}$$

where ρ is the density of water (1000 kg/m^3), g is the acceleration of gravity (9.81 m/s^2), η_o is the mechanical efficiency, and $\eta_t = \eta_h \cdot \eta_v \cdot \eta_o$ is the turbomachine's total efficiency.

3.5.4. Derivation of the velocity $V_{\theta 1}(r)$ through the Euler Equation

The Euler equation for the specific energy exchange in a turbine states:

$$g \cdot H_\infty = U_1(r) \cdot V_{\theta 1}(r) - U_2(r) \cdot V_{\theta 2}(r) \tag{3.31}$$

Where the inlet drag velocity $U_1(r)$ depends on the angular velocity ω and the radial position r_1:

$$U_1(r) = \omega \cdot r_1 \tag{3.32}$$

Assuming the optimal no-swirl outlet condition ($V_{\theta 2} = 0$), the equation simplifies to:

$$g \cdot H_\infty = U_1(r) \cdot V_{\theta 1}(r) \tag{3.33}$$

Substituting $H_\infty = H_n \cdot \eta_h$ and the expression for $U_1(r)$, we solve for $V_{\theta 1}(r)$ in terms of the net head and the primary parameters (the inverse dependence on the local radius r_1 dictated by the free-vortex law can be seen):

$$V_{\theta 1}(r) = \frac{g \cdot H_n \cdot \eta_h}{\omega \cdot r_1} \tag{3.34}$$

Therefore, the main considerations for the code implementation are:

- The meridional velocity component (V_{m1}) at the inlet will depend on the blades' flow-through area and on the useful flow rate $Q_{\text{runner,actual}} = Q_0 \cdot \eta_v$.
- The azimuthal component ($V_{\theta 1}$) depends exclusively on the net head H_n, the hydraulic efficiency η_h, gravity g, the angular velocity ω, and the local radius r_1.

3.5.5. Justification of the Calculation Procedure: Euler's Hypothesis

In the conceptual design of the turbomachine, calculating the inlet velocity triangles ($V_{\theta 1}, V_{m1}$) requires knowing the theoretical energy exchanged by the runner. To this end, the ideal Euler hypothesis ($z \to \infty$) is adopted, assuming an infinite number of blades. Under this assumption:

1. The flow is perfectly guided by the blades, so that $\beta = \beta_{\text{blade}}$.
2. The optimal design condition is imposed at the outlet with no residual rotation ($V_{\theta 2} = 0$), simplifying the Euler equation to:

$$g \cdot H_\infty = U_1 \cdot V_{\theta 1} \tag{3.35}$$

where the theoretical energy available is defined as $H_\infty = H_n \cdot \eta_h$.

In strict academic rigor, the $z \to \infty$ hypothesis should be corrected at a later stage to account for the effect of a finite number of blades (slip factor), which reduces the theoretical energy-exchange capacity to a real value $H_z < H_\infty$.

However, for the scope of this work it was decided to keep the formulation based on H_∞, omitting that iterative loop. This decision is based on the following considerations:

- Interdependence and computational complexity: There is a direct circular dependency in the sizing process: to apply the traditional slip corrections (such as the Stodola or Pfleiderer methods) [25, Cap. 7] and determine the theoretical head with a finite number of blades (H_z), the blade count (Z) must already be known beforehand. However, the chord length required to determine the blade count through the solidity criterion depends on the tangential velocity component (V_θ), obtained from the free-vortex hypothesis, as well as on the infinite theoretical head (H_∞). Resolving this interdependence would require a coupled iterative scheme for each radial and axial section, which would drastically increase the computational load and the complexity of the blade-design algorithm.

- Uncertainty for Deriaz turbines: These empirical formulations were originally developed for purely radial or axial turbomachines. In a diagonal-flow turbine such as the Deriaz, the simultaneous variation of radius and meridional component along the streamline means that no sufficiently validated or benchmarked H_z correction method exists in the technical literature.

Despite this simplification, it is acknowledged that accounting for the transition from H_∞ to H_z would be a necessary step toward achieving maximum rigor in the detailed geometric sizing of the runner.

3.6. Code developed for calculating and generating the three-dimensional geometry of the blade's mean surface through MATLAB programming

For the kinematic characterization and three-dimensional synthesis of the blade's mean surface for the reference axial turbomachine (Kaplan-type turbine), a custom numerical script has been implemented in the MATLAB programming environment. The algorithm carries out the discrete solution of the flow equations, the curvilinear integration of the 3D hydrodynamic chord, and the determination of the blade count (Z) using the Stepanoff solidity criterion. To facilitate the reproducibility of the results and allow their reuse in future projects, the complete script with its different calculation blocks is included in Appendix A, and is freely available for use and modification.

The most relevant functional blocks of the tool developed are analyzed below, breaking down the mathematical foundations and the physical justification of their key statements.

3.6.1. Block 1: Energy, geometric, and efficiency parameters

In this first stage, the hydroelectric plant's nominal operating conditions (Q_0, H_n, RPM) are parametrized, and the fundamental geometric dimensions of the annular flow-through channel ($R_{\text{hub}}, R_{\text{tip}}, L_z$) are defined. Internal losses are also introduced through the hydraulic (η_h), volumetric (η_v), and mechanical/organic (η_o) efficiencies, computing the reference hydraulic mean radius (R_m), which divides the annular ring into two sections of identical cross-sectional area and will be key to calculating the blade count.

CODE 3.1. Definition of energy, geometric, and internal efficiency parameters.

```
RPM = 450; % Runner rotational speed (rpm)
Q0 = 12; % Nominal volumetric flow rate drawn from the head (m^3/s)
Hn = 15; % Net head available at the plant (m)
g = 9.81; % Acceleration of gravity (m/s^2)
rho = 1000; % Density of water (1000kg/m^3)

% Turbomachine efficiencies (internal efficiencies)
eta_h = 0.90; % Hydraulic efficiency (friction and dissipation losses)
eta_v = 0.96; % Volumetric efficiency (leakage losses in the hub clearance)
eta_o = 0.98; % Organic/mechanical efficiency (bearing friction losses)
eta_t = eta_h * eta_v * eta_o; % Total turbine efficiency

% Main geometric dimensions of the flow-through channel
R_cubo = 0.3; % Runner inner radius (m)
R_punta = 0.65; % Runner outer radius (m)
L_z = 0.35; % Blade axial length (m)

R_m = sqrt((R_punta^2 + R_cubo^2) / 2); % Hydraulic mean radius (m)
```

Explicitly including η_v, η_h, and η_o from the initial stage is essential to accurately evaluate the actual power transferred from the fluid to the shaft without overestimating the machine's capacity.

3.6.2. Block 2: Fluid kinematics and effective velocities

In this second block, the useful-flow-rate correction (Q_{real}) is applied by discarding the volumetric leakage losses, and the meridional/axial velocity component ($V_z = V_m$) is evaluated, which is assumed uniform over the entire annular flow-through section. In addition, the available hydraulic power balance ($\dot{W}_n$), the mechanical shaft power ($\dot{W}_t$), and the theoretical Eulerian head (H_∞) corrected for viscous losses are determined.

CODE 3.2. Calculation of kinematic variables, effective velocities, and power balances.

```
omega = (2 * pi * RPM) / 60; % Angular rotational speed (rad/s)
Q_real = Q0 * eta_v; % Useful flow rate actually passing through the runner (m^3/
    s)
Area_paso = pi * (R_punta^2 - R_cubo^2); % Flow cross-sectional area (m^2)
V_z = Q_real / Area_paso; % Constant meridional/axial velocity (Vm1 = Vm2 = V_z)

% Theoretical and useful head powers
W_n = rho * g * Q0 * Hn; % Gross hydraulic power available (W)
W_t = W_n * eta_t; % Useful mechanical power delivered to the shaft (W)
H_inf = Hn * eta_h; % Theoretical energy per Euler's hypothesis (m)
```

By adopting the kinematic hypothesis of axisymmetric flow and uniform meridional velocity ($V_{m1} = V_{m2} = V_z$), the continuity equation imposes a constant value of V_z throughout the entire runner domain. Likewise, the reduced Eulerian head ($H_\infty = H_n \cdot \eta_h$) decouples the kinematic calculation from viscous wall losses, ensuring that the design of the inlet and outlet velocity triangles faithfully reflects the actual specific energy transfer.

3.6.3. Block 3: Spatial discretization and data matrices

In this block, the discrete domain infrastructure is established by generating the numerical-spatial mesh. The working space is discretized into N_{radios} concentric streamlines along the radius and N_{chord} axial integration stations. Likewise, the three-dimensional Cartesian coordinate matrices (X, Y, Z) and the axial (`z_vec`) and radial (`r_vec`) position vectors are pre-allocated.

CODE 3.3. Definition of the structural mesh and discretization of 3D space.

```
N_radios = 25; % Number of concentric streamlines
N_cuerda = 40; % Number of axial integration stations (dz)

X = zeros(N_cuerda, N_radios);
Y = zeros(N_cuerda, N_radios);
Z = zeros(N_cuerda, N_radios);

z_vec = linspace(0, -L_z, N_cuerda);
r_vec = linspace(R_cubo, R_punta, N_radios);
```

Establishing a 40×25 resolution mesh (1000 calculation nodes) represents an optimal trade-off between the precision of the curvilinear numerical integration and the script's computational cost. The uniform axial discretization in the negative direction ($0 \rightarrow -L_z$) ensures the correct physical sense of the downward flow, while the radial partition allows the hydrodynamic behavior in each streamline cylinder to be evaluated independently.

3.6.4. Block 4: Velocity triangles and 3D integration (Euler's Hypothesis)

This block constitutes the fundamental physical and mathematical core of the algorithm developed. For each concentric streamline at radius r, the velocity triangles at the inlet (β_1, $z = 0$) and outlet (β_2, $z = -L_z$) are solved by imposing the free-vortex condition at the inlet ($V_{\theta 1} \cdot r = \text{Const}$) and pure axial discharge at the outlet ($V_{\theta 2} = 0$). An angular distribution law is then applied through cubic interpolation $\beta(z)$, and the blade differential equation is solved through discrete numerical integration, first obtaining the profile's position in cylindrical coordinates (r, θ, z). The three-dimensional coordinates (X, Y, Z) that define the blade's mean surface are then determined through trigonometric conversion to Cartesian coordinates ($X = r\cos\theta$, $Y = r\sin\theta$, $Z = z$).

CODE 3.4. Velocity triangles, cubic blade law, and integration of the 3D differential equation.

```
beta1_vec = zeros(1, N_radios);
beta2_vec = zeros(1, N_radios);

for i = 1:N_radios
    r = r_vec(i);
    U = omega * r;

    % INLET: Required azimuthal component
    V_theta1 = (g * H_inf) / (omega * r);
    beta1 = atan2(V_z, (U - V_theta1));
    beta1_vec(i) = beta1;

    % OUTLET: Ideal nominal condition (V_theta2 = 0)
    beta2 = atan2(V_z, U);
    beta2_vec(i) = beta2;

    % --- CUBIC INTERPOLATION LAW FOR THE BLADE ---
    t_dim = abs(z_vec) / L_z;
    polinomio_cubico = 3*(t_dim.^2) - 2*(t_dim.^3);
    beta_z = beta1 + (beta2 - beta1) .* polinomio_cubico;

    % --- NUMERICAL INTEGRATION OF THE DIFFERENTIAL EQUATION ---
    theta_rel = 0; % Lag of the molecule relative to the blade
    for j = 1:N_cuerda
        if j > 1
            dz = abs(z_vec(j) - z_vec(j-1));
            d_theta_rel = (cot(beta_z(j)) / r) * dz;
            theta_rel = theta_rel - d_theta_rel; % Counterclockwise direction
        end

        X(j, i) = r * cos(theta_rel);
        Y(j, i) = r * sin(theta_rel);
        Z(j, i) = z_vec(j);
    end
```

```
end
```

- Free-Vortex Hypothesis: The formulation $V_{\theta 1} = \frac{g \cdot H_\infty}{\omega \cdot r}$ ensures a constant specific work along the entire blade span, which prevents the appearance of destructive radial pressure gradients and unwanted secondary flows.
- Cubic Blade Polynomial ($3t^2 - 2t^3$): By imposing the zero-derivative condition at both ends ($\frac{d\beta}{dz}\Big|_{z=0} = \frac{d\beta}{dz}\Big|_{z=-L_z} = 0$), the geometric curvature is smoothed near the leading and trailing edges. This reduces the risk of local boundary-layer separation and minimizes the propensity for cavitation.
- Integration of the Blade Equation: Discretizing the kinematic equation $d\theta_{\text{rel}} = \frac{\cot(\beta)}{r}\, dz$ makes it possible to transform the scalar variation of the relative flow angle into a continuous 3D curvilinear path, guaranteeing exact tangency between the blade and the relative velocity vector ($\vec{W}$) at each mesh node.

  Kinematic convention and direction of curvature: The blade is assumed to rotate counterclockwise, defined as the positive angular direction ($+\theta$). As the fluid travels along the chord, the fluid particle/molecule falls angularly behind the blade's motion owing to flow deflection. Therefore, although the particle continues to move counterclockwise, it does so at a lower angular velocity than the blade, generating a relative lag in the clockwise direction (negative, $-\theta$). To model this kinematic deflection and maintain the designed velocity triangle, the angle $d\theta_{\text{rel}}$ is accumulated with a negative sign at each iteration along the chord, causing the blade to progressively curve clockwise within the runner's reference frame.

3.6.5. Block 5: Calculation of the Blade Count (Z)

Once the blade's three-dimensional mean surface geometry has been generated, the blade count (Z) is determined using the hydrodynamic solidity criterion (σ). This procedure is implemented in the calculation environment through the following code block:

CODE 3.5. Calculation of the 3D chord and determination of the blade count

```
% 1. Locating the index corresponding to the mean radius R_m
[~, idx_Rm] = min(abs(r_vec - R_m));

% 2. Exact curvilinear integration of the 3D chord at R_m (L_cuerda_real)
dx = diff(X(:, idx_Rm));
dy = diff(Y(:, idx_Rm));
dz = diff(Z(:, idx_Rm));
dL = sqrt(dx.^2 + dy.^2 + dz.^2); % 3D Pythagorean theorem
L_cuerda_real = sum(dL); % Actual integrated 3D chord, in meters
```

```
% 3. Target hydrodynamic solidity (sigma = L_chord / tangential_pitch)
sigma_target = 1.25; % Value consistent with Kaplan turbines

% 4. Analytical determination of the blade count Z
paso_tangencial_req = L_cuerda_real / sigma_target;
Z_calculado = (2 * pi * r_vec(idx_Rm)) / paso_tangencial_req;
Z_optimo = round(Z_calculado);

% Physical safety limit
Z_optimo = max(3, min(Z_optimo, 12));

% Kinematic specific speed (nq) for the report
nq = RPM * sqrt(Q0) / ((g*Hn)^(3/4));
```

The process detailed in Code 3.5 is based on the following physical and geometric stages:

1. Integration of the actual three-dimensional chord (L_{chord}): Instead of using simplified two-dimensional projections, the code performs an exact curvilinear integration along the 3D profile located at the radius closest to the theoretical mean radius (R_m). After identifying the spatial index associated with R_m, the spatial differentials $(\Delta X, \Delta Y, \Delta Z)$ allow the arc differential dL to be evaluated through the Euclidean metric:
$$dL = \sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2} \tag{3.36}$$
The total length $L_{\text{chord_actual}}$ is obtained by summing these discrete increments along the chord.

2. Hydrodynamic solidity criterion (σ): The relationship between the blade chord and the tangential pitch ($t = \frac{2\pi R_m}{Z}$) is established:
$$\sigma = \frac{L_{\text{chord_actual}}}{t} = \frac{Z \cdot L_{\text{chord_actual}}}{2\pi R_m} \tag{3.37}$$
adopting a target value $\sigma_{\text{target}} = 1.25$, representative of Kaplan turbines, so as to optimize flow guidance while minimizing surface friction.

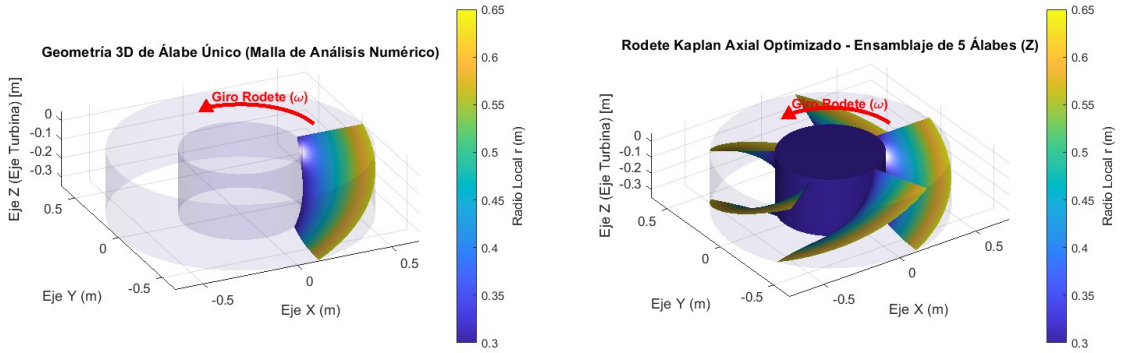
3. Calculation and bounding of Z_{optimal}: The continuous blade count ($Z_{\text{calculated}}$) is solved for and rounded to the nearest integer. To prevent possible geometric inconsistencies under drastic variations in the input data, a safety range $Z \in [3, 12]$ is imposed.

4. Specific speed (n_q): Finally, the kinematic specific speed (n_q) is calculated to verify the runner's dimensionless characterization and ensure the overall consistency of the design.

3.6.6. Blocks 6 and 7: 3D Visualization and Geometric Reconstruction

Finally, the algorithm's post-processing stage (Blocks 6 and 7) transforms the discrete coordinates obtained into three-dimensional geometric representations for visual inspection and validation of the design:

- Generation and rendering of the individual blade (Block 6): The blade's mean surface is reconstructed by extrapolating the Cartesian coordinates (X, Y, Z). Figure 3.7a shows the final geometry obtained after the blade-shaping and surface-smoothing process.
- Reconstruction of the complete axial runner (Block 7): Using the optimal blade count (Z_{optimal}) calculated in the previous block, an angular rotation matrix is applied about the machine's axis ($\Delta\theta = 2\pi/Z_{\text{optimal}}$). This makes it possible to clone and spatially position the Z blades around the hub, as seen in the three-dimensional reconstruction of the turbine in Figure 3.7b.



(a) Rendered individual blade.

(b) Complete axial turbine runner.

Fig. 3.7. Three-dimensional rendering results: (a) mean surface of the isolated blade, and (b) runner configuration with the distribution of Z_{optimal} blades in an axial turbomachine. Own elaboration.

3.7. Geometric optimization of the blade and considerations on the double-regulation mechanism

The numerical algorithm developed in this document adopts a three-dimensional inverse design method. This means that the blade's final geometry is not drawn arbitrarily, but is calculated mathematically as a direct consequence of imposing the desired fluid-dynamic behavior on the water. Unlike traditional, simplified methods that force the trailing edge to align strictly on a flat radial plane (θ_{rel} = constant) for the sole purpose of facilitating workshop manufacturing, this model prioritizes pure hydrodynamic optimization. By allowing a free, independent angular offset for each radius at the discharge section (z =

$-L_z$), the rigorous satisfaction of the pure axial outlet condition ($V_{\theta 2} = 0$) and of the free-vortex law at the inlet is guaranteed, thereby minimizing secondary-separation losses and maximizing the machine's efficiency.

As a direct consequence of this coupling of physical laws, the algorithm generates a natural, pronounced twist toward the runner's periphery. Although this twisting introduces a geometric offset known as blade lean, the potential increase in hydrodynamic torque on the blade orientation mechanism does not invalidate the feasibility of the channel obtained. In the development methodology for double-regulation turbomachines, the calculation of fluid behavior precedes and governs the structural analysis of the materials. The resulting residual torque is mitigated in later stages of the project through the spatial optimization of the pivot axis, making it optimally coincide with the blade's integrated center of pressure. From a constructive standpoint, this load is absorbed through the proper sizing of the hydraulic servomotor housed inside the hub, which is standard practice in today's hydroelectric industry.

In the same way that the trailing edge's manufacturing constraints are handled in later stages, the three-dimensional tilt or lag of the leading edge (leading edge sweep, or skew) is not introduced directly into the core of the inverse design algorithm developed here. In various commercial Kaplan- or Deriaz-type turbomachines, it is common to observe that the leading edge does not align on a strict horizontal plane at $z = 0$ for all radii, but rather that the peripheral sections (near the tip) are axially displaced downstream relative to the hub junction.

This geometric deviation follows advanced industrial optimization criteria aimed at mitigating tip-vortex cavitation in the peripheral gap, smoothing the initial pressure gradient, and sequencing the flow's hydrodynamic impact to reduce mechanical pulsations and structural fatigue at the hub. However, including these variable integration limits along the Z axis for each streamline would unnecessarily increase the mathematical and computational complexity of the applied Euler differential equations, without adding any further analytical value to the solution of the original velocity triangle.

Therefore, the methodology adopted is based on establishing the exact numerical solution on an orthogonal reference domain, precisely distinguishing the physical nature of each industrial geometric post-processing operation. In the case of leading-edge sweep, implementing it as a rigid axial translation of the entire streamline (displacing both the leading and trailing edges together by the same amount) introduces no physical alteration: since it is a local property of the curve, the angle β remains unchanged at every point, and the specific energy transferred at each radius, which depends on the peripheral velocity U and the change in V_θ between inlet and outlet rather than on its absolute axial position, is conserved exactly.

The radial alignment of the trailing edge, however, has a different geometric nature: it is not a translation but a trim, and as such it cannot be considered neutral from a hydrodynamic standpoint. By truncating a peripheral streamline before its integration reaches

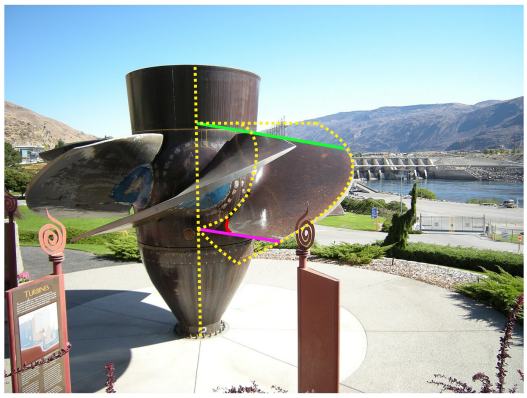
the design plane $z = -L_z$, that streamline leaves the region guided by the blade at a point where the numerical model has not yet completed the unwinding needed to cancel the outlet tangential component, $V_{\theta 2}$. As a result, the flow rate associated with that streamline leaves the runner retaining an undesigned residual tangential component, which translates into a local loss of extracted specific energy, directly quantifiable through the very Euler equation developed in this work, as well as into the introduction of residual rotational disturbances downstream of the runner, with the consequent risk of pressure pulsations and additional losses in the draft tube. This operation therefore constitutes a deliberate, quantifiable engineering trade-off between the trailing edge's constructive feasibility and the maximum theoretical hydraulic efficiency, and not a cost-free simplification, as is usually assumed in the detail-engineering phase of the hydroelectric industry when prioritizing the runner's mechanical robustness and manufacturability over the last percentage point of efficiency.

Consequently, it can be concluded that the numerical algorithm developed provides the fundamental physical and hydrodynamic core of the turbomachine. The transition toward the final industrial design does not require a reformulation of the energy-exchange equations, but rather the sequential application of two geometric post-processing operations of a different nature in the computer-aided design (CAD) environment: a rigid, physically neutral axial translation at the leading edge to configure the peripheral sweep, and a radial-alignment trim at the trailing edge which, while ensuring the runner's constructive feasibility, introduces a quantifiable hydraulic cost and should be understood as a detail-engineering trade-off decision, not as a simplification without consequences for the rigorous hydrodynamic model developed in this work. This workflow is completed in the detail-engineering phase together with hydrodynamic rounding operations at the profile's ends and the introduction of mechanical fillet radii at the blade root, a practice that mitigates fatigue stress concentration and consolidates the runner's structural feasibility.

In order to illustrate the transition between the pure numerical solution and the final commercial product, Figure 3.8 presents the comparison on a real runner. Figure 3.8a shows the as-manufactured physical geometry, while Figure 3.8b shows the methodological diagram of the geometric adjustments applied to the original theoretical blade (shown with dashed lines).



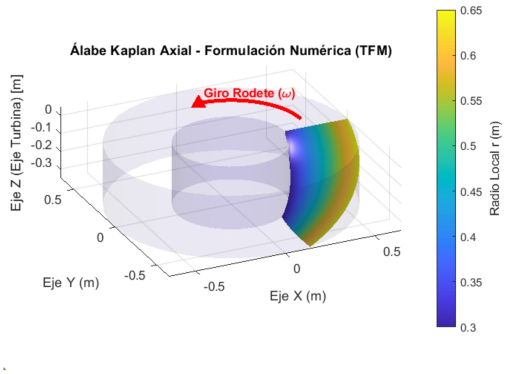
(a) Original geometry of a commercial Kaplan-type runner.



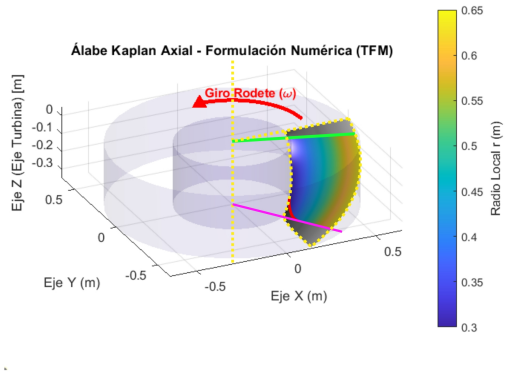
(b) Diagram of industrial transition geometric operations applied to the theoretical blade.

Fig. 3.8. Visual validation of the trimming operations on a real runner [27]. Dashed lines delimit the base theoretical volume; green traces the leading-edge translation (sweep); pink marks the flat radial trim of the trailing edge (constant θ_{rel}); and red highlights the mechanical root fillet. Own elaboration.

The reverse process is reproduced in Figure 3.9 from the results of the computational algorithm developed. Figure 3.9a shows the three-dimensional surface generated by the numerical script in its base orthogonal domain. Figure 3.9b, in turn, explicitly details the geometric post-processing actions that must be carried out on this point matrix to faithfully reproduce the features of the commercial design analyzed.



(a) Three-dimensional surface of the optimal blade obtained through the exact numerical algorithm.



(b) Identification of the cutting lines required for convergence to the commercial design.

Fig. 3.9. Three-dimensional visualization in the MATLAB environment of the formulated blade, and delimitation of the industrial cutting boundaries. Own elaboration.

Consequently, and given that the primary objective of this Master's Thesis is to parametrize and numerically obtain the optimal blade geometry for a reversible Deriaz-type turbine as

a function of its initial design variables, such as the hub and tip radii (R_{hub}, R_{tip}), the nominal flow rate (Q), the net head (H), the angular velocity (ω), among other parameters, the spatial modifications specific to commercial geometric post-processing will not be integrated at this initial stage. The algorithmic development will focus strictly on solving the inverse numerical model, prioritizing the optimization of the hydrodynamic profiles to maximize hydraulic power transfer and the machine's efficiency. This methodological delimitation of scope guarantees an exact, rigorous numerical basis, on which any detail-engineering phase or industrial constructive adaptation can be built with full theoretical assurance.

4. NUMERICAL DESIGN OF THE DERIAZ TURBINE

4.1. Transition Considerations toward the Deriaz Diagonal Model

It is essential to note that, while the above integral scheme accurately describes an axial flow, the transition toward designing a Deriaz turbine introduces notable fluid-mechanical and geometric complexities secondary to its diagonal nature:

- Kinematics of the Meridional Velocity ($\vec{V}_m$): Unlike the Kaplan turbine, where the meridional velocity is strictly vertical and parallel to the axis ($V_m = V_z$), in the Deriaz turbine the fluid moves along paths on a spherical surface, which implies the intrinsic appearance of a radial component (V_r) absent from the Kaplan's cylindrical analysis. Because of this condition, both the spatial direction of the vector $\vec{V}_m$ and its magnitude vary continuously along the channel. Since the flow rate remains constant, the fluid passes through cross-sections whose area corresponds to the lateral surface of coaxial truncated cones, causing the magnitude of V_m to change as the angle γ is traversed. Likewise, this kinematic distribution must satisfy the irrotational-flow condition and zero vorticity in the meridional plane.
- Distribution of the Inlet Tangential Velocity ($V_{\theta 1}$): Just before striking the runner's leading edge, the fluid is not yet guided by the blades, and its initial tangential velocity is set by the distributor according to the free-vortex law. Whereas in the Kaplan turbine this velocity depends solely on the cylindrical inlet radius, in the Deriaz the magnitude with which the water's tangential component meets the blade on each streamline is directly conditioned by the diagonal tilt angle (γ_1) at which the fluid enters the spherical surface.
- Need for Dual Angular Parametrization: The three-dimensional, spherically curved nature of the blade prevents the blade's orientation from being defined by a single angular parameter. For the Deriaz configuration, it becomes geometrically essential to calculate and coordinate two distinct spatial angles: the constructive angle with respect to the tangential plane (β), and a second meridional tilt angle (γ) that describes the blade shape in accordance with the streamline's local conical slope.

4.1.1. Transition in the construction of the velocity triangle's components

On the other hand, the major difference between a diagonal Deriaz turbine and an axial Kaplan turbine lies in its spherical surface of revolution and the intrinsic tilt of its streamlines in the meridional plane. In the Deriaz, fluid particles do not travel in purely longitudinal paths over straight cylinders, but diagonally, following the curvature of a

hemispherical wall, which substantially changes the definition and calculation of the tangential velocity owing to three main geometric factors:

1. Dependence on the channel's tilt angle (γ): Although conservation of free-vortex angular momentum remains unchanged at the runner inlet ($V_\theta \cdot r_c$ = Constant), the radius is no longer a flat coordinate r. In the Deriaz, the local cylindrical radius (r_c) is a trigonometric projection that depends on the machine's spherical radius (R_e) and on the channel's local tilt angle (γ) with respect to the horizontal. The tangential velocity at the leading edge is therefore calculated as:

$$V_{\theta 1}(R_e, \gamma_1) = \frac{\text{Constant}}{R_e \cdot \cos(\gamma_1)} \tag{4.1}$$

2. Two-dimensional velocity coupling: Unlike the Kaplan (where the tangential component interacts only with a purely axial velocity V_z), in the Deriaz the swirl coexists with the radial (V_r) and axial (V_z) components. When evaluating the kinematic triangles, the absolute tangential velocity must be vectorially superimposed on the native meridional velocity (V_m), which runs parallel to the tilt of the turbomachine's wall.

3. Variation of the flow-through radius between inlet and outlet: In a Kaplan turbine, the radius of a streamline is identical at the blade's inlet and outlet ($r_1 = r_2$). In the Deriaz, owing to the tilted contour of the hemisphere, the same streamline enters at a given cylindrical radius $r_{c1} = R_e \cdot \cos(\gamma_1)$ and discharges at a smaller radius $r_{c2} = R_e \cdot \cos(\gamma_2)$. As the local radius changes continuously along the blade's body, the fluid's tangential velocity changes in a coupled manner with the local drag velocity U, forcing the integration of the blade's path to be genuinely three-dimensional and to vary dynamically step by step over the spherical arc.

Effect of radius reduction along the path through the runner

As the fluid moves through the interior of the runner from the inlet section (γ_1) toward the outlet (γ_2), the conical path causes a progressive decrease in the local cylindrical radius $r_c(\gamma)$. This geometric variation significantly alters the flow's behavior and must be formally accounted for when constructing the velocity triangles along the blade:

- Acceleration of the meridional flow (V_m): The reduction in the conical flow-through cross-sections toward the outlet increases the magnitude of V_m, which changes the height of the velocity triangle between inlet and outlet.

- Variation of the runner's peripheral velocity (U): The machine's drag velocity decreases linearly with the cylindrical radius ($U = \omega \cdot r_c$). Since $U_2 < U_1$, the base of the velocity triangle narrows toward the outlet.

- Energy transfer and relative velocity (W): The coupled variation of V_m and U, together with the hydraulic work extraction imparted by the blade's curvature, which reduces V_θ, directly alters both the magnitude and the angle of the relative velocity β.

Consequently, the reduction in cylindrical radius drives a continuous three-dimensional evolution of the velocity triangles along each streamline, which is essential for precisely determining the twist and the angles of the blade's hydraulic profiles from the hub to the casing.

4.2. Distribution of the Meridional Velocity in Curved Channels and Diagonal Turbines

For the hydrodynamic sizing of axial and diagonal turbomachines, such as Kaplan and Deriaz turbines, it is essential to precisely characterize the behavior of the meridional velocity (V_m) along the flow-through cross-section.

In straight, axially symmetric ducts, such as a purely vertical or horizontal cylindrical channel, the streamlines are parallel and show no curvature in the meridional plane. Under the assumption of ideal, irrotational flow (omitting the effects of boundary-layer development at the walls), the meridional velocity remains strictly constant along the cylindrical radial direction:

$$\frac{\partial V_m}{\partial r} = 0 \quad \Longrightarrow \quad V_m = \text{constante} \tag{4.2}$$

However, when the duct undergoes a change in its path — as occurs in bends, in the transition from the distributor to the runner in a Kaplan turbine, or intrinsically throughout the entire channel of a diagonal Deriaz-type runner — the streamlines acquire meridional curvature. In these cases, V_m is no longer constant across the cross-section owing to the pressure gradient induced by the geometry.

4.3. Analytical calculation of the mean meridional velocity in the Deriaz runner

To model the behavior of the continuous flow through the flow-through channel of a Deriaz turbine, it is necessary to parametrize the runner's hemispherical geometry. The channel is bounded by two concentric physical boundaries: the hub (inner surface of spherical radius $R_{e,\text{int}}$) and the casing (outer surface of spherical radius $R_{e,\text{ext}}$). Both surfaces originate at the sphere's geometric center (C_e), located on the machine's axis of rotation.

The position of any particle in the meridional plane is determined by its spherical radius (R_e) and its diagonal, or azimuthal, tilt angle (γ), measured strictly positive downward from the horizontal plane intersecting C_e. In a real Deriaz turbine, the variable-pitch

blades do not sweep the full range of a hemisphere ($0° \leq \gamma \leq 90°$), since this would imply the fluid physically colliding with the axis of rotation (r_c (particle's cylindrical radius) = 0 for $\gamma = 90°$). Instead, the channel is restricted to an active angular band bounded by the interval:

$$\gamma \in [\gamma_1, \gamma_2] \tag{4.3}$$

The inlet angle is typically around $\gamma_1 \approx 15°$ or $30°$ (predominantly axial flow), and the outlet angle around $\gamma_2 \approx 45°$ or $60°$ (mixed/diagonal flow).

To calculate the magnitude of the mean meridional velocity ($V_{m,\text{mean}}$) at any section of the runner, a differential conical ring is chosen as the control surface (identical to the lateral surface of a coaxial truncated cone whose generatrix extends along the channel bounded by the inner ($R_{e,\text{int}}$) and outer ($R_{e,\text{ext}}$) spherical radii). This ring is obtained by evaluating a cross-section at a constant angle γ, which cuts the meridional streamlines orthogonally. The elemental area (dA) of an infinitesimal ring located at a spherical radial distance R_e with a differential thickness dR_e is determined by its lateral cylindrical perimeter:

$$dA = 2\pi \cdot r_c \cdot dR_e \tag{4.4}$$

Since the cylindrical radius's position coordinate (r_c) is the horizontal projection of the spherical radius for a given tilt γ, the fundamental trigonometric relation $r_c = R_e \cdot \cos(\gamma)$ is applied, substituting that term into the elemental-area equation:

$$dA = 2\pi \cdot R_e \cdot \cos(\gamma) \cdot dR_e \tag{4.5}$$

To obtain the total flow-through area perpendicular to the flow (A) at that specific angular section of the runner, the set of differential rings is integrated from the hub's lower boundary to the casing's upper limit:

$$A(\gamma) = \int_{R_{e,\text{int}}}^{R_{e,\text{ext}}} 2\pi \cdot R_e \cdot \cos(\gamma) \cdot dR_e \tag{4.6}$$

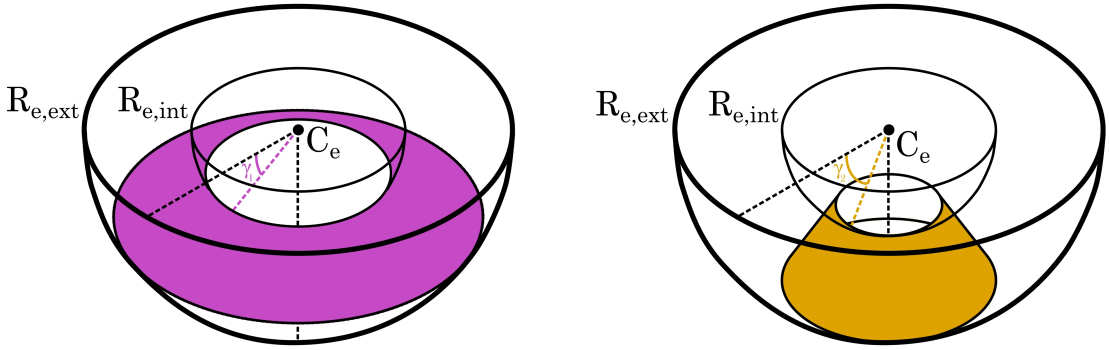
Pulling the constant terms out of the integration operator, since the angle γ is invariant along that transverse control band, the expression is solved directly as follows:

$$A(\gamma) = 2\pi \cdot \cos(\gamma) \left[\frac{R_e^2}{2} \right]_{R_{e,\text{int}}}^{R_{e,\text{ext}}} \tag{4.7}$$

$$A(\gamma) = \pi \cdot \cos(\gamma) \cdot \left(R_{e,\text{ext}}^2 - R_{e,\text{int}}^2 \right) \tag{4.8}$$

As illustrated in Figure 4.1, the flow channel between the hub ($R_{e,\text{int}}$) and the outer wall ($R_{e,\text{ext}}$) is geometrically defined with its origin at the sphere's center C_e. Since the meridional velocity $\vec{V}_m$ is orthogonal to the spherical radius at all times, the cross-sections through which the fluid passes correspond to lateral surfaces of coaxial truncated cones.

As the fluid advances along the meridional path, the tilt angle γ increases from a smaller value at the inlet (Figure 4.1a) to a larger value at the outlet (Figure 4.1b). This progression demonstrates the shrinking nature of the flow-through areas, since the cylindrical radius's projection decreases as the position vector approaches the vertical axis of rotation ($\gamma = 90°$). Consequently, in order to keep the flow rate passing through the runner constant, the magnitude of the mean meridional velocity ($V_{m,\text{mean}}$) must dynamically increase as the angle γ is traversed.



(a) Flow-through surface for a smaller tilt angle γ (near the channel inlet), bounded between the inner spherical radius $R_{e,\text{int}}$ and the outer one $R_{e,\text{ext}}$ from the sphere's center C_e.

(b) Flow-through surface for a larger angle γ (as the fluid advances toward the outlet), showing the drastic reduction in the truncated cone's lateral area.

Fig. 4.1. Geometric evolution of the flow-through cross-section along the meridional channel in a Deriaz turbine. The flow-through surface takes the form of a truncated cone orthogonal to the streamlines, centered at C_e and bounded by the spherical radii $R_{e,\text{int}}$ and $R_{e,\text{ext}}$. As the fluid advances and the angle γ increases (from the horizontal $\gamma = 0°$ toward the vertical $\gamma = 90°$), the cylindrical radius's projection decreases, which gives the flow area an intrinsically shrinking nature. Own elaboration.

Once the geometric area of the conical section has been analytically determined as a function of the angular coordinate, the principle of mass conservation for an incompressible fluid under steady-state conditions is invoked. Dividing the turbine's total design volumetric flow rate (Q) by the computed area gives the mean magnitude of the meridional velocity for each angular position along the flow-through channel:

$$V_{m,\text{mean}}(\gamma) = \frac{Q}{\pi \cdot \cos(\gamma) \cdot \left(R_{e,\text{ext}}^2 - R_{e,\text{int}}^2\right)} \tag{4.9}$$

This analytical result reveals the restrictive role the channel walls play on the flow's kinematics: as the fluid descends and progresses toward the outlet, increasing the value of γ, the term $\cos(\gamma)$ gradually decreases. If the radial spacing between the hub and the casing ($R_{e,\text{ext}} - R_{e,\text{int}}$) remained strictly constant, the cross-sectional area would decrease and the mean meridional velocity would be forced to accelerate markedly toward the outlet, a criterion that must be carefully balanced in the profile design laws to optimize hydraulic efficiency and avoid local cavitation phenomena.

4.3.1. Justification of the variation of velocity V_m with the spherical radius (R_e)

From what was presented in Section 4.3, it follows that the mean meridional velocity increases as the angle γ increases. However, for a rigorous analysis it is essential not to limit the discussion to this mean value, but to evaluate the local velocity at each point. This requires considering its dependence on both the angle γ and the spherical radius R_e, which makes it possible to precisely construct the velocity triangle at any coordinate within the runner's domain.

The equation used to describe the pressure distribution transverse to the flow direction is obtained from Euler's equation applied to an ideal fluid. Consider a differential fluid element of mass dm and volume $dV = dn \cdot dA$ moving at a meridional velocity V_m along a curved streamline characterized by a radius of curvature R_e.

Owing to the curvature of the path, the particle experiences a normal acceleration directed toward the geometric center of rotation, given in magnitude by:

$$a_n = \frac{V_m^2}{R_e} \tag{4.10}$$

According to Newton's second law ($dF_n = dm \cdot a_n$), there must be a net force in the normal direction capable of providing this centripetal acceleration. Since viscous stresses are neglected in an ideal-fluid model, the only mechanical force responsible for curving the path is that originating from the static pressure gradient p.

It is essential to clarify the conceptual distinction between static pressure and the forces it generates on the faces of the fluid element. Pressure p is a strictly positive scalar quantity that always acts in a state of compression toward the inside of the control volume. The resulting pressure force $\vec{F}_p$, by contrast, is a vector quantity oriented along the inward normal of each surface.

Figure 4.2 defines the normal coordinate n (equivalent to the local spherical radius R_e), whose origin is located at the geometric center of the path, which in this case coincides with the center of the Deriaz turbine's spherical chamber, and whose positive direction increases outward, away from that center. The algebraic sign of the forces therefore depends exclusively on their projection onto the n axis.

On one hand, the force on the inner face, denoted F_1, compresses the element's upper face at a position $R_e = n$. Since it pushes outward, away from the center of curvature, this

force points in the increasing direction of n, resulting in a positive contribution:

$$F_1 = +p\,dA \tag{4.11}$$

On the other hand, the force on the outer face, denoted F_2, compresses the lower face located at a distance $R_e = n + dn$. Since it pushes inward, toward the center of the path, its direction opposes the increase of n, so it takes a negative sign:

$$F_2 = -\left(p + \frac{\partial p}{\partial n} dn\right) dA \tag{4.12}$$

Geometrically, the differential control volume is determined by a thickness dn in the normal direction and a cross-sectional area dA corresponding to the faces perpendicular to that direction n. The product of both dimensions defines the element's differential volume, $dV = dA \cdot dn$.

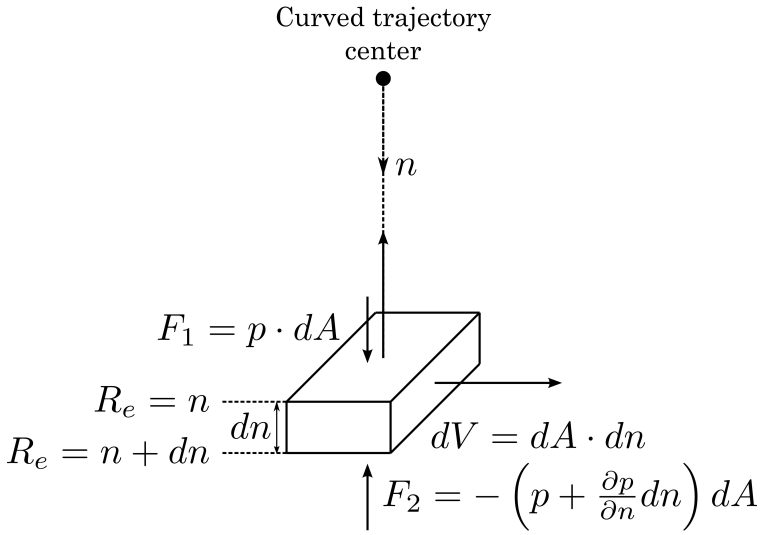


Fig. 4.2. Balance of pressure forces on a differential fluid element in the direction normal to the path. Own elaboration.

Performing the algebraic force balance in the normal direction n gives the net pressure force dF_n:

$$dF_n = F_1 + F_2 = p\,dA - \left(p + \frac{\partial p}{\partial n} dn\right) dA = -\frac{\partial p}{\partial n} \underbrace{dn\,dA}_{dV} \tag{4.13}$$

which leads directly to the expression in terms of the differential volume dV:

$$dF_n = -\frac{\partial p}{\partial n}\, dV \tag{4.14}$$

Substituting the mass of the differential element ($dm = \rho\, dV$, where ρ represents the fluid's density) into the equation of motion gives:

$$-\frac{\partial p}{\partial n}\, dV = \rho\, dV \left(-\frac{V_m^2}{R_e}\right) \tag{4.15}$$

Simplifying the differential volume dV and taking the spherical radius of curvature R_e as the normal spatial coordinate ($dn \equiv dR_e$), the force-balance expression in the meridional plane reduces to:

$$\frac{dp}{dR_e} = \rho \frac{V_m^2}{R_e} \tag{4.16}$$

However, the general radial equilibrium equation in the meridional plane of a turbomachine, formulated in a relative reference frame and including the effects of fluid swirl and rotation, can be expressed as [28]:

$$\frac{\partial p}{\partial R_e} = \rho \left(\frac{V_m^2}{R_e} + \frac{V_\theta^2}{r} \cos(\gamma) + 2\omega V_m f(\gamma) + V_m \frac{\partial V_m}{\partial s} \sin(\alpha) \right) \tag{4.17}$$

Where the term $\frac{V_m^2}{R_e}$ represents the normal acceleration due to channel curvature, $\frac{V_\theta^2}{r}\cos(\gamma)$ the centrifugal contribution due to swirl, $2\omega V_m f(\gamma)$ the apparent Coriolis component, and the term $V_m \frac{\partial V_m}{\partial s}\sin(\alpha)$ the projection onto the normal direction of the advective acceleration along the streamline.

For the development of this analytical model, this expression is simplified to its purely geometric form, equation 4.16, considering only the centripetal acceleration due to channel curvature and neglecting the effects of swirl, rotation, and gradients along the streamline.

This reduction is based on three justifications of a physical and methodological nature:

1. Adoption of an absolute reference frame (absence of Coriolis): By formulating the analysis of the spherical chamber and the flow channel in an inertial (fixed) coordinate system, the fictitious or apparent forces, such as the Coriolis acceleration, are rigorously zero, since the effect of rotation is fully captured within the kinematics of the absolute velocities.

2. Methodological sequencing of the inverse problem (decoupling of V_θ): At this design stage, the tangential velocity component (V_θ) is an a priori unknown, since it depends on the blades' final geometric layout (β) and on the velocity triangles along the runner. To avoid a circular dependency in the calculation, it is methodologically essential to first determine the meridional velocity field $V_m(R_e, \gamma)$ as a function of the channel's curvature. The resulting V_m profile will act as the fundamental boundary condition for the subsequent construction of the velocity triangles and the characterization of V_θ.

3. Third, the contribution of the gradient along the streamline is negligible, being a second-order term. Since the longitudinal variation of the velocity V_m with respect to the coordinate s is an effect of much smaller magnitude than the main normal-acceleration terms, the influence of the product $V_m \frac{\partial V_m}{\partial s} \sin(\alpha)$ is infinitesimal in the force balance.

Therefore, from a physical standpoint, this fundamental relation (4.16) establishes that the transverse pressure gradient is the mechanism responsible for supplying the centripetal force required for the fluid to maintain its curved path against the channel walls, avoiding boundary-layer separation.

As the streamlines curve, the fluid's inertia induces a tendency to move toward the outer casing, accumulating and compressing the fluid mass against that concave surface. Since all the terms on the right-hand side of equation 4.16 are strictly positive ($\rho > 0$, $V_m^2 > 0$, and $R_e > 0$), it follows that:

$$\frac{dp}{dR_e} > 0 \tag{4.18}$$

This mathematically demonstrates that the static pressure is not uniform across the cross-section, but increases monotonically in the radial direction from the hub ($R_{e,\min}$, lower-pressure zone) toward the casing, or shroud ($R_{e,\max}$, higher-pressure zone).

On the other hand, considering an incompressible, ideal meridional flow with no mechanical work exchange, and neglecting gravitational effects compared with the dynamic pressure forces in the region analyzed, the total specific energy remains constant along each streamline according to Bernoulli's principle [25, Cap. 2]:

$$p + \frac{1}{2}\rho V_m^2 = \text{constante} \tag{4.19}$$

Differentiating Equation (4.19) with respect to the radial coordinate R_e:

$$\frac{dp}{dR_e} + \rho V_m \frac{dV_m}{dR_e} = 0 \quad \Longrightarrow \quad \frac{dp}{dR_e} = -\rho V_m \frac{dV_m}{dR_e} \tag{4.20}$$

Equation (4.20) reveals the principle of energy conservation: since the static pressure increases outward ($\frac{dp}{dR_e} > 0$), the kinetic energy must undergo a compensating decrease in that same direction ($\frac{dV_m}{dR_e} < 0$).

Equating the pressure gradient obtained from the dynamic equilibrium (Equation 4.16) with the one derived from the energy balance (Equation 4.20):

$$\rho \frac{V_m^2}{R_e} = -\rho V_m \frac{dV_m}{dR_e} \tag{4.21}$$

Simplifying the common terms ρ and V_m (assuming $V_m \neq 0$):

$$\frac{V_m}{R_e} = -\frac{dV_m}{dR_e} \quad \Longrightarrow \quad \frac{dV_m}{dR_e} + \frac{V_m}{R_e} = 0 \tag{4.22}$$

Equation (4.22) represents the analytical irrotationality condition in the meridional plane. Its result confirms that the meridional velocity cannot be uniform along the radius, but must take a hyperbolic distribution such that the velocity is maximum near the hub (smaller R_e) and minimum near the casing (larger R_e).

In other words, since the fluid enters the channel with uniform total energy across its section, as it passes through the curved region the pressure naturally increases toward the casing and decreases toward the hub. For the Bernoulli constant to remain identical on each streamline without generating imbalances or vortices that force the fluid to rotate on itself, the differences in pressure energy must be compensated exactly by the kinetic energy. This requires the velocity V_m to increase in the lower-pressure regions and decrease in the higher-pressure regions, strictly satisfying the irrotational-flow equation in the meridional plane.

The first-order differential equation 4.22 has separable variables. Rearranging the terms to proceed with its integration gives:

$$\frac{1}{V_m} dV_m = -\frac{1}{R_e} dR_e \tag{4.23}$$

Integrating both sides of the equality between a reference streamline and a generic spherical radius gives the relation:

$$\ln(V_m) = -\ln(R_e) + \ln(K) \implies \ln(V_m) = \ln\left(\frac{K}{R_e}\right) \tag{4.24}$$

Applying the exponential function to both sides gives the hyperbolic distribution law of the meridional velocity strictly in terms of R_e:

$$V_m(R_e, \gamma) = \frac{K(\gamma)}{R_e} \tag{4.25}$$

Where $K(\gamma)$ is a coupling function, constant for each angular section, whose value depends on the total flow rate Q passing through the machine. To determine this constant analytically, the continuity-based definition of flow rate integrated across the conical section's area is used:

$$Q = \int_{R_{e,\text{int}}}^{R_{e,\text{ext}}} V_m(R_e, \gamma) \cdot dA \tag{4.26}$$

Substituting the expression for the differential area analyzed previously (Section 4.3) ($dA = 2\pi \cdot R_e \cdot \cos(\gamma) \cdot dR_e$) and the velocity law obtained, the integral is defined as:

$$Q = \int_{R_{e,\text{int}}}^{R_{e,\text{ext}}} \left(\frac{K(\gamma)}{R_e} \right) \cdot (2\pi \cdot R_e \cdot \cos(\gamma) \cdot dR_e) \tag{4.27}$$

Simplifying the spherical radius R_e in the integrand and pulling the constant quantities out of the operation gives an immediate integral:

$$Q = 2\pi \cdot \cos(\gamma) \cdot K(\gamma) \int_{R_{e,\text{int}}}^{R_{e,\text{ext}}} dR_e \tag{4.28}$$

$$Q = 2\pi \cdot \cos(\gamma) \cdot K(\gamma) \cdot (R_{e,\text{ext}} - R_{e,\text{int}}) \tag{4.29}$$

Solving for the coupling constant $K(\gamma)$ in terms of the known design variables (flow rate and channel geometry) gives:

$$K(\gamma) = \frac{Q}{2\pi \cdot \cos(\gamma) \cdot (R_{e,\text{ext}} - R_{e,\text{int}})} \tag{4.30}$$

Finally, substituting $K(\gamma)$ into the original distribution law gives the exact analytical expression for the meridional velocity at any point in the runner, strictly as a function of its spherical radius and angular position:

$$V_m(R_e, \gamma) = \frac{Q}{2\pi \cdot R_e \cdot \cos(\gamma) \cdot (R_{e,\text{ext}} - R_{e,\text{int}})} \tag{4.31}$$

The irrotational-flow condition in the meridional plane was imposed to ensure an optimal streamline design free of secondary vortices in the flow-through channel. By solving the zero-vorticity equation on the runner's spherical geometry, the mathematics naturally dictates that the velocity profile take a hyperbolic distribution. This is a direct consequence of mass conservation: since the outer peripheral perimeter is much larger than that of the hub, the fluid undergoes a natural deceleration toward the casing, a physical phenomenon that must be rigorously incorporated into the blade profile design to avoid hydraulic separation.

To facilitate understanding of the fluid's behavior in the hemispherical space, Figure 4.3 schematically illustrates the qualitative evolution of the meridional velocity profile $V_m(R_e, \gamma)$. It should be noted that the cross-section shown in the illustration corresponds strictly to the turbomachine's meridional plane, formally defined as any plane that longitudinally contains the axial axis of rotation and the fluid particle's meridional velocity vector.

In this representation in the meridional plane, two fundamental physical trends can be clearly observed:

- Effect of the spherical radius (R_e): At a given angular section γ, the magnitude of the velocity increases toward the inner boundary (hub) and decreases toward the

outer boundary (casing, or tip). This hyperbolic variation confirms that a smaller spherical radius corresponds to a higher meridional velocity, as required by the irrotationality condition in this plane.

- Effect of the meridional tilt angle (γ): As the flow advances toward the machine's outlet and the angle increases ($\gamma_2 > \gamma_1$), the velocity vectors undergo a general increase in magnitude. This is due to the progressive reduction in the available flow-through cross-sectional area, which drives the velocity increase through the continuity equation.

It should be clarified that the geometric representation in Figure 4.3 is a didactic idealization used to visualize the full domain of the mathematical model. In a real Deriaz turbine, the cavity does not span a full semicircle from 0° to 90°, since this would prevent the continuous discharge of water. In practice, the operating angle range spans roughly between 15° and 60°, connecting at the lower end with the draft-tube piping. Nevertheless, this graphical simplification confirms that the entire blading of the runner is contained within this hemispherical cavity, ensuring flow that is perfectly guided between the concentric walls.

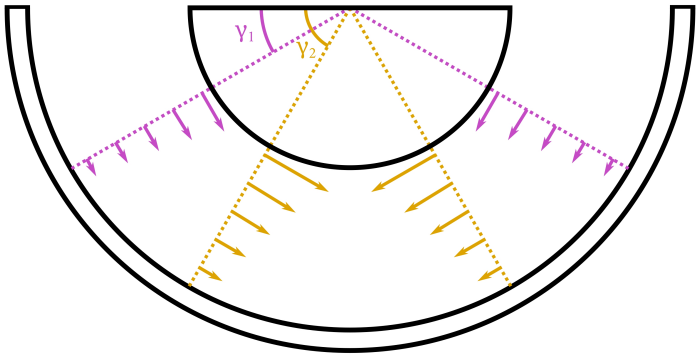


Fig. 4.3. Schematic representation, in the meridional plane, of the evolution of the meridional velocity profile $V_m(R_e, \gamma)$ through the Deriaz turbine's spherical flow-through channel. Own elaboration.

Additionally, the methodology of Morabito et al. [8] incorporates into the meridional velocity calculation a blade blockage factor (τ) that reduces the effective flow-through area as a function of the blades' local thickness and tilt: this factor grows near the hub, where the blade thickness is greater, and also as γ advances toward the outlet, since a similar blade thickness blocks an increasing fraction of an ever-smaller flow-through area. In that methodology, incorporating τ is straightforward because the blade count is fixed as a starting input. In this work, however, the blade count (Z) is not an input but a result obtained a posteriori through the solidity criterion (Section 3.4), so incorporating τ would introduce a circular dependency between the blade geometry, Z itself, and the meridional velocity, requiring an iterative solution scheme seeking convergence under a predefined tolerance. For this reason, within the scope of this work it was decided not to incorporate

this factor, with its implementation noted as an essential line of future work to improve the software's accuracy (Section 6.4).

Once the local magnitude $V_m(R_e, \gamma)$ has been determined, the direction and sense of the meridional velocity vector $\vec{V}_m$ are defined by the flow-through channel's geometric constraint. In three-dimensional space, the vector $\vec{V}_m$ is strictly confined to the meridional plane; that is, the orthogonal coordinate plane formed by the axial axis of rotation ($\vec{e}_z$) and the local cylindrical radius ($\vec{e}_r$).

Likewise, since the hub and casing contours impose concentric spherical boundaries of constant radius with respect to the geometric center C_e, the meridional velocity vector is at every point orthogonal to the local spherical radius ($\vec{R}_e$). Taking as a reference the spherical coordinate system centered at the machine, and since the fluid undergoes continuous displacement from the upper inlet section (γ_1) toward the lower outlet section (γ_2), the flow direction coincides with the direction of increasing meridional tilt angle γ. The meridional velocity vector can therefore be formally expressed in spherical components as:

$$\vec{V}_m(R_e, \gamma) = V_m(R_e, \gamma) \cdot \vec{e}_\gamma \tag{4.32}$$

Where $\vec{e}_\gamma$ is the unit vector tangent to the meridional streamline, oriented downward toward the runner outlet. This vector definition excludes any twisting component in the meridional plane, isolating the fluid's through-flow behavior before superimposing the effects of the tangential velocity or of the runner's rotation.

4.3.2. Expression of the meridional velocity vector in cylindrical components

To facilitate the kinematic integration of the flow with the runner's rotational motion, it is convenient to project the meridional velocity vector onto the usual cylindrical coordinate system ($\vec{e}_{r_c}, \vec{e}_\theta, \vec{e}_z$). Taking as a reference the orientation of the spherical channel, where the axial axis z takes increasingly negative values downward and the cylindrical radius r_c increases outward toward the casing, the through-flow tangent unit vector $\vec{e}_\gamma$ is trigonometrically decomposed as follows:

$$\vec{e}_\gamma = -\sin(\gamma)\,\vec{e}_{r_c} - \cos(\gamma)\,\vec{e}_z \tag{4.33}$$

Applying this decomposition to the local magnitude of the velocity, the vector $\vec{V}_m$ is determined by its cylindrical radial (V_r) and axial (V_z) components:

$$\vec{V}_m(R_e, \gamma) = V_r(R_e, \gamma)\,\vec{e}_{r_c} + V_z(R_e, \gamma)\,\vec{e}_z \tag{4.34}$$

Where the scalar velocity components are formally defined as:

$$V_r(R_e, \gamma) = -V_m(R_e, \gamma) \cdot \sin(\gamma) \tag{4.35}$$

$$V_z(R_e, \gamma) = -V_m(R_e, \gamma) \cdot \cos(\gamma) \tag{4.36}$$

The negative sign in both components has a strict physical meaning. On one hand, the negative radial component indicates that as the fluid advances through the spherical channel toward the outlet, increasing the meridional tilt angle γ, the streamlines contract toward the machine's central axis of rotation. On the other hand, the negative sign in the axial component reflects consistency with the chosen reference system, in which the flow moves toward increasingly negative values of z.

Introducing the analytical definition of the local cylindrical radius, expressed in terms of the spherical radius and the angular position as $r_c = R_e \cdot \cos(\gamma)$, it is possible to fully reformulate the kinematic velocity triplet at any point in space.

First, substituting the spherical radius $R_e = r_c / \cos(\gamma)$ into the magnitude of the meridional velocity, the angular terms in the denominator cancel reciprocally, reducing the meridional velocity to an expression dependent on the cylindrical radial position:

$$V_m(r_c) = \frac{Q}{2\pi \cdot r_c \cdot (R_{e,\text{ext}} - R_{e,\text{int}})} \tag{4.37}$$

From this fundamental simplification, the orthogonal components of the velocity vector in the cylindrical system are derived explicitly and completely through their respective trigonometric projections:

$$V_r(r_c, \gamma) = \frac{-Q \cdot \sin(\gamma)}{2\pi \cdot r_c \cdot (R_{e,\text{ext}} - R_{e,\text{int}})} \tag{4.38}$$

$$V_z(r_c, \gamma) = \frac{-Q \cdot \cos(\gamma)}{2\pi \cdot r_c \cdot (R_{e,\text{ext}} - R_{e,\text{int}})} \tag{4.39}$$

This integral analytical formulation couples the particle's linear behavior with the physical limits of the spherical channel of revolution. With the complete expressions for the radial, axial, and meridional components available, the mathematical basis needed to subsequently superimpose the absolute tangential velocity induced by the power transfer in the runner blades is established.

It is important to note that, although the work transfer and the three-dimensional effects associated with the runner's rotation locally alter the purely irrotational condition in the real flow, the hyperbolic distribution $V_m(R_e, \gamma)$ derived in Equation (4.31) is adopted as the fundamental kinematic design law. This simplifying hypothesis allows the meridional velocity to be fixed at each cross-section and the subsequent construction of the velocity triangles along all the streamlines of the flow-through channel.

4.3.3. Justification of the kinematic decoupling in the transition toward the draft tube

A fundamental aspect of the flow model's formulation is the analysis of the fluid's behavior as it leaves the runner's conical-spherical cavity and enters the cylindrical draft-tube piping. Although the channel's geometry forces a reorientation of the streamlines from the diagonal direction to the axial direction, it is not necessary to incorporate corrections or cross-interference effects into the runner's velocity triangles, by virtue of the following principles of fluid mechanics and turbomachinery design:

1. Continuity and incompressibility of the medium: Water behaves as a continuous, incompressible medium. For this reason, the spatial convergence does not produce particle "collisions" at the micro/macroscopic level, but rather a continuous, smooth redistribution of pressures and velocities. The kinematic transition follows a gradual variation of the pressure field governed by the continuity equation and energy conservation.

2. Geometric proximity and curvature in the central zone: The streamlines closest to the inner boundary ($R_{e,\text{int}}$) travel along the runner's hub surface. Since they are already at a small cylindrical radius close to the axis of rotation, their transition to the axial direction requires only a minimal radius of curvature. To avoid discontinuities or separation zones at this central point, the hub's geometric design is extended with an aerodynamic nose cone that progressively guides the flow, preventing the formation of central stagnation zones or turbulence.

3. Outlet design condition ($V_{\theta 2} \approx 0$): The blade profile at the outlet section (γ_2) is sized under the premise of discharge with zero, or near-zero, angular momentum ($V_{\theta 2} \to 0$). By eliminating the tangential velocity component at the runner outlet, the formation of a reactive helical vortex (vortex rope) in the draft tube is avoided. This ensures that the flow enters the diffuser with an essentially axial, uniform path, canceling out any hydrodynamic interaction that could affect the blading upstream.

4. Hydro-kinematic decoupling principle: Under the subcritical, incompressible regime characteristic of hydraulic turbomachines, the boundary conditions set downstream (in the draft tube) do not alter the momentum exchange occurring inside the runner. The blading's function is to extract hydraulic energy, while the draft tube's function is to convert the residual outlet kinetic energy ($V_2^2/2g$) into static pressure energy.

For all of the above, it is concluded that the velocity law $V_m(R_e, \gamma)$ and the kinematic distribution formulated are fully valid and self-sufficient for sizing the runner blades up to the outlet section γ_2, with no additional terms required for convergence effects in the draft tube.

4.4. Geometric Transformation of the Velocity Triangles in the Deriaz Diagonal Turbine

To understand the three-dimensional sizing of the runner in the Deriaz turbine, it is necessary to analyze how the velocity triangles evolve compared with conventional axial turbomachines (Kaplan type).

In a purely axial turbomachine, the flow runs along cylindrical streamlines of constant radius ($r_{c1} = r_{c2} = r_c$). This condition simplifies the fluid's kinematics:

- The runner's drag velocity remains unchanged between the inlet and outlet of the same streamline: $U_1 = U_2 = \omega \cdot r_c$.
- The meridional velocity is purely axial ($V_m = |V_z|$) and remains practically constant along the flow path ($V_{m1} \approx V_{m2}$).
- The relative flow angle β relates only the axial displacement (dz) to the relative angular displacement ($r_c \cdot d\theta_{\text{rel}}$) on the cylindrical surface:

$$\tan\beta = \frac{V_m}{U - V_\theta} = \frac{dz}{r_c \cdot d\theta_{\text{rel}}} \tag{4.40}$$

However, in the conical-spherical geometry of the Deriaz turbine, the water particle's path is no longer cylindrical but becomes a three-dimensional diagonal streamline. This geometric change alters the components of the velocity triangle along the blading:

1. Variation of the drag velocity ($U_1 \neq U_2$): As the fluid advances from the inlet section (γ_1) toward the outlet (γ_2), the cylindrical radius progressively decreases ($r_{c2} < r_{c1}$). Consequently, the base of the velocity triangle contracts linearly toward the outlet:

$$U(R_e, \gamma) = \omega \cdot r_c = \omega \cdot R_e \cdot \cos(\gamma) \quad \Longrightarrow \quad U_2 < U_1 \tag{4.41}$$

2. Acceleration of the meridional velocity ($V_{m1} \neq V_{m2}$): The reduction in the cross-sectional area of the conical flow-through sections requires, by the continuity equation, an increase in the magnitude of the meridional velocity toward the outlet ($V_{m2} > V_{m1}$), increasing the height of the velocity triangle.

3. Differential decomposition of the meridional arc (ds_m): The meridional velocity $\vec{V}_m$ follows the tangent to the spherical path, projecting onto the cylindrical radial and axial components as a function of the meridional tilt angle γ:

$$dr_c = -ds_m \cdot \sin(\gamma) \tag{4.42}$$

$$dz = -ds_m \cdot \cos(\gamma) \tag{4.43}$$

Consequently, the kinematic relation defining the relative velocity angle and the blade's spatial orientation (β) is expressed differentially in terms of the meridional arc ds_m and the angular deformation $r_c \cdot d\theta_{\text{rel}}$:

$$\tan\beta = \frac{V_m}{U - V_\theta} = \frac{ds_m}{r_c \cdot d\theta_{\text{rel}}} = \frac{\sqrt{dr_c^2 + dz^2}}{r_c \cdot d\theta_{\text{rel}}} \tag{4.44}$$

Solving for the meridional arc differential from the previous trigonometric relations gives $ds_m = -\frac{dr_c}{\sin(\gamma)}$ and $ds_m = -\frac{dz}{\cos(\gamma)}$. Substituting these equivalences directly into the numerator of the kinematic relation gives the fundamental geometric coupling law of the diagonal blade:

$$\tan\beta = \frac{-dr_c}{\sin(\gamma) \cdot r_c \cdot d\theta_{\text{rel}}} = \frac{-dz}{\cos(\gamma) \cdot r_c \cdot d\theta_{\text{rel}}} \tag{4.45}$$

Equation (4.45) demonstrates the fundamental role of the meridional tilt angle γ: as the flow advances toward the outlet (increasing γ), a larger fraction of the blade's relative slope $\tan\beta$ translates into movement toward the axis of rotation (reduction of the cylindrical radius r_c), at the expense of purely axial advance (z). This three-dimensional coupling requires designing blades with a complex double spatial curvature to ensure correct energy transfer along the entire runner.

4.5. Three-Dimensional Geometric Construction of the Blade

4.5.1. Fundamental Differential Equation of the Diagonal Blade

To convert the local kinematic relations obtained in the previous section into a three-dimensional physical surface, it is necessary to express the blade's orientation in a coordinate system adapted to the channel's spherical geometry.

In the previous section, the differential relation for the relative flow angle was defined as:

$$\tan\beta = \frac{ds_m}{r_c \cdot d\theta_{\text{rel}}} \tag{4.46}$$

Since the streamlines in the spherical chamber trace circular arcs of constant radius R_e, the meridional arc differential ds_m is directly linked to the meridional angle differential $d\gamma$ through the fundamental geometric relation of the circle:

$$ds_m = R_e \cdot d\gamma \tag{4.47}$$

Likewise, recalling that the local cylindrical radius is the projection $r_c = R_e \cdot \cos(\gamma)$, the denominator of the velocity triangle is expressed as $r_c \cdot d\theta_{\text{rel}} = R_e \cdot \cos(\gamma) \cdot d\theta_{\text{rel}}$.

Substituting both equivalences into the expression for $\tan\beta$, the spherical radius R_e cancels in the numerator and denominator, removing the need to work with the coordinates dr_c and dz separately:

$$\tan\beta = \frac{R_e \cdot d\gamma}{R_e \cdot \cos(\gamma) \cdot d\theta_{\text{rel}}} = \frac{d\gamma}{\cos(\gamma) \cdot d\theta_{\text{rel}}} \tag{4.48}$$

Solving for the azimuthal angular variation $d\theta_{\text{rel}}$ with respect to the meridional advance $d\gamma$ gives the fundamental differential equation for tracing the blade's skeleton lines in three-dimensional space:

$$\frac{d\theta_{\text{rel}}}{d\gamma} = \frac{1}{\tan(\beta(\gamma)) \cdot \cos(\gamma)} = \frac{\cot(\beta(\gamma))}{\cos(\gamma)} \tag{4.49}$$

Where:

- θ_{rel} is the relative rotation angle of the water particle with respect to the blade as a moving reference.
- γ is the independent angular integration variable in the meridional plane (ranging, for example, from the inlet $\gamma_1 = 30°$ to the outlet $\gamma_2 = 60°$).
- $\cos(\gamma)$ acts as the projection factor of the spherical radius onto the axis of rotation, accounting for the diagonal channel's spatial contraction.
- $\beta(\gamma)$ is the continuous distribution of the blade angle obtained through cubic Hermite interpolation.

4.5.2. Blade Angle Distribution: Cubic Hermite Interpolation (C^1)

Once the velocity triangles at the inlet (γ_1) and outlet (γ_2) have been solved for each streamline R_e, the flow's kinematic angles $\beta_1(R_e)$ and $\beta_2(R_e)$ are available. However, to define the blade's physical surface in 3D space, it is necessary to establish how the blade angle $\beta(\gamma)$ evolves along the entire meridional path and to solve the blade's differential equation.

To ensure optimal hydraulic behavior and avoid flow separation or shock at the runner's inlet and outlet, the transition between β_1 and β_2 cannot be done linearly. A cubic Hermite interpolation is used over the dimensionless domain $t \in [0, 1]$, defined as:

$$t = \frac{\gamma - \gamma_1}{\gamma_2 - \gamma_1} \tag{4.50}$$

The interpolation polynomial takes the form:

$$P(t) = 3t^2 - 2t^3 \tag{4.51}$$

In this way, the local variation of the blade angle $\beta(\gamma)$ is given by:

$$\beta(\gamma) = \beta_1 + (\beta_2 - \beta_1) \cdot P(t) \tag{4.52}$$

Properties of the Hermite Polynomial

- Boundary Conditions: At the ends of the flow path, $P(0) = 0$ and $P(1) = 1$ hold strictly, ensuring that $\beta(\gamma_1) = \beta_1$ and $\beta(\gamma_2) = \beta_2$.
- Derivative Continuity (C^1): The derivatives at the ends are zero:

$$\left.\frac{dP}{dt}\right|_{t=0} = 0 \quad \text{and} \quad \left.\frac{dP}{dt}\right|_{t=1} = 0 \tag{4.53}$$

  This ensures smooth tangency at the inlet (leading edge) and at the outlet (trailing edge), minimizing shock losses and decoupling the blade's central curvature from its edges.

4.5.3. Discretization and Step-by-Step Numerical Integration

To solve the differential equation along each streamline $R_e \in [R_{e,\text{int}}, R_{e,\text{ext}}]$, the meridional interval is discretized into N_{chord} equally spaced steps.

The angular increment $\Delta\theta_{\text{rel}}$ between two consecutive points $j-1$ and j is calculated using the midpoint method (centered scheme) to maximize numerical accuracy:

1. Step midpoint:

$$\gamma_{\text{mid}} = \frac{\gamma_j + \gamma_{j-1}}{2}, \quad \beta_{\text{mid}} = \frac{\beta(\gamma_j) + \beta(\gamma_{j-1})}{2} \tag{4.54}$$

2. Incremental angular differential:

$$\Delta\theta_{\text{rel}} = \frac{\cot(\beta_{\text{mid}})}{\cos(\gamma_{\text{mid}})} \cdot (\gamma_j - \gamma_{j-1}) \tag{4.55}$$

3. Cumulative integration (Counterclockwise Rotation): Setting the leading edge at the inlet as the angular origin ($\theta_{\text{rel}}(\gamma_1) = 0$), the local angular position is updated as:

$$\theta_{\text{rel},j} = \theta_{\text{rel},j-1} - \Delta\theta_{\text{rel}} \tag{4.56}$$

   (The negative sign represents the molecule's relative lag with respect to the turbine's direction of rotation).

4.5.4. Generation of the 3D Cartesian Mesh

Once the coordinate pair $(\gamma_j, \theta_{\text{rel},j})$ has been obtained for each streamline i and point along the chord j, the conversion to 3D Cartesian coordinates (X, Y, Z) is carried out for the CAD reconstruction of the surface:

$$\begin{cases} r_c(j,i) = R_e(i) \cdot \cos(\gamma_j) \\ X(j,i) = r_c(j,i) \cdot \cos(\theta_{\text{rel}}(j,i)) \\ Y(j,i) = r_c(j,i) \cdot \sin(\theta_{\text{rel}}(j,i)) \\ Z(j,i) = -R_e(i) \cdot \sin(\gamma_j) \end{cases} \tag{4.57}$$

The result is a three-dimensional array of points that fully defines the Deriaz turbine's blade mean surface, ensuring a smooth, continuous transition consistent with the calculated kinematic velocity triangles.

4.6. Code developed for calculating and generating the three-dimensional geometry of the blade's mean surface through MATLAB programming

Following and adapting the programming procedure previously developed for axial turbomachines (Kaplan type), its equivalent formulated for diagonal Deriaz-type turbines is presented below. This adaptation incorporates all of the theoretical foundations, hydraulic hypotheses, and flow dynamics explained in the previous sections. To facilitate the reproducibility of the results and allow their reuse in future projects, the complete script with its different calculation blocks is included in Appendix B, and is freely available for use and modification.

Unlike the predominantly axial flow of Kaplan turbines, Deriaz technology operates on a conical-spherical meridional path in which the flow direction has both axial and radial components. The algorithm developed therefore modifies the geometric discretization and the governing equations to incorporate the characteristic spherical radii of curvature and conical angles, ensuring rigorous three-dimensional modeling of the blade's mean surface.

4.6.1. Block 1: Energy, geometric, and efficiency parameters

In this first stage, the hydroelectric plant's nominal operating conditions (Q_0, H_n, RPM) are parametrized, and the fundamental geometric dimensions of the Deriaz runner's conical-spherical meridional path are defined. The flow channel is bounded by the hub's spherical radius ($R_{e,\text{int}}$) and the casing's spherical radius ($R_{e,\text{ext}}$), as well as by the inlet (γ_1) and outlet (γ_2) conical angles. Internal losses are also introduced through the hydraulic (η_h), volumetric (η_v), and mechanical/organic (η_o) efficiencies.

CODE 4.1. Block 1: Energy, geometric, and efficiency parameters.

```
RPM = 150; % Runner rotational speed (rpm)
Q0 = 110; % Nominal volumetric flow rate drawn from the head (m^3/s)
Hn = 60; % Net head available at the plant (m)
g = 9.81; % Acceleration of gravity (m/s^2)

% Turbomachine efficiencies (internal efficiencies)
eta_h = 0.91; % Hydraulic efficiency (friction and dissipation losses)
eta_v = 0.97; % Volumetric efficiency (leakage losses in the hub clearance)
eta_o = 0.98; % Organic/mechanical efficiency (bearing friction losses)
eta_t = eta_h * eta_v * eta_o; % Total turbine efficiency

% Boundary spherical radii (Hub and Casing):
Re_int = 2; % Hub spherical radius (m)
Re_ext = 2.6; % Casing spherical radius (m)

% Meridional path limits (conical angles):
gamma1_deg = 30; % Runner inlet (degrees)
gamma2_deg = 60; % Runner outlet (degrees)
```

Explicitly accounting for the internal efficiencies (η_h, η_v, η_o) from the initial stage is critical for accurately determining the actual flow-through rate and the Eulerian energy transferred. The parametrization through the spherical framework (R_e, γ), in turn, constitutes the essential geometric basis for modeling the diagonal flow characteristic of Deriaz technology.

4.6.2. Block 2: Kinematics and conversions

In this second block, the kinematic and dynamic quantities governing the energy transfer between the fluid and the runner are evaluated. From the rotational speed and the system's efficiencies, the effective useful flow rate (Q_{real}), the available hydraulic power ($\dot{W}_n$), the mechanical power delivered to the shaft ($\dot{W}_t$), and the ideal Eulerian head (H_∞) are determined.

One of the most important particularities in the kinematic characterization of Deriaz turbines is the definition of the flow-through geometric framework. Owing to the divergence of the conical-spherical flow channel, the representative radius is not evaluated using a simple arithmetic mean, but through the equivalent mean spherical radius ($R_{e,\text{mean}}$), based on the exact logarithmic mean of the physical boundaries of the hub ($R_{e,\text{int}}$) and the casing ($R_{e,\text{ext}}$).

CODE 4.2. Block 2: Kinematics and conversions

```
omega = (2 * pi * RPM) / 60; % Angular velocity (rad/s)
```

```
Q_real = Q0 * eta_v; % Effective useful flow rate (m^3/s)
gamma1 = deg2rad(gamma1_deg); % Inlet tilt angle (rad)
gamma2 = deg2rad(gamma2_deg); % Outlet tilt angle (rad)

% Theoretical and useful head powers
W_n = 1000 * g * Q0 * Hn; % Gross hydraulic power available (W)
W_t = W_n * eta_t; % Useful mechanical power delivered to the shaft (W)
H_inf = Hn * eta_h; % Theoretical energy per Euler's hypothesis (m)

% Equivalent mean spherical radius (Exact Logarithmic Mean):
Re_medio = (Re_ext - Re_int) / log(Re_ext / Re_int);
```

Applying the logarithmic mean for $R_{e,\text{mean}}$ makes it possible to properly weight the continuous distribution of the meridional area along the channel. This mean spherical radius will later be used to calculate the blade's mean chord length and thereby determine the runner's blade count (Z).

4.6.3. Block 3: Spatial discretization and data matrices

Once the design conditions and the global spherical framework have been fixed, the three-dimensional flow-channel domain is meshed and discretized. In this block, the runner's spatial resolution is defined through the number of radial spherical streamlines (N_{radios}), spanning the continuous space between the hub and the casing, and the number of integration steps (N_{chord}) along the flow's path in the conical direction (γ).

From these dimensions, the coordinate vectors (R_e, γ) are structured, and the three-dimensional storage matrices (X, Y, Z) and the cylindrical radius-of-curvature matrix (RC) are initialized. This memory pre-allocation is key to optimizing the efficiency of the matrix calculations in MATLAB during the subsequent numerical integration of the blade shape.

CODE 4.3. Block 3: Spatial discretization and data matrices.

```
N_radios = 30; % Number of streamlines between hub and casing
N_cuerda = 50; % Number of steps in the flow direction (gamma)

Re_vec = linspace(Re_int, Re_ext, N_radios);
gamma_vec = linspace(gamma1, gamma2, N_cuerda);

X = zeros(N_cuerda, N_radios);
Y = zeros(N_cuerda, N_radios);
Z = zeros(N_cuerda, N_radios);
RC = zeros(N_cuerda, N_radios);
```

This matrix structuring sets the structure of the calculation domain, making it possible to independently evaluate each of the 30 streamlines as a decoupled one-dimensional entity during the solution phase, while remaining perfectly integrated into the three-dimensional reconstruction of the complete blade.

4.6.4. Block 4: 3D numerical integration in the conical-spherical channel (Methodological core)

This fourth block constitutes the fundamental conceptual and mathematical core of the work, where the blade's differential equation, adapted to the diagonal conical-spherical geometry of Deriaz turbines, is solved numerically. As stated earlier, unlike a simple axial blade (where the radial component is negligible), in the Deriaz geometry the fluid travels a path over spherical caps of constant radius R_e, simultaneously varying its conical angle γ and its cylindrical position $r_c = R_e \cos(\gamma)$.

The numerical procedure is structured into four interconnected phases for each streamline:

1. Boundary velocity triangles and boundary conditions

At the runner inlet ($\gamma = \gamma_1$), the local drag velocity U_1 and the meridional velocity V_{m1} are related to the theoretical Eulerian head H_∞ through the transfer of angular momentum:

$$U_1 = \omega \cdot R_e \cos(\gamma_1), \quad V_{\theta 1} = \frac{g \cdot H_\infty}{U_1} \tag{4.58}$$

A strict physical feasibility check ($V_{\theta 1} < U_1$) is established to ensure there are no dynamic inconsistencies or unwanted recirculation at the inlet. The blades' kinematic angle β_1 is then calculated:

$$\beta_1 = \arctan\left(\frac{V_{m1}}{U_1 - V_{\theta 1}}\right) \tag{4.59}$$

At the outlet ($\gamma = \gamma_2$), under the hypothesis of zero tangential flow at the outlet under design conditions ($V_{\theta 2} = 0$), the velocity triangle simplifies, giving the angle β_2:

$$\beta_2 = \arctan\left(\frac{V_{m2}}{U_2}\right) \tag{4.60}$$

2. Blade angle distribution law $\beta(\gamma)$

As mentioned earlier, to reduce the risk of boundary-layer separation and minimize cavitation, the transition of the blade angle between the inlet (β_1) and the outlet (β_2) is not done linearly. A smooth, third-order cubic polynomial interpolation is applied, whose derivatives at the endpoints are zero ($\frac{d\beta}{d\gamma} = 0$ at γ_1 and γ_2). This guarantees a smooth change in the blade's physical curvature:

$$t = \frac{\gamma - \gamma_1}{\gamma_2 - \gamma_1}, \quad \beta(\gamma) = \beta_1 + (\beta_2 - \beta_1)(3t^2 - 2t^3) \tag{4.61}$$

3. Derivation of the Conical-Spherical Blade Differential Equation

Considering a differential element on the spherical surface of radius R_e, the flow-guidance condition relates the meridional component $ds_m = R_e\, d\gamma$ to the particle's tangential relative lag with respect to the blade, $ds_{\theta,\mathrm{rel}} = r_c\, d\theta_{rel} = R_e \cos(\gamma)\, d\theta_{rel}$, through the cotangent of the local angle $\beta(\gamma)$:

$$\cot(\beta) = \frac{ds_{\theta,\mathrm{rel}}}{ds_m} = \frac{R_e \cos(\gamma)\, d\theta_{rel}}{R_e\, d\gamma} \implies \frac{d\theta_{rel}}{d\gamma} = \frac{\cot(\beta(\gamma))}{\cos(\gamma)} \tag{4.62}$$

This ordinary differential equation is integrated numerically using the midpoint method (Euler/Trapezoidal midpoint) to ensure greater accuracy in advancing the wrap angle θ_{rel}.

4. Reconstruction of the three-dimensional surface

Finally, integrating the azimuthal rotation (θ_{rel}) makes it possible to map the local curvilinear coordinates onto the global Cartesian system (X, Y, Z) through the transformation:

$$X = R_e \cos(\gamma)\cos(\theta_{rel}), \quad Y = R_e \cos(\gamma)\sin(\theta_{rel}), \quad Z = -R_e \sin(\gamma) \tag{4.63}$$

CODE 4.4. Block 4: 3D numerical integration in the conical-spherical channel.

```
for i = 1:N_radios
    Re = Re_vec(i); % Spherical radius of the current streamline

    % --- 1. VELOCITY TRIANGLE AT THE INLET (gamma1) ---
    rc1 = Re * cos(gamma1); % Inlet cylindrical radius
    U1 = omega * rc1; % Inlet drag velocity
    Vm1 = Q_real / (2 * pi * Re * cos(gamma1) * (Re_ext - Re_int)); % Meridional
        Vm

    % Azimuthal component required by Euler (V_theta1 * U1 = g * H_inf)
    Vtheta1 = (g * H_inf) / U1;

    % Physical safety check:
    if Vtheta1 >= U1
        error('Physical error at Re = %.3f m: V_theta1 (%.2f m/s) >= U1 (%.2f m/s)
            .', Re, Vtheta1, U1);
    end

    beta1 = atan2(Vm1, (U1 - Vtheta1)); % Inlet kinematic angle

    % --- 2. VELOCITY TRIANGLE AT THE OUTLET (gamma2) ---
    rc2 = Re * cos(gamma2); % Outlet cylindrical radius
    U2 = omega * rc2; % Outlet drag velocity
    Vm2 = Q_real / (2 * pi * Re * cos(gamma2) * (Re_ext - Re_int)); % Accelerated
        Vm
```

```
    beta2 = atan2(Vm2, U2); % Outlet kinematic angle (Vtheta2 = 0)

    % --- 3. CUBIC INTERPOLATION LAW FOR THE ANGLE BETA(gamma) ---
    t = (gamma_vec - gamma1) / (gamma2 - gamma1);
    polinomio_cubico = 3*(t.^2) - 2*(t.^3);
    beta_gamma = beta1 + (beta2 - beta1) .* polinomio_cubico;

    % --- 4. NUMERICAL INTEGRATION OF THE DIFFERENTIAL EQUATION ---
    theta_rel = 0;

    for j = 1:N_cuerda
        gamma_actual = gamma_vec(j);

        if j > 1
            d_gamma = gamma_vec(j) - gamma_vec(j-1);

            % Evaluation at the step midpoint
            gamma_mid = (gamma_vec(j) + gamma_vec(j-1)) / 2;
            beta_mid = (beta_gamma(j) + beta_gamma(j-1)) / 2;

            % Exact differential equation of the Deriaz diagonal blade:
            d_theta_rel = (cot(beta_mid) / cos(gamma_mid)) * d_gamma;

            % Counterclockwise rotation (fluid lag)
            theta_rel = theta_rel - d_theta_rel;
        end

        % Local cylindrical coordinates:
        rc_local = Re * cos(gamma_actual);
        z_local = -Re * sin(gamma_actual);

        % Conversion to 3D Cartesian coordinates:
        X(j, i) = rc_local * cos(theta_rel);
        Y(j, i) = rc_local * sin(theta_rel);
        Z(j, i) = z_local;
        RC(j, i) = rc_local;
    end
end
```

Implementing the explicit differential scheme in the conical domain ensures that the curvature of the blade surface responds appropriately to the local kinematic conditions, minimizing friction losses and ensuring correct transfer of Eulerian energy along all the flow streamlines of the meridional channel.

Kinematic discussion: Complexity in the evolution of the angle β compared with axial turbomachines

It is especially noteworthy how the diagonal nature of the conical-spherical channel alters the dynamics of the velocity triangles compared with axial turbomachines (Kaplan type). In a purely axial machine, the cylindrical radius remains constant along each streamline ($r_{c1} = r_{c2}$), which implies that the drag velocity $U = \omega \cdot r_c$ and the meridional velocity V_m (under the constant flow-through section hypothesis) remain unchanged between the runner's inlet and outlet. Under this axial assumption, the evolution of the flow's kinematic angle $\beta = \arctan\left(\frac{V_m}{U-V_\theta}\right)$ depends almost exclusively on the reduction in the tangential velocity component (V_θ), which decreases as the fluid gives up its angular momentum, causing the angle β to decrease monotonically from inlet to outlet ($\beta_2 < \beta_1$).

In the Deriaz runner, by contrast, the conical path introduces a simultaneous variation of three independent variables along the meridional path:

1. Decrease in drag velocity (U): As the flow advances from γ_1 to γ_2, the local cylindrical radius $r_c(\gamma) = R_e \cos(\gamma)$ changes, directly affecting the peripheral velocity $U(\gamma) = \omega \cdot R_e \cos(\gamma)$.

2. Acceleration/deceleration of the meridional velocity (V_m): The flow-through cross-section is not uniform, so $V_m(\gamma)$ is coupled to the convergence or divergence of the spherical channel.

3. Transfer of angular momentum (V_θ): The tangential component progressively decreases owing to the Eulerian energy exchange.

Consequently, in the Deriaz turbomachine the direct rule that β should decrease systematically due solely to the vanishing of V_θ does not hold a priori. The local kinematic angle $\beta(\gamma)$ is defined by a decoupled function dependent on the conical angle, for a given spherical radius R_e:

$$\beta(\gamma) = \arctan\left(\frac{V_m(\gamma)}{U(\gamma) - V_\theta(\gamma)}\right) \tag{4.64}$$

This multivariable dependence fully justifies the need to adopt a smooth interpolation law (such as the cubic polynomial implemented) to explicitly control the blade's curvature and avoid boundary-layer separation induced by adverse velocity gradients.

4.6.5. Block 5: Calculation of the blade count (Z)

The fifth block addresses the numerical determination of the blade count (Z) needed for the runner, applying to the conical-spherical geometry of the Deriaz channel the same

hydrodynamic solidity criterion (σ) introduced in Section 3.4 for the axial case, now referred to the local cylindrical radius r_c and its tangential pitch ($t = \frac{2\pi r_c}{Z}$):

$$\sigma = \frac{L_{\text{chord}}}{t} = \frac{L_{\text{chord}} \cdot Z}{2\pi r_c} \tag{4.65}$$

To evaluate this parameter in a physically representative way, the three-dimensional chord integration is not carried out at the boundaries (hub or casing), but on the representative streamline associated with the equivalent mean spherical radius ($R_{e,\text{mean}}$) determined in Block 2 (Section 4.6.2).

The numerical procedure is carried out in the following stages:

1. Spatial localization: The discrete streamline closest to $R_{e,\text{mean}}$ is identified by searching for the minimum absolute difference.

2. 3D curvilinear integration of the chord: The actual Euclidean length along the three-dimensional profile is calculated by summing the spatial differentials ($\Delta X, \Delta Y, \Delta Z$):

$$dL = \sqrt{dx^2 + dy^2 + dz^2} \implies L_{\text{chord, actual}} = \sum_{j=1}^{N_{\text{chord}}-1} dL_j \tag{4.66}$$

3. Calculation of the blade count (Z): Imposing a target hydrodynamic solidity ($\sigma_{\text{target}} = 1.4$, suitable for minimizing friction losses and preventing separation and interference phenomena in diagonal Deriaz runners), Z is solved for and rounded to the nearest integer:

$$Z_{\text{calculated}} = \frac{2\pi \bar{r}_c \cdot \sigma_{\text{target}}}{L_{\text{chord, actual}}} \implies Z_{\text{optimal}} = \text{round}(Z_{\text{calculated}}) \tag{4.67}$$

   Physical safety limits are applied ($4 \leq Z \leq 12$) in accordance with industrial design practice. The lower limit is set at 4, compared with the 3 allowed in the Kaplan's axial case (Section 3.4), reflecting the fact that, naturally, the more pronounced a turbomachine's radial character, the greater the number of blades it requires to guide the flow efficiently.

4. Kinematic characterization (n_q): The dimensionless/kinematic specific speed is evaluated to safely classify the machine within the family of diagonal turbines:

$$n_q = \frac{\text{RPM} \cdot \sqrt{Q_0}}{(g \cdot H_n)^{3/4}} \tag{4.68}$$

CODE 4.5. Block 5: Calculation of the blade count (Z).

```
% 1. Locating the streamline closest to Re_medio
[~, idx_Rmedio] = min(abs(Re_vec - Re_medio));
```

```
% 2. Exact curvilinear integration of the 3D chord at Re_medio
dx = diff(X(:, idx_Rmedio)); % diff subtracts each element from the next (dX1,
    dX2)
dy = diff(Y(:, idx_Rmedio));
dz = diff(Z(:, idx_Rmedio));
dL = sqrt(dx.^2 + dy.^2 + dz.^2);
L_cuerda_real = sum(dL); % Actual integrated 3D chord (m)

% 3. Equivalent mean cylindrical radius for the pitch calculation
rc_promedio_medio = mean(RC(:, idx_Rmedio));

% 4. Target hydrodynamic solidity (sigma = L_chord / tangential_pitch)
sigma_target = 1.4; % Value consistent with Deriaz turbines

% 5. Determination of the blade count Z
paso_tangencial_req = L_cuerda_real / sigma_target;
Z_calculado = (2 * pi * rc_promedio_medio) / paso_tangencial_req;
Z_optimo = round(Z_calculado);

% Physical safety limits for Deriaz turbines
Z_optimo = max(4, min(Z_optimo, 12));

% Kinematic specific speed (nq) for the technical report
nq = RPM * sqrt(Q0) / ((g*Hn)^(3/4));
```

This methodology ensures that the blade density obtained guarantees effective guidance of the fluid in accordance with Euler's theory, preventing both blockage from excessive friction (associated with a high blade count) and unguided flow (associated with an insufficient blade count).

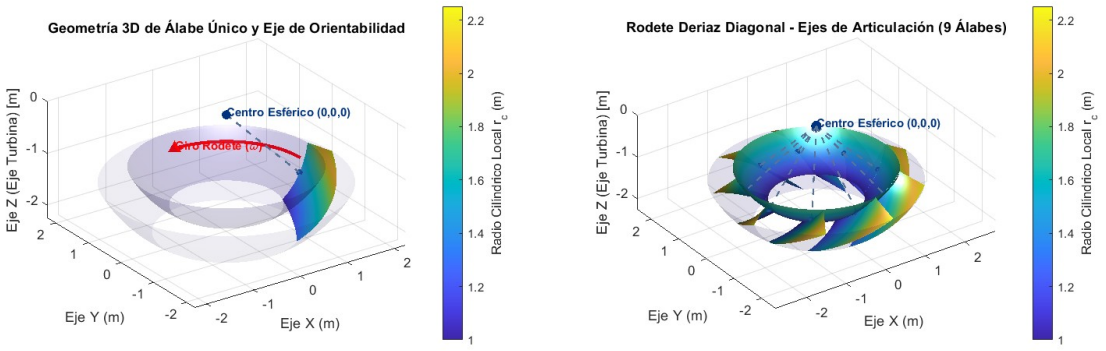
4.6.6. Blocks 6 and 7: Three-dimensional rendering of the blade's mean surface and assembly of the complete runner

The program's last two blocks are devoted to the graphical post-processing and three-dimensional spatial reconstruction of the results obtained. Rather than stopping at the algebraic calculation, the algorithm generates the geometric models corresponding to the blade's mean surface:

- Block 6 (Rendering of a single blade): From the coordinate matrices (X, Y, Z) obtained in Block 4 (Section 4.6.4), the individual blade's mean surface is rendered using three-dimensional mesh plotting commands (`surf`). This makes it possible to closely inspect the curvature, the evolution of the blade angle $\beta(\gamma)$, and the spatial blade shape between the hub and casing streamlines.
- Block 7 (Generation of the complete runner): Using the calculated blade count

(Z_{optimal}) determined in Block 5 (Section 4.6.5), a rotation matrix about the Z axis is applied with an angular step $\Delta\theta = \frac{2\pi}{Z_{\text{optimal}}}$. This clones and cyclically positions the mean-surface geometry to reconstruct the complete Deriaz runner, making it possible to visually verify the hydrodynamic solidity (σ), the absence of undue overlaps, and the consistency of the flow-through channel.

Figure 4.4 shows the three-dimensional representations generated by the MATLAB script for the individual mean surface and the circular arrangement of the complete runner.



(a) Three-dimensional mean surface of a single Deriaz blade.

(b) Spatial assembly of the complete runner (Z blades).

Fig. 4.4. 3D rendering of the blade mean-surface geometry and of the diagonal Deriaz runner generated in MATLAB. Own elaboration.

The visual finish obtained validates the robustness of the conical-spherical blade's numerical integration, providing the point clouds and reference surfaces needed for subsequent export to computer-aided design (CAD) environments and computational fluid dynamics (CFD) analysis.

4.7. Geometric Basis and Blade Trimming Post-processing

The numerical method implemented in this work solves the three-dimensional blade differential equation over pure conical-spherical streamlines:

$$\frac{d\theta}{d\gamma} = \frac{R_e \cdot \cot\beta(\gamma)}{r_c(\gamma)} \tag{4.69}$$

Where γ represents the conical elevation angle with respect to the equatorial plane, R_e the radius of the theoretical reference sphere, $r_c(\gamma) = R_e \cos\gamma$ the local cylindrical radius, and $\beta(\gamma)$ the flow's angular distribution law.

This formulation guarantees strict compliance with the theoretical velocity triangles along the Deriaz turbine's streamlines. As a direct result of the differential integration

in spherical coordinates, the raw blade surface generated has leading and trailing edges aligned with the conical wavefronts of the limit angles γ_1 and γ_2.

4.7.1. Technical Justification of CAD/CAM Trimming

In the industrial practice and manufacturing of diagonal turbomachinery, the surfaces resulting from hydrodynamic integration are subjected to a geometric post-processing, or trimming, stage, for the following fundamental reasons:

1. Compatibility with the Distributor: The leading edges must have a cut fitted to the cylindrical-horizontal or conical planes of the wicket-gate distributor, minimizing losses from distortion of the incoming flow.
2. Orientability Clearances (Tip & Hub Clearance): Since the Deriaz turbine's blades rotate mechanically about a conical pivot axis to change their pitch angle, the ends adjacent to the spherical hub and to the outer casing must be trimmed with specific tolerances to avoid mechanical collisions during pivoting.
3. Machining and Structural Integrity: Flat or cylindrical truncation of the edges smooths the sharp hyperbolic edges at the root and tip, reducing dynamic stress concentration and facilitating multi-axis (CNC) machining.

The geometry calculated through this code therefore represents the theoretical primary hydrodynamic surface, on which the spatial post-processing equivalent to the trimming performed in industrial CAD environments must be applied.

5. PRELIMINARY PROPOSAL TO REPLACE A FRANCIS UNIT WITH A DERIAZ RUNNER

As discussed in the initial sections, severe regulatory and environmental-impact restrictions drastically limit the construction of new hydroelectric power plants domestically. Faced with this limitation, the most efficient alternative is to modernize already-installed infrastructure. In this context, the search is motivated for a hydroelectric power plant in Spain (traditionally equipped with Francis turbines) whose operating conditions are suitable for the installation of a double-regulation Deriaz turbine, thereby optimizing the plant's dynamic response to the new flexibility demands of the electricity market.

5.1. Site selection

Given that Deriaz runners operate within their ideal range at heads between 40 m and 100 m, the Saucelle II Hydroelectric Power Plant, located on the Duero River in the province of Salamanca, stands out as one of the most suitable options for replacing conventional Francis units with highly flexible diagonal turbines.

At this site, the existing turbine units operate under sharp flow-rate fluctuations in the international stretch of the Duero and reservoir-level variations that directly affect the available head. These variations drastically reduce the efficiency of fixed-geometry Francis turbines when operation moves away from the nominal point. Installing Deriaz runners would mitigate off-design flow losses and would allow operation at partial load with maximum efficiency under modulated flow rates, optimizing generation and the plant's integration into the electricity market.

5.2. Current site characteristics

The plant has a set of operating parameters defined for its current configuration:

- Nominal net head: 57 m (with a gross head of 63 m).
- Total installed capacity: 285 MW, split between 2 Francis units of 142.5 MW each.
- Nominal design flow rate per unit: 110 m^3/s in turbine mode (220 m^3/s in total for the plant).
- Rotational speed: 125 rpm.

The public sources consulted on Saucelle II are not fully consistent with one another, and none of them explicitly documents the rotational speed of its units. For this reason, the

hydraulic and synchronism parameters given here (net head, flow rate per unit, and rotational speed) have been adopted as a reasoned estimate, obtained by interpreting the available sources on Saucelle II together with a comparison against other better-documented plants of similar type and scale, rather than as figures taken literally from a single official technical datasheet.

5.2.1. Justification of the Designated Parameters and Runner Design Criteria

The geometric and hydrodynamic sizing of the diagonal Deriaz-type runner for the Saucelle II hydroelectric power plant is based on a set of interconnected technical decisions. These variables were calculated numerically using the code developed in MATLAB from the plant's starting hydraulic parameters (H_n, Q_0, n) and from the hydraulic, volumetric, and organic efficiencies assumed as input data (η_h, η_v, η_o).

In particular, a design net head of $H_n = 60$ m is taken, from which, considering the assumed hydraulic efficiency ($\eta_h = 0.91$), the theoretical Euler head actually exploited by the runner results:

$$H_\infty = H_n \cdot \eta_h = 60 \cdot 0.91 = 54.6 \text{ m} \tag{5.1}$$

Nominal Rotational Speed ($n = 150$ rpm)

It should be noted that, although the Saucelle II hydroelectric power plant is estimated to currently operate Francis turbines at a nominal speed of 125 RPM ($p = 24$ pole pairs), in this work it was decided to size the diagonal Deriaz alternative for an operating regime of 150 RPM ($p = 20$ pole pairs). The choice of 150 rpm follows a strict synchronous-coupling criterion with the European 50 Hz electrical grid. A generator's kinematic speed is defined as:

$$n = \frac{60 \cdot f}{p} \tag{5.2}$$

For $f = 50$ Hz and $n = 150$ rpm, the alternator requires $p = 20$ pole pairs (40 total poles), a highly standardized, efficient configuration for high-power synchronous generators ($W_t \approx 56$ MW).

From a fluid-dynamic perspective, 150 rpm increases the peripheral drag velocity at the root ($U_1 = \omega \cdot r_{c1}$), guaranteeing the physical stability condition $V_{\theta 1} < U_1$ throughout the entire domain. This avoids the formation of anomalous relative angles ($\beta_1 > 90°$) at the hub inlet, eliminating the risk of flow separation and shock at the leading edge. It also places the dimensionless kinematic specific speed at:

$$n_q = \frac{\omega \cdot \sqrt{Q_0}}{(g \cdot H_n)^{3/4}} = 13.17 \text{ rpm} \quad (\approx 73 \text{ metric rpm}) \tag{5.3}$$

placing the operating point at the sweet spot of maximum efficiency for diagonal turbomachines.

Spherical Geometry and Radius Ratio (R_{int} = 2.0 m, R_{ext} = 2.6 m, ν = 1.30)

The spherical radii of the meridional contour were sized by setting the inlet meridional velocity to a suitable range under the effective useful flow rate $Q_{\text{real}} = Q_0 \cdot \eta_v = 110 \cdot 0.97 = 106.70\ \text{m}^3/\text{s}$, lower than the plant's nominal gross flow rate ($Q_0 = 110\ \text{m}^3/\text{s}$) precisely because of the assumed volumetric efficiency ($\eta_v = 0.97$):

- Hub radius ($R_{\text{int}} = 2.00$ m): Provides an outer hub diameter of $D_{\text{hub}} = 4.00$ m, a volume that is structurally essential to house the seating discs, trunnions, bearings, and the hydraulic servomechanism crosshead that operates the blades synchronously.
- Casing radius ($R_{\text{ext}} = 2.60$ m): Defines a maximum envelope diameter of $D_{\text{tip}} = 5.20$ m.
- Radius ratio ($\nu = R_{\text{ext}}/R_{\text{int}} = 1.30$): This work adopts as its own design criterion that the radius ratio in Deriaz runners must be confined to the interval $\nu \in [1.25, 1.40]$. A controlled ratio of $\nu = 1.30$ avoids an excessive increase in the tangential pitch at the tip, limiting the thickening of the boundary layer over the profile and suppressing losses from bulky wakes at the trailing edge.

Hydrodynamic Solidity (σ = 1.40) and Blade Count (Z = 9)

The blade count is determined analytically, integrated over the mean streamline ($R_{\text{mean}} = 2.29$ m). A target hydrodynamic solidity criterion is imposed of:

$$\sigma = \frac{L_{\text{chord}}}{t} = 1.40 \tag{5.4}$$

Where L_{chord} is the integrated 3D curvilinear length ($L_{\text{chord}} = 1.574$ m) and t is the tangential pitch ($t = 2\pi r_c/Z$). Solving for Z, the code yields an exact value of $Z = 9$ blades:

1. Hydrodynamic anti-stall guarantee: Provides a 40% overlap between consecutive blades, distributing the aerodynamic load evenly and mitigating the risk of cavitation in the draft tube.
2. Internal mechanical feasibility: Maintains sufficient angular clearance ($\Delta\theta = 40°$) to avoid physical interference between the trunnion articulation linkages inside the hub.

Meridional Path Limits ($\gamma_1 = 30°$, $\gamma_2 = 60°$)

Finally, the envelope's conical angles define a meridional angular span of $\Delta\gamma = 30°$, placing the blade trunnion's pivot axis at a mean angle of $\gamma_{\text{mean}} = 45°$ with respect to the turbine's vertical axis. This range has been adopted in this work as a reasonable value consistent with the usual constructive type of high-power Deriaz turbines:

- Inlet ($\gamma_1 = 30°$): Receives the diagonal flow coming from the distributor with the maximum inlet acceleration component.
- Outlet ($\gamma_2 = 60°$): Gradually orients the flow toward the discharge tube in a conical-divergent direction, ensuring a smooth hydrodynamic transition.

5.3. Preliminary nature of the design and future validation directions

It is essential to emphasize that the blade geometry obtained within this proposal corresponds to a preliminary design stage. The kinematic variables generated constitute a first approximation aimed at feeding the initial stages of the iterative turbomachinery design cycle.

To consolidate the proposal's feasibility and define the final geometry for manufacturing, numerical analyses using Computational Fluid Dynamics (CFD) will be required. Such studies must validate the flow's hydrodynamic behavior and will make it possible to give the blade a final physical thickness based on specific hydrodynamic profiles. In particular, it is of special interest to implement reversible profiles (such as symmetric or elliptical profiles, usually applied in reversible turbomachines), which guarantee an optimal hydrodynamic response in both turbine and pump regimes. Likewise, this refinement phase must incorporate the post-processed geometry trims discussed previously to avoid mechanical interference and optimize the clearance within the casing.

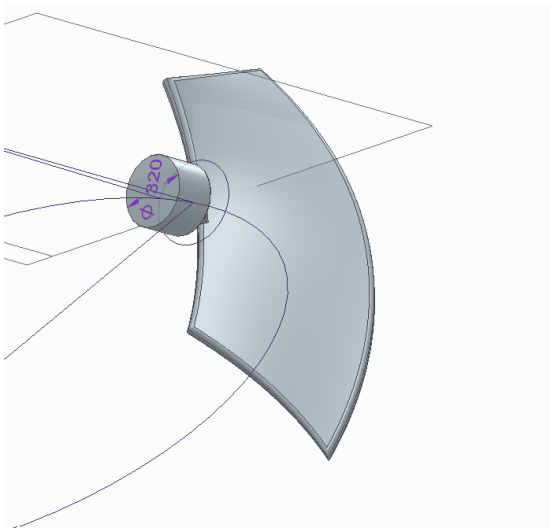
Notwithstanding these necessary validation stages, the data and models presented in this thesis provide a robust, consistent geometric and kinematic basis, fully valid for initiating advanced optimization iterations.

5.4. CAD modeling of the Deriaz turbine's three-dimensional blade shape for the chosen site

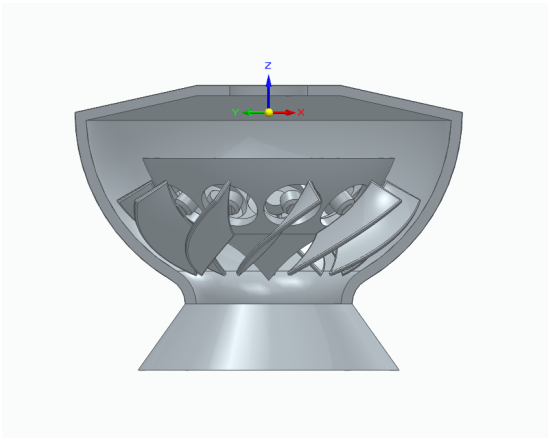
In order to verify the technical plausibility of the geometry generated by the numerical algorithm, beyond its purely mathematical consistency, a proof of concept has been carried out in a professional computer-aided design (CAD) environment. It is important to stress that the geometry used in this validation does not correspond to a generic case, but specifically to running the MATLAB code described in Chapter 4 with the specific set of design parameters obtained in Section 5.2.1 for the Saucelle II Hydroelectric Power Plant ($n = 150$ rpm, $R_{\text{int}} = 2.0$ m, $R_{\text{ext}} = 2.6$ m, $Z = 9$ blades, $\gamma_1 = 30°$, and $\gamma_2 = 60°$). The CAD model presented below is, therefore, the specific replacement proposal put forward for that site, and not an independent illustrative geometry. The blade's mean surface thus obtained (a zero-thickness surface resulting from the integration of the blade's differential equation) was exported in STL format and imported into Siemens Solid Edge.

From that surface, the *Thicken* tool was used to generate a workable solid from the

zero-thickness surface, confirming that the resulting geometry was topologically correct and operable within a standard CAD environment (Figure 5.1a), with no self-intersections or surface errors that would prevent its further processing. Based on this, the complete runner assembly (blade, spherical hub, and enclosing casing) was also modeled using a circular pattern, replicating the blade count obtained by the algorithm (Figure 5.1b).



(a) Individual blade with thickness generated from the exported STL surface.



(b) Assembly of the complete runner (blades, spherical hub, and casing) prior to 3D printing.

Fig. 5.1. Solid Edge modeling of the geometry generated by the numerical algorithm. Own elaboration.

As additional validation of the design's mechanical feasibility, a physical prototype of the assembly was manufactured through 3D printing (Figure 5.2), incorporating the blade articulation on bearings that make it possible to physically reproduce their orientation rotation (the double-regulation mechanism described in Chapter 4). This physical test made it possible to tangibly verify that, for the range of pitch angles considered in the design, the blades can pivot without interfering with or colliding against the conical-spherical walls of the hub and casing, confirming the geometric consistency of the clearance gap considered in the formulation.

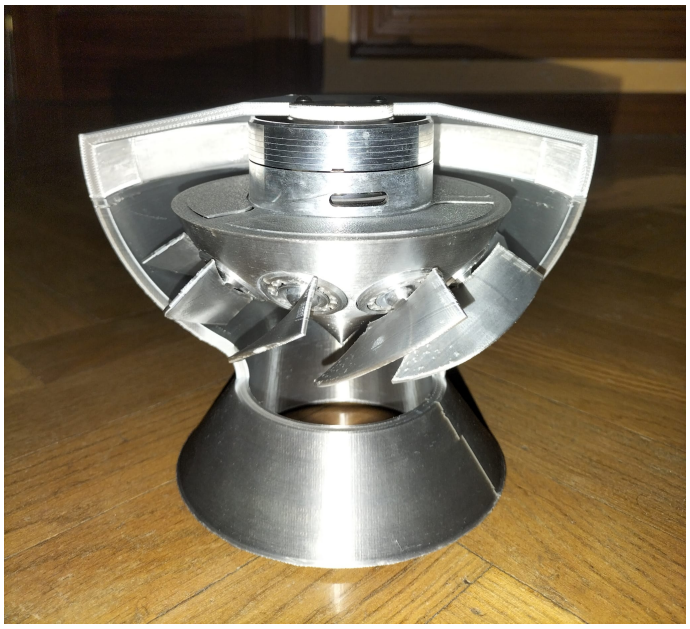


Fig. 5.2. 3D-printed physical prototype of the Deriaz runner, showing the blade articulation on bearings and the absence of interference with the spherical walls of the hub and casing during orientation rotation. Own elaboration.

In summary, this verification validates exclusively two aspects of the design: its geometric consistency (the surface generated by the algorithm is a real geometry, with no self-intersections or topological errors, convertible into a solid operable in CAD) and its kinematic feasibility (the blade orientation mechanism rotates freely, with no interference with the hub or casing, within the range of angles considered in the design). It does not validate, under any circumstances, the hydraulic behavior (efficiency, pressure distribution, cavitation), nor the structural or mechanical behavior under real service load, aspects that remain pending analysis through CFD and structural study, respectively, as discussed in Section 6.4.

6. CONCLUSIONS, BUDGET, AND FUTURE WORK

6.1. General conclusions

This Master's Thesis has developed, from first principles, a complete hydrodynamic design methodology for diagonal-flow, Deriaz-type hydraulic turbines, culminating in its implementation as a functional software tool, conceptually validated and applied to a real site.

The work starts from the rigorous analytical derivation of the Euler equation for turbomachines, built from the cylindrical coordinate reference frame and the physical concept of angular momentum, without assuming any starting kinematic relation. On this theoretical basis, the case of an axial turbomachine (Kaplan type) was solved first as a pedagogical step and methodological validation, developing the cubic interpolation law of the angle $\beta(r, z)$, the blade differential equation, and the hydrodynamic solidity criterion for determining the optimal blade count, all implemented and verified through a custom MATLAB script.

Building on this basis, the formulation was extended to the diagonal-flow case characteristic of the Deriaz turbine, incorporating the logarithmic mean spherical radius, the parametrization through the conical angle γ, the derivation of the meridional velocity in conical-spherical channels through the irrotationality condition in the meridional plane, and the corresponding blade differential equation adapted to the diagonal geometry. This formulation was likewise implemented as custom code, capable of generating the runner's complete three-dimensional geometry solely from the global design hydraulic parameters (net head, flow rate, rotational speed, and efficiencies), without requiring a starting geometry or CFD-based optimization loops.

This methodology was finally applied to a real case study, the Saucelle II Hydroelectric Power Plant, yielding a specific, dimensionally consistent design (synchronous speed of 150 rpm, spherical radii of 2.0 and 2.6 m, 9 blades according to the solidity criterion, and a radius ratio within the range recommended by the literature), demonstrating the practical applicability of the algorithm developed beyond the purely theoretical case.

As argued in the state of the art, none of the software tools or published methodologies reviewed offers an open, automated, publicly accessible procedure capable of solving this inverse design problem specifically for Deriaz turbines. This work's contribution lies not only in the mathematical formulation, but in having turned it into a computational tool reusable by the community, as described in the following section.

It should be honestly noted that the model developed relies on simplifying hypotheses explicitly assumed and documented throughout the work (infinite blade count, absence of viscous effects at certain stages, efficiencies fixed as input parameters), whose correction

and refinement are proposed as future lines of work in Section 6.4. The value of this thesis lies in establishing a solid, open numerical and computational basis on which to build such refinements.

6.2. Release as free software: the KaplanDeriaz3D project

With the aim of maximizing the impact and usefulness of the methodology developed, the entire code generated in this Master's Thesis has been released as open-source software under the name KaplanDeriaz3D, publicly available in the GitHub organization https://github.com/KaplanDeriaz3D.

The project is structured into three independent repositories:

- kaplanderiaz3d-matlab: a complete application developed with MATLAB App Designer, distributed both as source code and as an installable package (`.mlappinstall`) for direct integration into the MATLAB Add-Ons environment.
- kaplanderiaz3d-python: a complete port of the application in Python (PySide6 + PyVista), intended for users without access to a MATLAB license, replicating the full functionality of the original version.
- kaplanderiaz3d-theory: the calculation notebooks (MATLAB Live Scripts) and the theoretical documentation underpinning the development, serving as a bridge between this thesis and the published code.

Figures 6.1 and 6.2 show the graphical interface of the kaplanderiaz3d-matlab and kaplanderiaz3d-python applications, respectively.

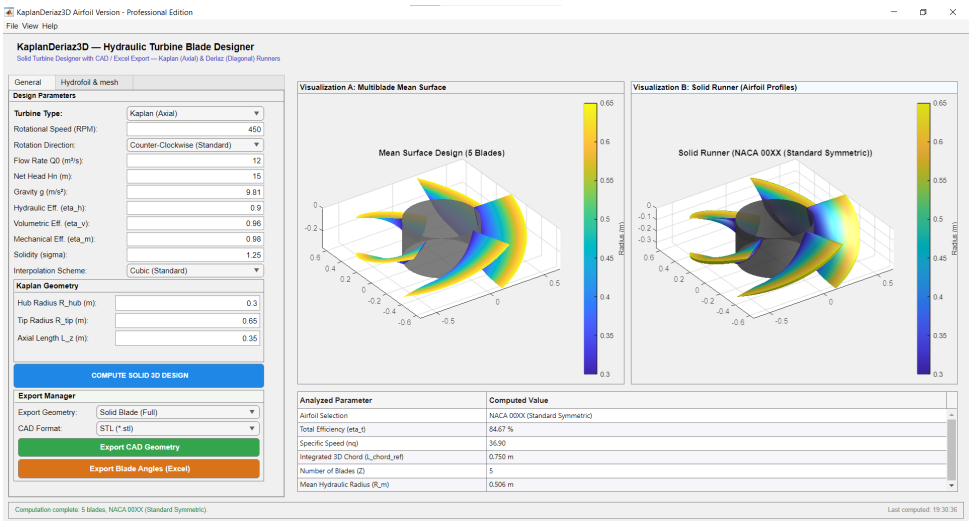


Fig. 6.1. Graphical interface of the KaplanDeriaz3D application for MATLAB, developed with MATLAB App Designer. Own elaboration.

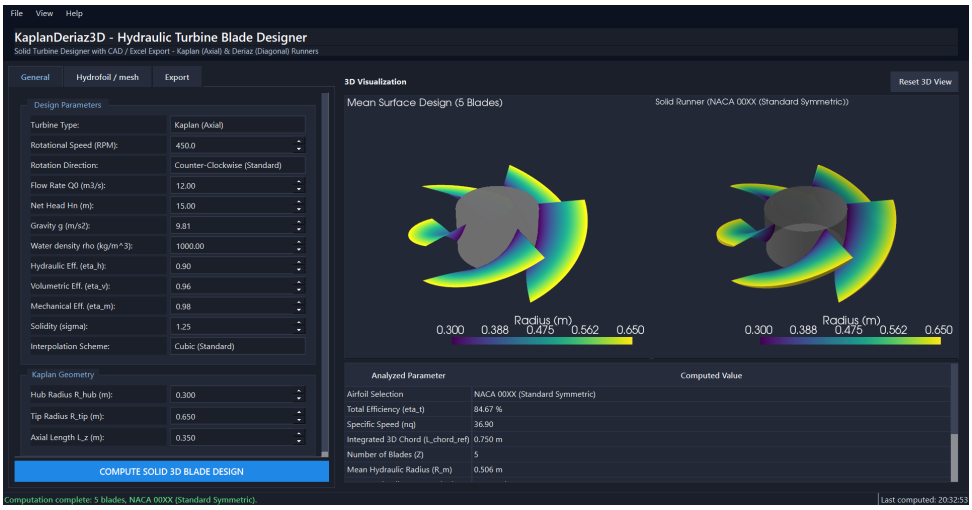


Fig. 6.2. Graphical interface of the KaplanDeriaz3D application for Python (PySide6 + PyVista). Own elaboration.

Unlike the calculation scripts developed in Chapters 3 and 4 of this thesis, focused on generating the blade's mean surface, the published application incorporates additional features aimed at turning the methodology into a professional-grade design tool:

- Generation of hydrodynamic profiles: unlike the mean-surface (skeleton line) modeling developed in the body of this thesis, the application incorporates the generation of complete hydrodynamic profiles based, among others, on NACA families [29] at each radial station, giving the blade real thickness and a solid three-dimensional geometry suitable for manufacturing, rather than a zero-thickness surface.
- CAD export: export of the generated solid geometry in STL and IGES formats, verified as operable in computer-aided design software (Solid Edge) for subsequent processing in the detail-engineering phase.
- Excel export: tabulated export of the blade's inlet and outlet angles (β_1, β_2) by radial station.
- Configuration management: saving and loading of complete design configurations in JSON format, allowing a specific design to be reproduced or shared immediately.
- Interactive 3D visualization: dual visualization panel (multi-blade mean surface and complete solid runner), with camera control and light/dark theme options.

The project is distributed entirely under the GNU General Public License v3.0 (GPLv3), which guarantees that anyone can freely use, study, modify, and redistribute the code, with the sole condition that any derivative work remains equally free software under the same license. Additionally, the MATLAB version has been linked to MATLAB File Exchange,

integrating it into the MathWorks community add-on ecosystem to maximize its visibility and make it easier for other researchers and industry engineers to discover.

It is important to stress that all of the software published is built entirely on the mathematical methodology developed in this Master's Thesis: it is not a generic tool adapted after the fact, but the direct, original implementation of the equations and algorithms described in Chapters 2 through 4 of this thesis. The declared intent of this release is for the project to be able to grow through community contributions, incorporating in the future the improvement lines described in Section 6.4, without necessarily depending on its original author for its maintenance and evolution.

6.3. Budget

This section breaks down an estimate of the cost associated with developing this Master's Thesis, considering 3 months of full-time dedication (8 hours per day), comparable to the hiring cost of a junior engineer in Spain, together with the costs of the software licenses used and other associated expenses.

6.3.1. Personnel costs

A dedication of 3 months at a rate of 8 hours per day, 5 days a week (13 weeks), is estimated, equivalent to 520 hours of work. A reference gross annual salary for a junior industrial engineer in Spain of €25,000 is assumed, to which the Social Security cost borne by the employer (approximately 30%) is added, resulting in a total cost to the employer of €32,500/year. Considering a standard annual working schedule of 1,800 hours, the hourly cost amounts to €18.06/h.

6.3.2. Software license costs

The cost, prorated over 3 months, of the software licenses used is included: MATLAB (standard individual license, approximately €800/year according to MathWorks' public rate) and Solid Edge (standard license, estimated cost of €1,600/year, used to verify the operability of the STL geometry exported by the code in a professional CAD environment).

6.3.3. Other costs

The proportional depreciation of the computer equipment used is included (professional-grade laptop, estimated acquisition cost of €1,200 and a useful life of 4 years), along with an allowance for indirect costs (internet connection, electricity supply, and other general workstation expenses).

Item	Quantity	Unit cost	Amount
Personnel costs (Junior Engineer)	520 h	€18.06/h	€9,391.20
MATLAB license (Individual, prorated 3 months)	1	€800/year	€200.00
Solid Edge license (Standard, prorated 3 months)	1	€1,600/year	€400.00
Computer equipment depreciation (3 months of 4 years)	1	€1,200	€75.00
Indirect costs (connection, electricity, etc.)	3 months	€50/month	€150.00
Subtotal direct costs			**€10,216.20**
Indirect structural costs (15%)			€1,532.43
Tax base			**€11,748.63**
VAT (21%)			€2,467.21
TOTAL BUDGET			**€14,215.84**

TABLE 6.1. ESTIMATED BUDGET FOR THE MASTER'S THESIS.

The project's total estimated budget amounts to fourteen thousand two hundred fifteen euros and eighty-four cents (€14,215.84), VAT included.

6.4. Future work

Throughout the development of this Master's Thesis, numerous lines of continuation have been identified that, for reasons of scope and available time, could not be addressed in this document. They are grouped below by subject area.

6.4.1. Numerical and experimental validation

- CFD simulation of the efficiency of the Deriaz geometry obtained, for different combinations of net head, flow rate, and rotational speed, including calculation of the flow's Reynolds number.
- CFD study of the optimal radius ratio between the hub and the casing/tip, and of the optimal range of inlet and outlet angles γ.
- Study of hydrodynamic profiles alternative to those already implemented in the application (beyond the current NACA family), and numerical validation of the existing profiles.
- Search for β-angle interpolation laws alternative to the cubic interpolation used (and that could vary with radius), aimed at suitably distributing pressure and reducing the chord at larger radii.
- Study of the maximum admissible blade thickness and width.

- Study of how the simplifying hypotheses adopted in the model (in particular, the absence of viscosity) actually influence reality, and determination of the corrections needed to bring the model closer to the fluid's real behavior.
- Specific cavitation studies and calculation of the available and required NPSH for the design obtained.

6.4.2. Refinement of the hydrodynamic model

- Correction of the theoretical Euler net head for a finite blade count ($H_{z\infty} \rightarrow H_z$) and the contribution of V_θ to the meridional velocity, currently not implemented owing to the circular interdependence it introduces into the algorithm.
- Incorporation of the blade blockage coefficient (τ) into the meridional velocity calculation as proposed by Morabito et al. [8], currently not implemented owing to the circular interdependence it introduces into the algorithm (see next point).
- Resolution of the circular-reference problem that arises when simultaneously applying the Pfleiderer correction, the contribution of V_θ to the meridional velocity, and the blade blockage factor (τ): these corrections depend on parameters (blade count and V_θ) that in turn require, for their calculation, quantities (corrected net head and meridional velocity) that depend on those same corrections. It would be of interest to develop an iterative or coupled solution scheme that robustly resolves this mutual dependency.
- Obtaining the hydraulic, volumetric, and organic efficiencies through more rigorous estimation methods specific to the Deriaz type, instead of the reference values adopted in this work.

6.4.3. Pump-mode behavior and regulation

- Detailed study of the Deriaz turbine's behavior in pump mode, quantifying the actual contribution of centrifugal force to the generated head.
- Analysis of how varying the blades' pitch angle can be used to regulate flow rate and power, both in turbine mode and in pump mode.
- Determination of the optimal location of the blades' pivot axis (the β-angle regulation pivot) so that the blade's mass is balanced, minimizing the torque required by the orientation servomechanism.

6.4.4. Structural analysis

- Structural study of the blades, in particular of the wake generated at the trailing edge, and assessment of whether removing it through geometric trimming is strictly necessary or could be improved.

6.4.5. Control and automation

- Implementation of a PLC control system that jointly regulates the distributor angle and the runner blade angle as a function of the reservoir level, the available flow rate, and the power demanded by the electrical grid.

6.4.6. Extension of the software tool

- Automatic calculation of the hub and casing cylindrical/spherical radii from the net head, the flow rate, the rotational speed, and the radius ratio ν, eliminating the need to enter them as input parameters.
- Incorporation of Francis turbine design into the KaplanDeriaz3D application, extending its scope beyond the Kaplan and Deriaz types.
- Integrated distributor design within the application, calculating the optimal angle of its vanes as a function of the flow rate and design head, which would allow the tool to be used for the actual control of any hydroelectric plant's distributor under different operating flow rates.
- Three-dimensional rendering of the distributor and calculation of the optimal number of guide vanes according to different design criteria.
- Selection of the most suitable hydrodynamic profile for the distributor's guide vanes.

BIBLIOGRAPHY

[1] Red Eléctrica de España, *Estructura de generación y almacenamiento acumulado progresivo – 1 de abril de 2026*, https://demanda.ree.es/visiona/peninsula/nacionalau/acumulada/2026-04-01, Accedido en agosto de 2026.

[2] Energy Encyclopedia, *Kaplan turbine*, https://www.energyencyclopedia.com/en/renewable-energy/water-energy/kaplan-turbine, Accedido en agosto de 2026.

[3] CNC TVAR, *Francis runners*, https://www.cnctvar.cz/products/hydro_turbines/francis_runners/, Accedido en agosto de 2026.

[4] Energy Education, University of Calgary, *Pelton turbine*, https://energyeducation.ca/encyclopedia/Pelton_turbine, Accedido en agosto de 2026.

[5] Jahobr, *Esquema de una turbina kaplan [ilustración en línea]*, 2016. [Online]. Available: https://commons.wikimedia.org/wiki/File:KaplanSketch.svg

[6] Jahobr, *Esquema de una turbina francis [ilustración en línea]*, 2016. [Online]. Available: https://commons.wikimedia.org/wiki/File:FrancisSketch.svg

[7] Jahobr, *Esquema de una turbina pelton [ilustración en línea]*, 2016. [Online]. Available: https://commons.wikimedia.org/wiki/File:PeltonSketch.svg

[8] A. Morabito, G. de Oliveira e Silva, and P. Hendrick, "Deriaz pump-turbine for pumped hydro energy storage and micro applications," *Journal of Energy Storage*, vol. 24, p. 100 788, 2019. DOI: 10.1016/j.est.2019.100788 [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S2352152X19302750

[9] Jahobr, *Esquema de una turbina deriaz [ilustración en línea]*, 2019. [Online]. Available: https://commons.wikimedia.org/wiki/File:Deriaz_Turbine_Sketch.svg

[10] A. E. Aeberli, "Deriaz type reversible pump-turbine installation at sir adam beck-niagara forebay pumped storage project," *Journal of Basic Engineering*, vol. 81, no. 4, pp. 505–515, 1959. DOI: 10.1115/1.4008540

[11] Fuji Electric Co., Ltd., *Site test of 2-speed deriaz type pump-turbine for kuromatagawa no. 2 power station, electric power development co., ltd.* https://www.fujielectric.com/about/technology/detail/review_10-04.html, Consultado el 17 de agosto de 2026, 1964.

[12] Mitsubishi Heavy Industries, Ltd., *Water turbine plant, reference projects*, https://www.mhi.com/business/products-services/energy-environment/renewable-energy/water-turbine-plant, Turbinas bomba Deriaz verticales de 2 x 103 MW suministradas a Hokkaido Electric Power Co. para la Central de Takami. Consultado el 17 de agosto de 2026.

[13] Voith GmbH & Co. KGaA, *Voith receives order for the modernization of pump turbines in two spanish hydropower plants*, https://www.voith.com/corp-en/news-room/press-releases/2020-09-08-vh-Voith-to-modernize-pump-turbines-for-two-Spanish-hydropower-plants.html, Consultado el 16 de agosto de 2026, Sep. 2020.

[14] P. Deriaz, "Control of feathering-blade runner," US2801068A, 1957. [Online]. Available: https://worldwide.espacenet.com/textdoc?DB=EPODOC&IDX=US2801068

[15] I. Kassanos, V. Alexopoulos, and J. Anagnostopoulos, "Numerical design methodology for reversible deriaz turbine with high energy performance and reduced fish impacts," *IOP Conference Series: Earth and Environmental Science*, vol. 1079, p. 012 076, 2022. DOI: 10.1088/1755-1315/1079/1/012076 [Online]. Available: https://iopscience.iop.org/article/10.1088/1755-1315/1079/1/012076

[16] H. Kikuta, R. Suzuki, S. Nakamura, and D. Anciger, "A runner modernization in a deriaz turbine hydropower station," *IOP Conference Series: Earth and Environmental Science*, vol. 240, p. 022 059, 2019. DOI: 10.1088/1755-1315/240/2/022059 [Online]. Available: https://www.researchgate.net/publication/332029606_A_runner_modernization_in_a_Deriaz_turbine_hydropower_station

[17] J. Schiffer, H. Jaberg, H. Benigni, and G. Pretis, "Double regulated mixed flow turbine for small hydro applications – design and operational experience," in *Proceedings of HYDRO 2022*, Strasbourg, France, 2022. [Online]. Available: https://www.jabergundpartner.com/fileadmin/user_upload/jabergundpartner/website_wissenswertes/2024_optimierung_diagonalturbine/Hydro2022_Paper_Double_Regulated_Mixed_Flow_Turbine_For_Small_Hydro_Applications.pdf

[18] NASA Glenn Research Center, *Turbo-design: Radial equilibrium code to design turbomachinery*, GitHub repository, 2024. [Online]. Available: https://github.com/nasa/turbo-design

[19] M. Gadalla, M. Tezzele, A. Mola, and G. Rozza, "BladeX: Python blade morphing," *Journal of Open Source Software*, vol. 4, no. 34, p. 1203, 2019. DOI: 10.21105/joss.01203 [Online]. Available: https://doi.org/10.21105/joss.01203

[20] B. P. Epps, M. J. Stanway, and R. W. Kimball, "OpenProp: An open-source design tool for propellers and turbines," in *SNAME Propeller and Shafting Symposium*, Williamsburg, VA, USA, 2009. [Online]. Available: https://www.epps.com/openprop

[21] OpenCFD Ltd, *Openfoam: The open source cfd toolbox*, https://www.openfoam.com/, Accedido en agosto de 2026, 2024.

[22] B. S. Thapa, B. Thapa, M. Eltvik, K. Gjosater, and O. G. Dahlhaug, "Optimizing runner blade profile of francis turbine to minimize sediment erosion," *IOP Conference Series: Earth and Environmental Science*, vol. 15, no. 3, p. 032 052, 2012. DOI: 10.1088/1755-1315/15/3/032052

[23] S. Leguizamón and F. Avellan, "Open-source implementation and validation of a 3D inverse design method for Francis turbine runners," *Energies*, vol. 13, no. 8, p. 2020, 2020. DOI: 10.3390/en13082020 [Online]. Available: https://infoscience.epfl.ch/handle/20.500.14299/168413

[24] Alab, *Alab – the hydropower workbench: Turbine designer module*, https://alab.no/, Accessed August 2026.

[25] S. L. Dixon, *Fluid Mechanics and Thermodynamics of Turbomachinery*, 4th. Oxford: Butterworth-Heinemann, 1998.

[26] A. Rusrl, *Exp 3: Free vortex and forced vortex*, Department of Mechanical Engineering, University of Malaya, [Online]. Available: Scribd. Accessed: Jul. 23, 2026, 2016. [Online]. Available: https://es.scribd.com/document/326842858/Exp-3-Free-Vortex-n-Forced-Vortex

[27] Wikimedia Commons, *Turbine and Rocky Reach Dam*, Accedido el 9 de julio de 2026, 2009. [Online]. Available: https://commons.wikimedia.org/wiki/File:Turbine_and_Rocky_Reach_Dam.jpg

[28] C.-H. Wu, "A general theory of three-dimensional flow in subsonic and supersonic turbomachines of axial-, radial-, and mixed-flow types," National Advisory Committee for Aeronautics, Washington, D.C., USA, Technical Note NACA TN 2604, 1952.

[29] I. H. Abbott, A. E. Von Doenhoff, and L. S. Stivers, "Summary of airfoil data," National Advisory Committee for Aeronautics (NACA), Tech. Rep. NACA Report 824, 1945, Consultado en agosto de 2026. [Online]. Available: https://ntrs.nasa.gov/citations/19930090976

A. APPENDIX A: COMPLETE MATLAB CODE FOR KAPLAN BLADE GENERATION

Note on code usage: The complete source code of the KaplanDeriaz3D software (MATLAB and Python versions) is distributed as free software under the GNU General Public License v3.0 (GPLv3). The code excerpts included in these appendices are reproduced for illustrative purposes of the work presented here. This MATLAB script contains the complete algorithmic implementation described in this document for the design, blade shaping, and 3D integration of an axial turbomachine's runner. This code is made available to the academic and technical community for free use, adaptation, distribution, or derivative development, with no commercial or academic restrictions.

```
% MATLAB SCRIPT: 3D KAPLAN DESIGN (THESIS)
% RUNNER ROTATION: COUNTERCLOCKWISE (viewed from above)
% PART 1: HYDRODYNAMIC CALCULATION AND GEOMETRY

clear; clc; close all;

% BLOCK 1: ENERGY AND GEOMETRIC PARAMETERS
RPM = 450; % Runner rotational speed (rpm)
Q0 = 12; % Nominal volumetric flow rate from the head (m^3/s)
Hn = 15; % Net head available at the plant (m)
g = 9.81; % Acceleration of gravity (m/s^2)
rho = 1000; % Density of water (kg/m^3)

% Turbomachine efficiencies
eta_h = 0.90; % Hydraulic efficiency
eta_v = 0.96; % Volumetric efficiency
eta_o = 0.98; % Organic/mechanical efficiency
eta_t = eta_h * eta_v * eta_o; % Total efficiency

% Main geometric dimensions
R_cubo = 0.3; % Runner inner radius (m)
R_punta = 0.65; % Runner outer radius (m)
L_z = 0.35; % Blade axial length (m)

R_m = sqrt((R_punta^2 + R_cubo^2) / 2); % Mean radius (m)

% BLOCK 2: FLUID KINEMATICS
omega = (2 * pi * RPM) / 60; % Angular velocity (rad/s)
Q_real = Q0 * eta_v; % Useful flow rate in the runner (m^3/s)
Area_paso = pi * (R_punta^2 - R_cubo^2); % Flow-through area (m^2)
V_z = Q_real / Area_paso; % Constant axial velocity
```

```
% Theoretical and useful power values
W_n = rho * g * Q0 * Hn; % Gross hydraulic power (W)
W_t = W_n * eta_t; % Useful mechanical power at the shaft (W)
H_inf = Hn * eta_h; % Theoretical Euler energy (m)

% BLOCK 3: SPATIAL DISCRETIZATION
N_radios = 25; % Number of streamlines
N_cuerda = 40; % Number of axial stations (dz)

X = zeros(N_cuerda, N_radios);
Y = zeros(N_cuerda, N_radios);
Z = zeros(N_cuerda, N_radios);

z_vec = linspace(0, -L_z, N_cuerda);
r_vec = linspace(R_cubo, R_punta, N_radios);

% BLOCK 4: VELOCITY TRIANGLES AND 3D INTEGRATION
beta1_vec = zeros(1, N_radios);
beta2_vec = zeros(1, N_radios);

for i = 1:N_radios
    r = r_vec(i);
    U = omega * r;

    % INLET: Azimuthal component
    V_theta1 = (g * H_inf) / (omega * r);
    beta1 = atan2(V_z, (U - V_theta1));
    beta1_vec(i) = beta1;

    % OUTLET: Ideal nominal condition (V_theta2 = 0)
    beta2 = atan2(V_z, U);
    beta2_vec(i) = beta2;

    % Cubic interpolation law for the blade
    t_dim = abs(z_vec) / L_z;
    polinomio_cubico = 3*(t_dim.^2) - 2*(t_dim.^3);
    beta_z = beta1 + (beta2 - beta1) .* polinomio_cubico;

    % Numerical integration of the differential equation
    theta_rel = 0;
    for j = 1:N_cuerda
        if j > 1
            dz = abs(z_vec(j) - z_vec(j-1));
            d_theta_rel = (cot(beta_z(j)) / r) * dz;
            theta_rel = theta_rel - d_theta_rel; % Counterclockwise
        end

        X(j, i) = r * cos(theta_rel);
        Y(j, i) = r * sin(theta_rel);
        Z(j, i) = z_vec(j);
    end
```

```
end

% PART 2: OPTIMAL BLADE SIZING AND REPORT

% BLOCK 5: CALCULATION OF THE OPTIMAL BLADE COUNT (Z)
% Index of the mean radius R_m
[~, idx_Rm] = min(abs(r_vec - R_m));

% Curvilinear integration of the 3D chord at R_m
dx = diff(X(:, idx_Rm));
dy = diff(Y(:, idx_Rm));
dz = diff(Z(:, idx_Rm));
dL = sqrt(dx.^2 + dy.^2 + dz.^2);
L_cuerda_real = sum(dL); % Actual integrated 3D chord (m)

sigma_target = 1.25; % Target hydrodynamic solidity

% Analytical determination of Z
paso_tangencial_req = L_cuerda_real / sigma_target;
Z_calculado = (2 * pi * r_vec(idx_Rm)) / paso_tangencial_req;
Z_optimo = round(Z_calculado);

% Physical safety limit
Z_optimo = max(3, min(Z_optimo, 12));

% Kinematic specific speed (nq)
nq = RPM * sqrt(Q0) / ((g*Hn)^(3/4));

% Results report
fprintf('================================================\n');
fprintf(' SIZING RESULTS (THESIS) \n');
fprintf('================================================\n');
fprintf('Specific angular speed (nq) : %.2f rpm\n', nq);
fprintf('Useful flow rate in the runner (Q_real) : %.2f m^3/s\n', ...
    Q_real);
fprintf('Useful shaft power (Wt) : %.2f kW\n', ...
    W_t / 1000);
fprintf('Integrated 3D chord at Rm (L_cuerda) : %.3f m\n', ...
    L_cuerda_real);
fprintf('Optimal blade count (Z) : %d blades\n', ...
    Z_optimo);
fprintf('================================================\n\n');

% PART 3: 3D VISUAL RENDERING

% BLOCK 6: FIGURE 1 - SINGLE REFERENCE BLADE
figure('Name', ...
    'Figure 1: Single Blade - Integration Mesh', ...
    'Color', 'w');

surf(X, Y, Z, r_vec .* ones(N_cuerda, 1), ...
```

```
    'FaceColor', 'interp', ...
    'EdgeColor', [0.1 0.1 0.1], ...
    'EdgeAlpha', 0.2);
hold on;

% Inner hub
[X_cubo, Y_cubo, Z_cubo] = cylinder(R_cubo, 40);
Z_cubo = Z_cubo * (-L_z);
mesh(X_cubo, Y_cubo, Z_cubo, ...
    'FaceColor', [0.5 0.5 0.5], ...
    'FaceAlpha', 0.15, ...
    'EdgeColor', [0.4 0.4 0.4]);

% Outer casing
[X_punta, Y_punta, Z_punta] = cylinder(R_punta, 40);
Z_punta = Z_punta * (-L_z);
mesh(X_punta, Y_punta, Z_punta, ...
    'FaceColor', [0.3 0.6 0.9], ...
    'FaceAlpha', 0.10, ...
    'EdgeColor', [0.6 0.6 0.6], ...
    'EdgeAlpha', 0.1);

% Direction-of-rotation arrow (Counterclockwise)
ang_flecha = linspace(0, pi/2, 20);
r_flecha = R_punta * 0.6;
z_flecha = 0.05;
plot3(r_flecha*cos(ang_flecha), ...
    r_flecha*sin(ang_flecha), ...
    z_flecha*ones(size(ang_flecha)), ...
    'r-', 'LineWidth', 3);
plot3(0, r_flecha, z_flecha, 'r^', ...
    'MarkerSize', 10, ...
    'MarkerFaceColor', 'r');
text(0.05, r_flecha + 0.05, z_flecha + 0.03, ...
    'Runner Rotation (\omega)', 'Color', 'r', ...
    'FontSize', 10, 'FontWeight', 'bold');

title('3D Geometry of a Single Blade');
xlabel('X Axis (m)');
ylabel('Y Axis (m)');
zlabel('Z Axis (Turbine Axis) [m]');
colormap(parula(256));
h_bar = colorbar;
ylabel(h_bar, 'Local Radius r (m)', 'FontSize', 10);
clim([R_cubo, R_punta]);
axis equal; grid on; view(-25, 25);
camlight('headlight'); lighting gouraud; shading interp;

% BLOCK 7: FIGURE 2 - COMPLETE OPTIMIZED RUNNER
figure('Name', ...
    sprintf('Figure 2: Complete Runner (%d Blades)', Z_optimo), ...
```

```
    'Color', 'w');
hold on;

delta_angle = (2 * pi) / Z_optimo;
[R_grid, ~] = meshgrid(r_vec, z_vec);

% Angular cloning of the blade
for k = 1:Z_optimo
    current_angle = (k - 1) * delta_angle;

    X_rot = X * cos(current_angle) - Y * sin(current_angle);
    Y_rot = X * sin(current_angle) + Y * cos(current_angle);
    Z_rot = Z;

    surf(X_rot, Y_rot, Z_rot, R_grid, ...
        'FaceColor', 'interp', ...
        'EdgeColor', 'none');
end

% Central mechanical hub
[X_cubo_sol, Y_cubo_sol, Z_cubo_sol] = cylinder(R_cubo, 60);
Z_cubo_sol = Z_cubo_sol * (-L_z);

C_cubo = R_cubo * ones(size(Z_cubo_sol));
surf(X_cubo_sol, Y_cubo_sol, Z_cubo_sol, C_cubo, ...
    'FaceColor', [0.4 0.4 0.4], ...
    'EdgeColor', 'none', ...
    'FaceAlpha', 0.95);

% Hub end caps
X_tapa_sup = [zeros(1, 61); X_cubo_sol(1,:)];
Y_tapa_sup = [zeros(1, 61); Y_cubo_sol(1,:)];
Z_tapa_sup = zeros(2, 61);
C_tapa = R_cubo * ones(2, 61);
surf(X_tapa_sup, Y_tapa_sup, Z_tapa_sup, C_tapa, ...
    'FaceColor', [0.3 0.3 0.3], ...
    'EdgeColor', 'none');

Z_tapa_inf = -L_z * ones(2, 61);
surf(X_tapa_sup, Y_tapa_sup, Z_tapa_inf, C_tapa, ...
    'FaceColor', [0.3 0.3 0.3], ...
    'EdgeColor', 'none');

% Outer casing
mesh(X_punta, Y_punta, Z_punta, ...
    'FaceColor', [0.3 0.6 0.9], ...
    'FaceAlpha', 0.10, ...
    'EdgeColor', [0.6 0.6 0.6], ...
    'EdgeAlpha', 0.1);

% Direction-of-rotation indicator
```

```
plot3(r_flecha*cos(ang_flecha), ...
    r_flecha*sin(ang_flecha), ...
    z_flecha*ones(size(ang_flecha)), ...
    'r-', 'LineWidth', 3);
plot3(0, r_flecha, z_flecha, 'r^', ...
    'MarkerSize', 10, ...
    'MarkerFaceColor', 'r');
text(0.05, r_flecha + 0.05, z_flecha + 0.03, ...
    'Runner Rotation (\omega)', 'Color', 'r', ...
    'FontSize', 10, 'FontWeight', 'bold');

title(sprintf('Axial Kaplan Runner - %d-Blade Assembly', ...
    Z_optimo));
xlabel('X Axis (m)');
ylabel('Y Axis (m)');
zlabel('Z Axis (Turbine Axis) [m]');
colormap(parula(256));
h_bar2 = colorbar;
ylabel(h_bar2, 'Local Radius r (m)', 'FontSize', 10);
clim([R_cubo, R_punta]);
axis equal; grid on; view(-35, 30);
camlight('headlight'); lighting gouraud; shading interp;
```

B. APPENDIX B: COMPLETE MATLAB CODE FOR DERIAZ BLADE GENERATION

Note on code usage: This MATLAB script contains the complete algorithmic implementation described in this document for the design, blade shaping, and 3D integration of a diagonal turbomachine's runner. This code is made available to the academic and technical community for free use, adaptation, distribution, or derivative development, with no commercial or academic restrictions.

```
% MATLAB SCRIPT: 3D DESIGN OF THE DERIAZ TURBINE BLADE (THESIS)
% RUNNER ROTATION: COUNTERCLOCKWISE (viewed from above)
%
% This code numerically solves the differential equation
% for a conical-spherical channel considering efficiencies.
clear; clc; close all;

%% BLOCK 1: ENERGY AND GEOMETRIC PARAMETERS
RPM = 150; % Runner rotational speed (rpm)
Q0 = 110; % Nominal volumetric flow rate (m^3/s)
Hn = 60; % Net head available (m)
g = 9.81; % Acceleration of gravity (m/s^2)

% Turbomachine efficiencies
eta_h = 0.91; % Hydraulic efficiency
eta_v = 0.97; % Volumetric efficiency
eta_o = 0.98; % Organic/mechanical efficiency
eta_t = eta_h * eta_v * eta_o; % Total efficiency

% Boundary spherical radii (Hub and Casing):
Re_int = 2; % Hub spherical radius (m)
Re_ext = 2.6; % Casing spherical radius (m)

% Meridional path limits (conical angles):
gamma1_deg = 30; % Runner inlet (degrees)
gamma2_deg = 60; % Runner outlet (degrees)

%% BLOCK 2: KINEMATICS AND CONVERSIONS
omega = (2 * pi * RPM) / 60; % Angular velocity (rad/s)
Q_real = Q0 * eta_v; % Useful flow rate (m^3/s)
gamma1 = deg2rad(gamma1_deg); % Inlet tilt angle (rad)
gamma2 = deg2rad(gamma2_deg); % Outlet tilt angle (rad)

% Theoretical and useful head powers
W_n = 1000 * g * Q0 * Hn; % Gross power available (W)
W_t = W_n * eta_t; % Useful shaft power (W)
H_inf = Hn * eta_h; % Theoretical Euler energy (m)
```

```
% Equivalent mean spherical radius:
Re_medio = (Re_ext - Re_int) / log(Re_ext / Re_int);

%% BLOCK 3: SPATIAL DISCRETIZATION AND MATRICES
N_radios = 30; % Streamlines between hub and casing
N_cuerda = 50; % Steps in the flow direction (gamma)

Re_vec = linspace(Re_int, Re_ext, N_radios);
gamma_vec = linspace(gamma1, gamma2, N_cuerda);

X = zeros(N_cuerda, N_radios);
Y = zeros(N_cuerda, N_radios);
Z = zeros(N_cuerda, N_radios);
RC = zeros(N_cuerda, N_radios);

%% BLOCK 4: 3D NUMERICAL INTEGRATION
for i = 1:N_radios
    Re = Re_vec(i); % Spherical radius of the current streamline

    % --- 1. VELOCITY TRIANGLE AT THE INLET (gamma1) ---
    rc1 = Re * cos(gamma1); % Inlet cylindrical radius
    U1 = omega * rc1; % Inlet drag velocity
    Vm1 = Q_real / (2 * pi * Re * cos(gamma1) * ...
          (Re_ext - Re_int)); % Meridional Vm

    % Azimuthal component required by Euler:
    Vtheta1 = (g * H_inf) / U1;

    % Physical safety check:
    if Vtheta1 >= U1
        error('Error at Re = %.3f m: Vtheta1 >= U1.', ...
              Re);
    end

    beta1 = atan2(Vm1, (U1 - Vtheta1)); % Inlet angle

    % --- 2. VELOCITY TRIANGLE AT THE OUTLET (gamma2) ---
    rc2 = Re * cos(gamma2); % Outlet cylindrical radius
    U2 = omega * rc2; % Outlet drag velocity
    Vm2 = Q_real / (2 * pi * Re * cos(gamma2) * ...
          (Re_ext - Re_int)); % Accelerated Vm

    beta2 = atan2(Vm2, U2); % Outlet angle (Vtheta2 = 0)

    % --- 3. CUBIC INTERPOLATION LAW BETA(gamma) ---
    t = (gamma_vec - gamma1) / (gamma2 - gamma1);
    polinomio_cubico = 3*(t.^2) - 2*(t.^3);
    beta_gamma = beta1 + (beta2 - beta1) .* ...
                 polinomio_cubico;
```

```
    % --- 4. NUMERICAL INTEGRATION OF THE ODE ---
    theta_rel = 0;

    for j = 1:N_cuerda
        gamma_actual = gamma_vec(j);

        if j > 1
            d_gamma = gamma_vec(j) - gamma_vec(j-1);

            % Evaluation at the midpoint
            gamma_mid = (gamma_vec(j) + ...
                         gamma_vec(j-1)) / 2;
            beta_mid = (beta_gamma(j) + ...
                        beta_gamma(j-1)) / 2;

            % Differential equation of the Deriaz blade:
            d_theta_rel = (cot(beta_mid) / ...
                           cos(gamma_mid)) * d_gamma;

            % Counterclockwise rotation
            theta_rel = theta_rel - d_theta_rel;
        end

        % Local cylindrical coordinates:
        rc_local = Re * cos(gamma_actual);
        z_local = -Re * sin(gamma_actual);

        % Conversion to 3D Cartesian coordinates:
        X(j, i) = rc_local * cos(theta_rel);
        Y(j, i) = rc_local * sin(theta_rel);
        Z(j, i) = z_local;
        RC(j, i) = rc_local;
    end
end

%% BLOCK 5: CALCULATION OF THE OPTIMAL BLADE COUNT (Z)
% 1. Streamline closest to Re_medio
[~, idx_Rmedio] = min(abs(Re_vec - Re_medio));

% 2. Exact curvilinear integration of the 3D chord
dx = diff(X(:, idx_Rmedio));
dy = diff(Y(:, idx_Rmedio));
dz = diff(Z(:, idx_Rmedio));
dL = sqrt(dx.^2 + dy.^2 + dz.^2);
L_cuerda_real = sum(dL); % Actual 3D chord (m)

% 3. Equivalent mean cylindrical radius
rc_promedio_medio = mean(RC(:, idx_Rmedio));

% 4. Target hydrodynamic solidity
sigma_target = 1.4; % Recommended value for Deriaz
```

```
% 5. Determination of the blade count Z
paso_tangencial_req = L_cuerda_real / sigma_target;
Z_calculado = (2 * pi * rc_promedio_medio) / ...
            paso_tangencial_req;
Z_optimo = round(Z_calculado);

% Physical safety limits
Z_optimo = max(4, min(Z_optimo, 12));

% Kinematic specific speed (nq)
nq = RPM * sqrt(Q0) / ((g*Hn)^(3/4));

fprintf('=========================================\n');
fprintf(' DERIAZ SIZING RESULTS (THESIS) \n');
fprintf('=========================================\n');
fprintf('Specific speed (nq) : %.2f rpm\n', nq);
fprintf('Useful flow rate (Q_real) : %.2f m^3/s\n', Q_real);
fprintf('Shaft power (Wt) : %.2f kW\n', W_t / 1000);
fprintf('3D chord (L_cuerda) : %.3f m\n', L_cuerda_real);
fprintf('Blade count (Z) : %d blades\n', Z_optimo);
fprintf('=========================================\n\n');

%% BLOCK 6: 3D VISUAL RENDERING - SINGLE BLADE
figure('Name', 'Figure 1: Single Deriaz Blade', ...
       'Color', 'w');

surf(X, Y, Z, RC, 'FaceColor', 'interp', ...
     'EdgeColor', [0.1 0.1 0.1], 'EdgeAlpha', 0.25);
hold on;

% Spherical contour mesh (Hub and Casing)
[theta_grid, gamma_grid] = meshgrid(...
    linspace(0, 2*pi, 60), linspace(gamma1, gamma2, 30));

% Inner contour (Hub):
X_cubo = Re_int * cos(gamma_grid) .* cos(theta_grid);
Y_cubo = Re_int * cos(gamma_grid) .* sin(theta_grid);
Z_cubo = -Re_int * sin(gamma_grid);
mesh(X_cubo, Y_cubo, Z_cubo, 'FaceColor', [0.5 0.5 0.5], ...
     'FaceAlpha', 0.20, 'EdgeColor', [0.4 0.4 0.4]);

% Outer contour (Casing):
X_carcasa = Re_ext * cos(gamma_grid) .* cos(theta_grid);
Y_carcasa = Re_ext * cos(gamma_grid) .* sin(theta_grid);
Z_carcasa = -Re_ext * sin(gamma_grid);
mesh(X_carcasa, Y_carcasa, Z_carcasa, ...
     'FaceColor', [0.2 0.5 0.8], 'FaceAlpha', 0.08, ...
     'EdgeColor', [0.3 0.7 0.9], 'EdgeAlpha', 0.20);

% Spherical Center (0,0,0)
```

```
plot3(0, 0, 0, 'o', 'MarkerSize', 7, ...
      'MarkerFaceColor', [0.0 0.2 0.5], ...
      'MarkerEdgeColor', 'none');

% Root attachment point at the hub
j_mid = round(N_cuerda / 2);
x_raiz = X(j_mid, 1);
y_raiz = Y(j_mid, 1);
z_raiz = Z(j_mid, 1);

% Subtle dashed axis
color_eje_sutil = [0.35 0.45 0.60 0.45];
plot3([0, x_raiz], [0, y_raiz], [0, z_raiz], '--', ...
      'Color', color_eje_sutil, 'LineWidth', 1.5);

plot3(x_raiz, y_raiz, z_raiz, 'o', 'MarkerSize', 5, ...
      'MarkerFaceColor', [0.1 0.3 0.6], ...
      'MarkerEdgeColor', 'none');

text(0.02, 0.02, 0.05, 'Spherical Center (0,0,0)', ...
     'Color', [0.0 0.2 0.5], 'FontSize', 9, ...
     'FontWeight', 'bold');

% Direction-of-rotation indicator
ang_flecha = linspace(0, pi/2, 20);
r_flecha = Re_ext * cos(gamma1) * 0.7;
z_flecha = -Re_ext * sin(gamma1) + 0.05;

plot3(r_flecha*cos(ang_flecha), r_flecha*sin(ang_flecha), ...
      z_flecha*ones(size(ang_flecha)), 'r-', ...
      'LineWidth', 2.5);
plot3(0, r_flecha, z_flecha, 'r^', 'MarkerSize', 8, ...
      'MarkerFaceColor', 'r');
text(0.05, r_flecha + 0.05, z_flecha + 0.03, ...
     'Runner Rotation (\omega)', 'Color', 'r', ...
     'FontSize', 9, 'FontWeight', 'bold');

title('3D Geometry of a Single Blade and Axis');
xlabel('X Axis (m)'); ylabel('Y Axis (m)');
zlabel('Z Axis (m)');
colormap(parula(256));
clim([min(RC(:)), max(RC(:))]);
h_bar = colorbar;
ylabel(h_bar, 'Cylindrical Radius r_c (m)', 'FontSize', 10);
axis equal; grid on; view(-35, 25);
camlight('headlight'); lighting gouraud; shading interp;

%% BLOCK 7: 3D VISUAL RENDERING - COMPLETE RUNNER
figure('Name', sprintf('Figure 2: Runner (%d Blades)', ...
       Z_optimo), 'Color', 'w');
hold on;
```

```
delta_angle = (2 * pi) / Z_optimo;

% Spherical Center (0,0,0)
plot3(0, 0, 0, 'o', 'MarkerSize', 8, ...
     'MarkerFaceColor', [0.0 0.2 0.5], ...
     'MarkerEdgeColor', 'none');
text(0.03, 0.03, 0.06, 'Spherical Center (0,0,0)', ...
    'Color', [0.0 0.2 0.5], 'FontSize', 9, ...
    'FontWeight', 'bold');

% Angular cloning of blades
for k = 1:Z_optimo
    current_angle = (k - 1) * delta_angle;

    % Rotation of the blade surface
    X_rot = X * cos(current_angle) - ...
            Y * sin(current_angle);
    Y_rot = X * sin(current_angle) + ...
            Y * cos(current_angle);
    Z_rot = Z;

    surf(X_rot, Y_rot, Z_rot, RC, 'FaceColor', 'interp', ...
         'EdgeColor', 'none');

    % Root coordinate for blade k
    x_raiz_k = X_rot(j_mid, 1);
    y_raiz_k = Y_rot(j_mid, 1);
    z_raiz_k = Z_rot(j_mid, 1);

    plot3([0, x_raiz_k], [0, y_raiz_k], [0, z_raiz_k], ...
          '--', 'Color', color_eje_sutil, ...
          'LineWidth', 1.3);

    plot3(x_raiz_k, y_raiz_k, z_raiz_k, 'o', ...
          'MarkerSize', 4.5, ...
          'MarkerFaceColor', [0.1 0.3 0.6], ...
          'MarkerEdgeColor', 'none');
end

% Solid rendering of the hub
surf(X_cubo, Y_cubo, Z_cubo, Re_int * cos(gamma_grid), ...
    'FaceColor', [0.4 0.4 0.4], 'EdgeColor', 'none', ...
    'FaceAlpha', 0.95);

% Outer casing contour
mesh(X_carcasa, Y_carcasa, Z_carcasa, ...
    'FaceColor', [0.2 0.5 0.8], 'FaceAlpha', 0.15, ...
    'EdgeColor', [0.4 0.7 0.9], 'EdgeAlpha', 0.15);

title(sprintf('Diagonal Deriaz Runner (%d Blades)', Z_optimo));
```

```
xlabel('X Axis (m)'); ylabel('Y Axis (m)');
zlabel('Z Axis (m)');
colormap(parula(256));
clim([min(RC(:)), max(RC(:))]);
h_bar2 = colorbar;
ylabel(h_bar2, 'Cylindrical Radius r_c (m)', 'FontSize', 10);
axis equal; grid on; view(-35, 30);
camlight('headlight'); lighting gouraud; shading interp;
```

C. APPENDIX C: DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GAI)

Note: This appendix reproduces the content of the Declaration of Use of Generative Artificial Intelligence (GAI) completed by the author, in accordance with the official Universidad Carlos III de Madrid template for Master's Theses.

Have I used GAI in my Master's Thesis? YES.

Having marked YES, the three parts of this declaration are completed below.

C.1. Part 1: declaration on legal, ethical, and responsible behavior

The use of GAI carries risks and can lead to serious academic consequences: the Master's Thesis would not be evaluated by the University if such use involved the use of confidential data, copyrighted materials, or personal data without meeting the conditions required in each case (authorization from the parties concerned, authorization from the rights holders, compliance with the University's instructions). In this regard, the following is declared:

- *In my interaction with GAI tools, I have provided confidential data always with the due authorization of the parties concerned.* Confidentiality covers all information that a person or organization wishes to protect for legal, commercial, privacy, or strategic reasons (such as patents or trade secrets).
  Answer: NO, I have not used confidential data.

- *In my interaction with GAI tools, I have provided copyrighted materials always with the authorization of the respective rights holders.*
  Answer: NO, I have not used copyrighted materials.

- *In my interaction with GAI tools, I have provided personal data with the due authorization of the parties concerned.*
  Answer: NO, I have not used personal data.

- *My use of the GAI tool has respected its terms of use, as well as the essential ethical principles, not directing it maliciously to obtain a result inappropriate for the work presented, that is, one that produces an impression or understanding contrary to the reality of the results obtained, that supplants my own work, or that could result in harm to other people.*
  Answer: YES.

C.2. Part 2: declaration of technical use

I declare having made use of GAI systems (Claude Sonnet and Gemini Flash) for:

C.2.1. Documentation and writing

- **Review or rewriting of previously drafted paragraphs:** For example, when writing the introduction I used Gemini and Claude to refine the explanation of the flow directions in Kaplan and Francis turbines. I had already written an explanation of how the flow entered and left axially in the Kaplan, and similarly in the Francis (content covered in the master's classes), but I felt I had not expressed it in the best way, so I asked both AIs to rewrite it more clearly and then selected whichever of the two versions I liked best.
- **Information search or answers to specific questions:** For example, I used Gemini Flash to tell me in which chapters of S. L. Dixon's book, *Fluid Mechanics and Thermodynamics of Turbomachinery* (a key reference in my bibliography), concepts such as the free vortex, solidity, or the Reynolds Transport Theorem could be found. These were concepts I already knew but needed to reference; the AI indicated their approximate location and, after checking it, it sometimes got the chapter wrong and other times got it right, so I manually corrected the discrepancies. Even so, it was useful for getting an approximate idea of where the concepts might be found.

C.2.2. Developing specific content

GAI has been used as a support tool for developing the Master's Thesis's specific content, including:

- **Assistance in developing lines of code (programming):** For example, I used Claude Sonnet to port the application from MATLAB to Python, since I do not have as much knowledge of Python.
- **Optimization processes:** I requested help to fix and explain errors in code that was not working correctly. Errors often occurred and I could not identify why (sometimes they were trivial, such as a missing semicolon or an unclosed parenthesis, but other times they were deeper errors).

C.3. Part 3: reflection on usefulness

For me, AIs have two main problems: the first is that users often do not know how to explain clearly and unambiguously what they want from the AI, and therefore the artificial

intelligence gives an unsatisfactory result, but purely through the user's fault, for not having phrased the request well. The second problem I have noticed is that the AI cannot access the full range of documents that a student, by having a university email address, can access, so it is currently impractical to delegate autonomy in bibliographic research to it, at least for now; for that reason I did not use it for that purpose except as a duplicate of the review I was carrying out in parallel myself, always questioning its answers.

Its advantages are that it can program in any language with much more fluency and optimization than I can, and surely than most students, although again often not autonomously. Also, when writing, it is very convenient to give it a vaguely written paragraph (without using the impersonal form, without accents, without attention to style) so that it can refine it and adapt it to the document's style, always with the user reviewing it. It is also quite useful for questioning the work's methodology and conclusions, as if it were another reviewer, although in my opinion the author's judgment should always prevail over the AI's criticisms.